## INTRODUCTION

This Maintenance Procedure currently exists as a part of a Gas Turbine Service Manual, and is intended to establish a consistent approach to the performance of flange-to-flange maintenance inspections for a MS6001B gas turbine. This procedure is intended to consolidate actual field experience and other maintenance related information normally available only from widely diverse sources. As maintenance experience is gained on this unit, this procedure will be supplemented with additional information and descriptions of the work to be performed. This article currently provides inspection information for Combustion Inspection workscope. The procedures described in the text are not intended to supersede or replace specific product department documentation.

## COMBUSTION INSPECTION

**CAUTION: Before proceeding with a combustion inspection ensure that the gas turbine electrical power is tagged-out; CO2 system is deactivated, liquid fuel system is purged, deactivated and/or the gas supply is blanked off.**

## DISASSEMBLY

### OPERATION 1 - PREPARE TURBINE COMPARTMENT ROOF FOR REMOVAL

1. Disconnect the blower(s) motor leads, piping flanges and tubing connections at the topside and underside of the roof.

2. Break flanges on exhaust frame cooling piping and all tubing connections.

3. Disconnect VA14 valve electrical wiring and conduit (if applicable to unit).

4. Disconnect CO2 piping and any other piping which may interfere with roof removal.

5. Unbolt the roof from the sidewall frames.

6. Check the topside and underside to ensure roof (with blowers attached) is mechanically free to be lifted.

### OPERATION 2 - REMOVE TURBINE COMPARTMENT ROOF AND SIDE PANELS

1. Attach four eyebolts to the roof section threaded receptacles provided for lifting.

2. Rig the roof (with blower(s) attached) for lifting with a chainfall or come-along in the rigging.

3. Lift the roof off and set on suitable cribbing.

4. Remove turbine compartment side panels to gain access to turbine compartment. Store panels and/or doors in an upright position in a protected area.

**NOTE:** To gain easier access to the turbine area to remove bolting and piping, it may be desirable to remove some of the side panel vertical support beams.

**OPERATION 3 - REMOVE LIQUID FUEL LINES**

1. Identify and tag each fuel line so it can be reassembled to the same combustor from which it was removed.

2. Prior to loosening and/or removal of the tube fittings, non-destructively matchmark the nut and fitting body. This matchmark should be different from any other existing matchmark on these fitting parts.

**Note:** On the new Swagelok fittings, a stop collar is incorporated between the nut and the fitting body. Matchmarking is not required on fittings having the stop collar. See Fig. CI-D3.

3. Disconnect the fuel lines from the fuel check valves and from the fuel supply fittings. Hold the fittings steady with a backup wrench to prevent twisting the fitting on the tube. See Figs. CI-D2 and CI-D3.

**CAUTION: Do not use pipe wrenches or adjustable wrenches which will result in fitting damage.**

a. Always use open-end, or tubing box wrenches.

b. Cap the open end of the fuel lines to prevent entry of dirt, insects or other contaminants.

**CAUTION: Do not use improper size caps or rags to cover pipe openings, they may be accidentally pushed inside the tubing.**

c. Where applicable, liquid fuel purge lines will have to be removed. Follow the same general directions as stated above.

**Note:** When fuel purge lines are clamped to the atomizing air lines. It is advisable to leave the clamps intact and remove both lines together.

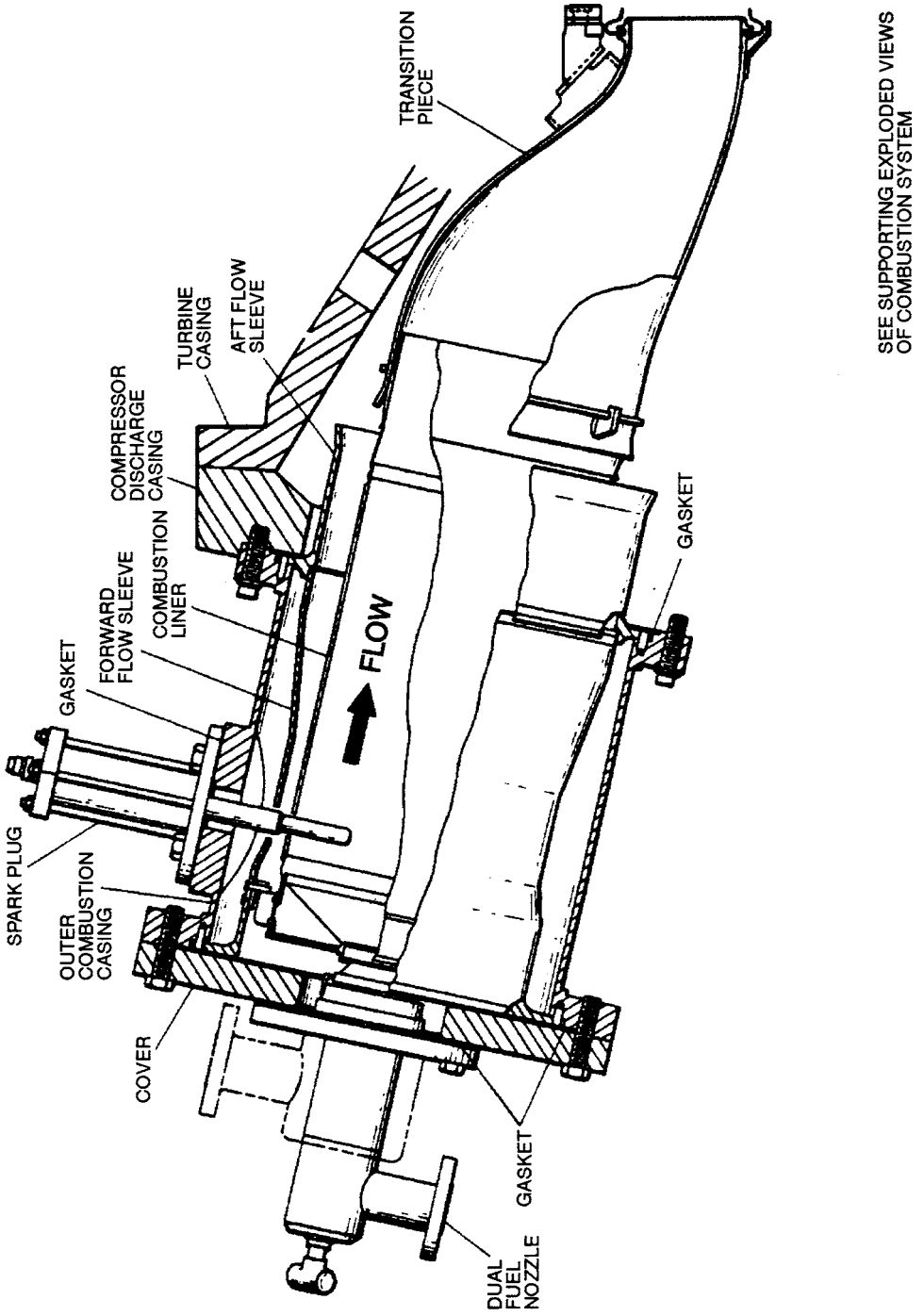



**Figure CI-D1. Combustion System Components (1 of 5)**

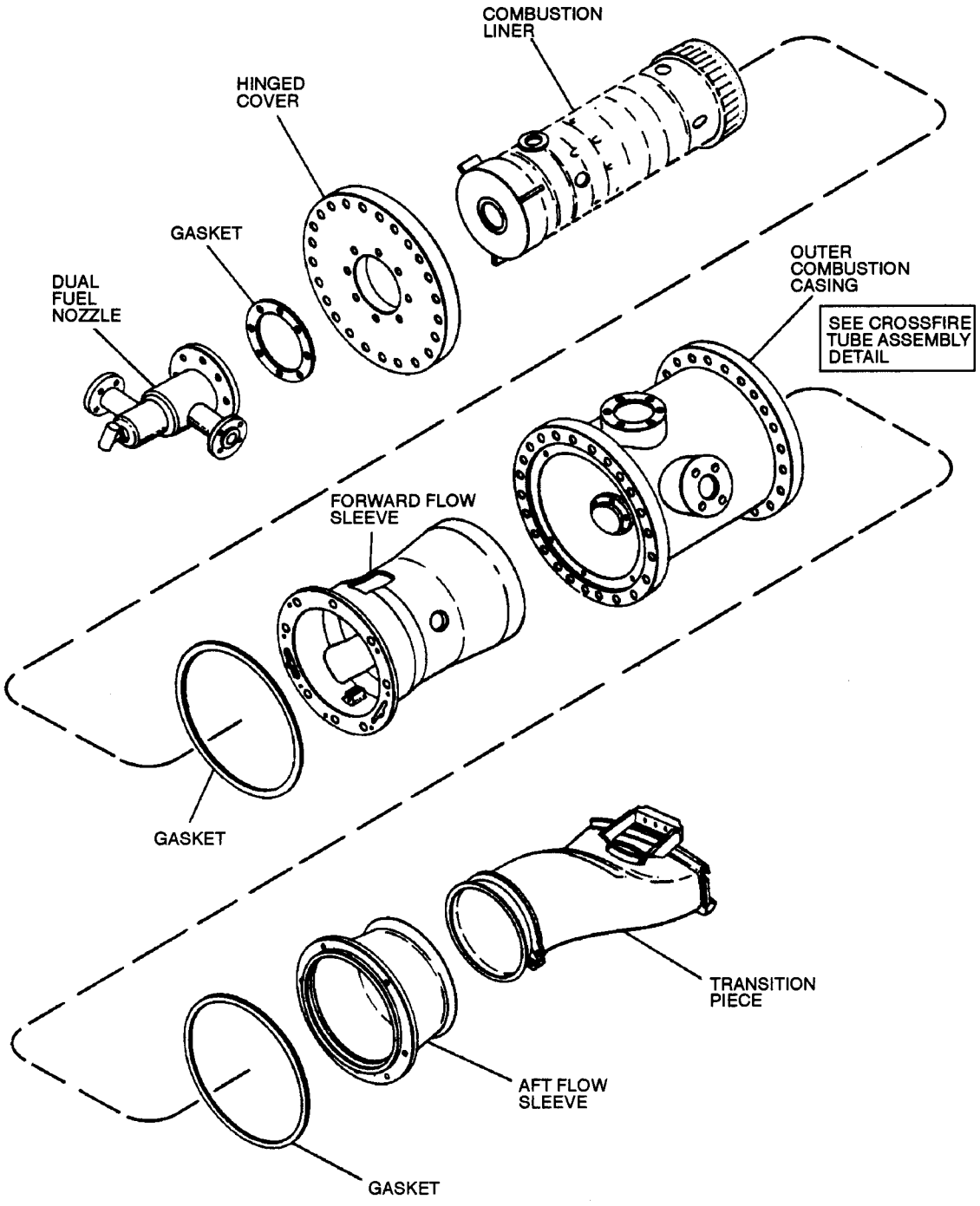



**Figure CI-D1. Combustion System Components (2 of 5)**

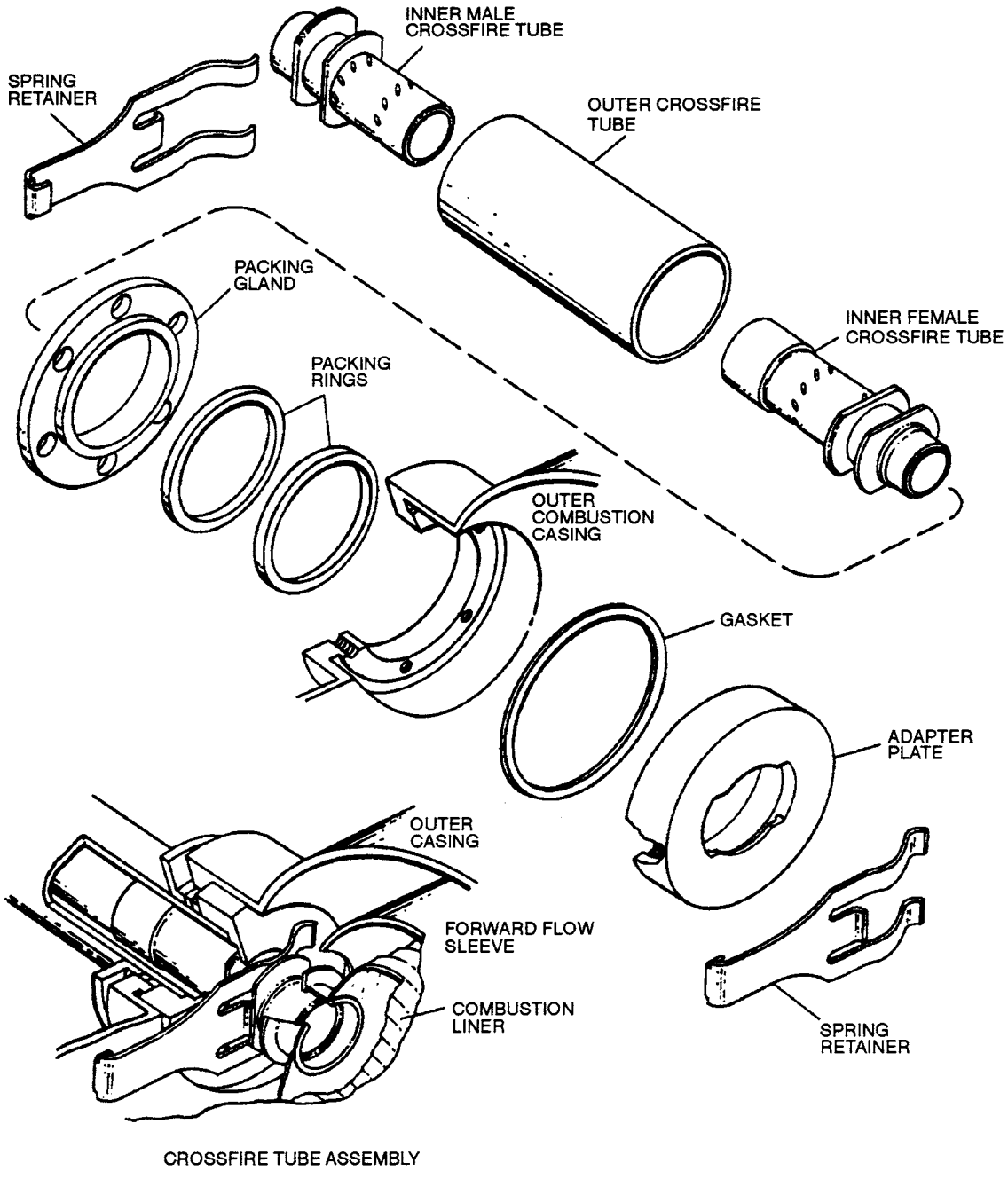



**Figure CI-D1. Combustion System Components (3 of 5)**

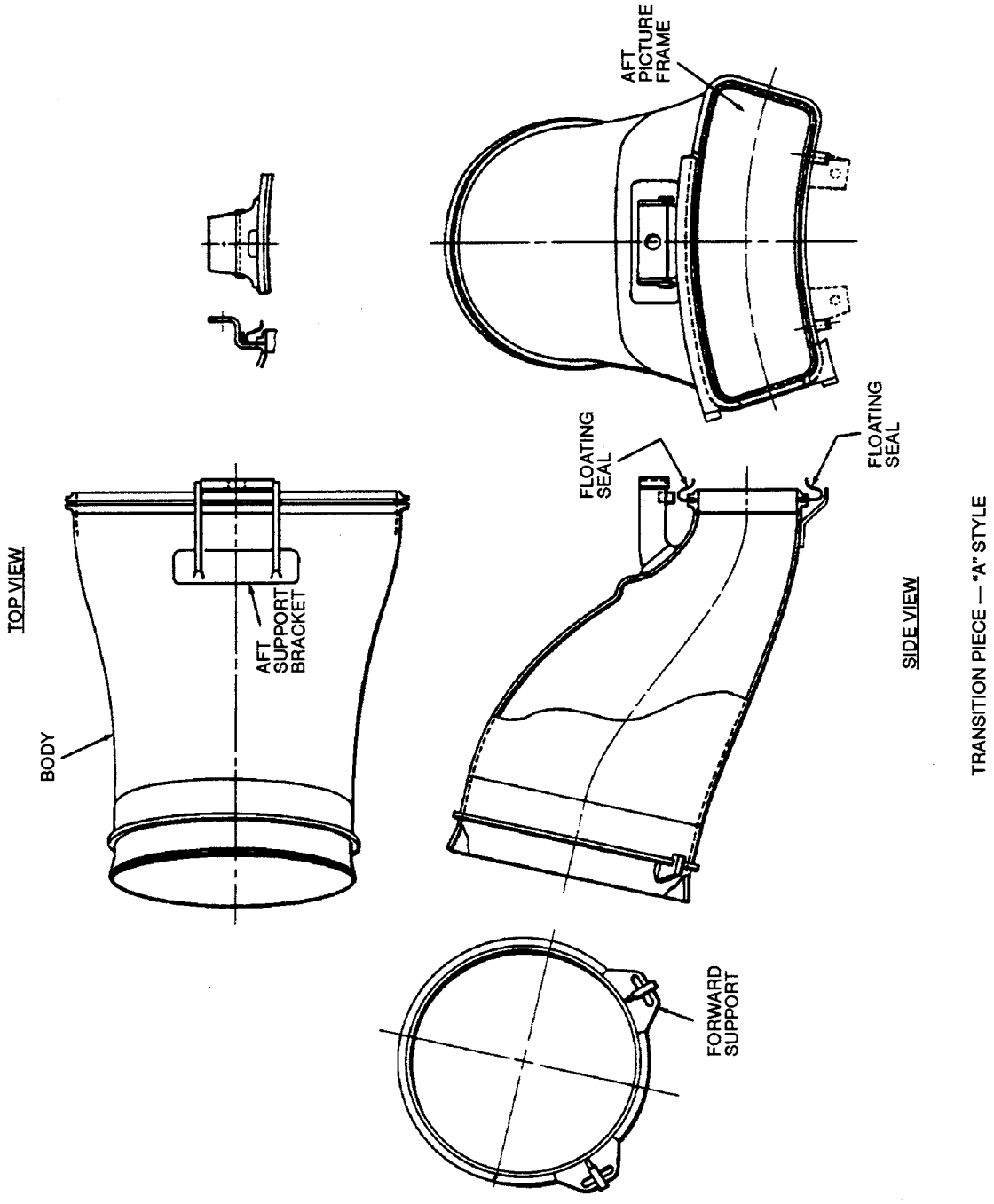



**Figure CI-D1. Combustion System Components (4 of 5)**

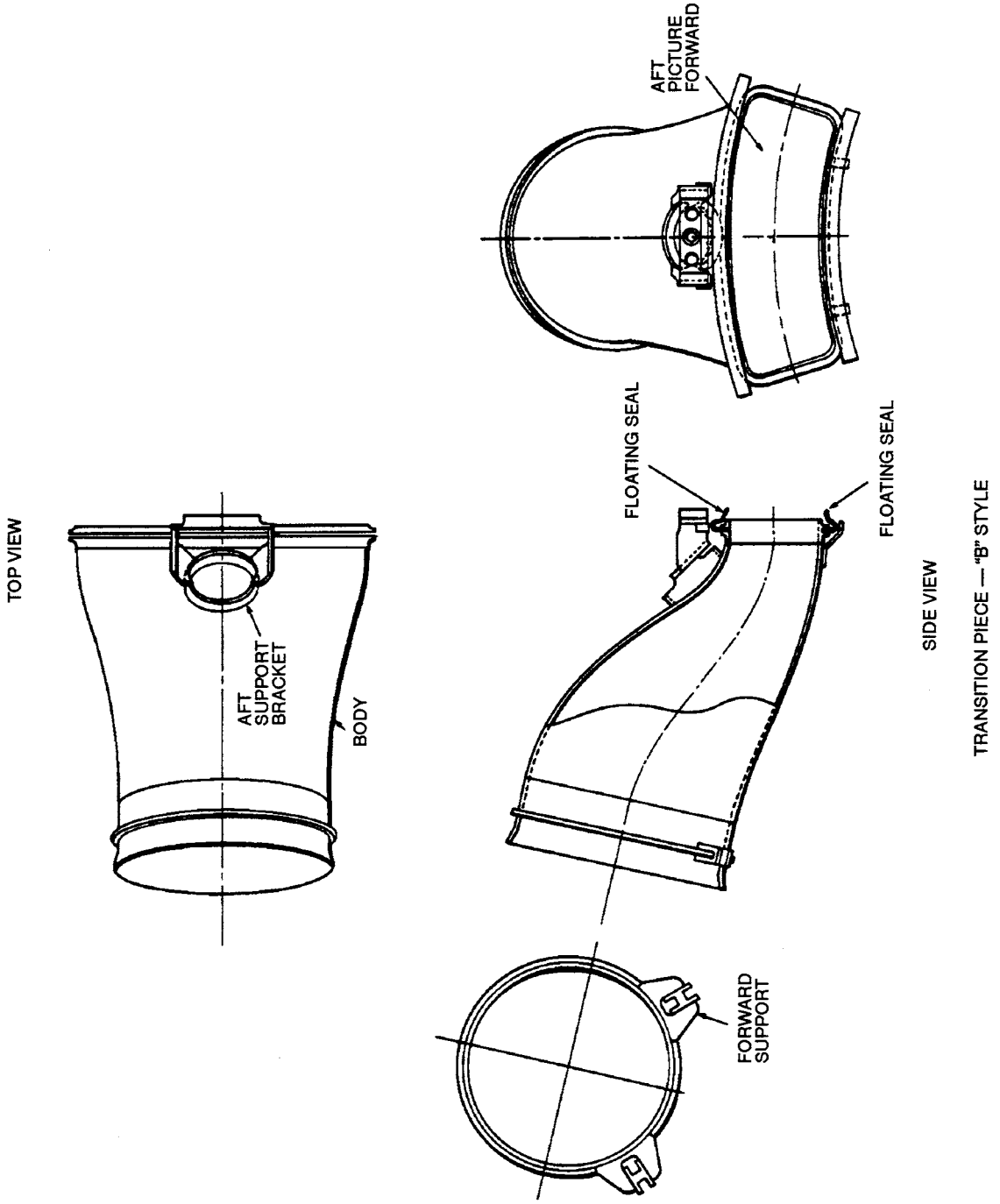



**Figure CI-D1. Combustion System Components (5 of 5)**

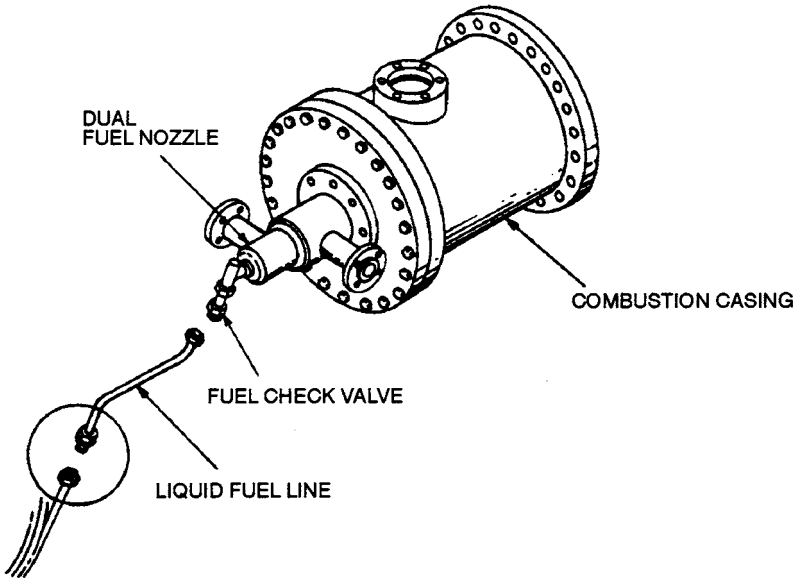



**Figure CI-D2. Removal of Liquid Fuel Lines**

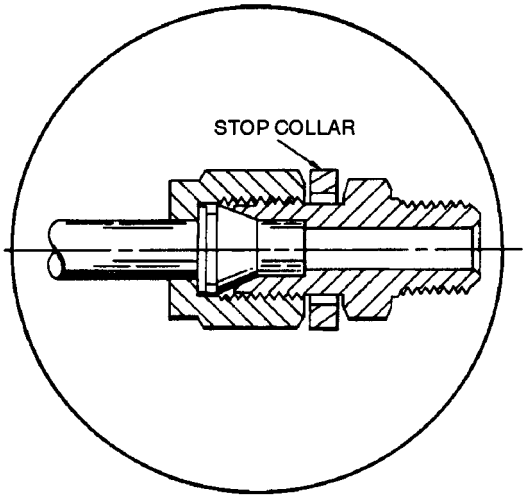



**Figure CI-D3. Swagelok Fitting with Stop Collar**

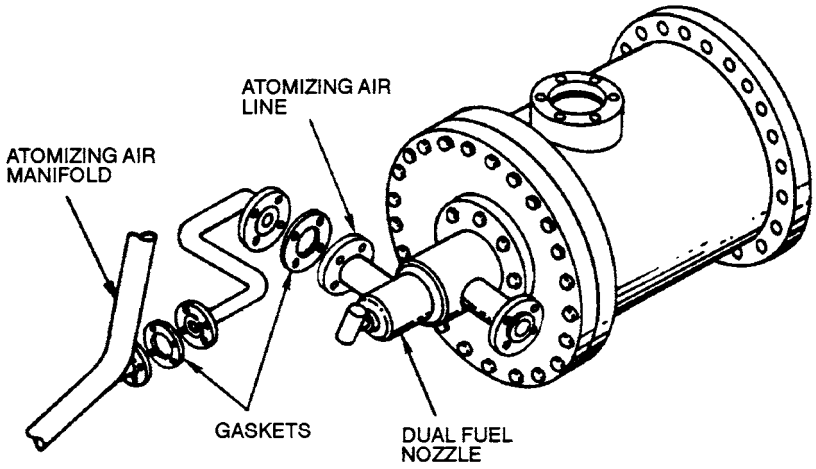



**Figure CI-D4. Removal of Atomizing Air Piping**

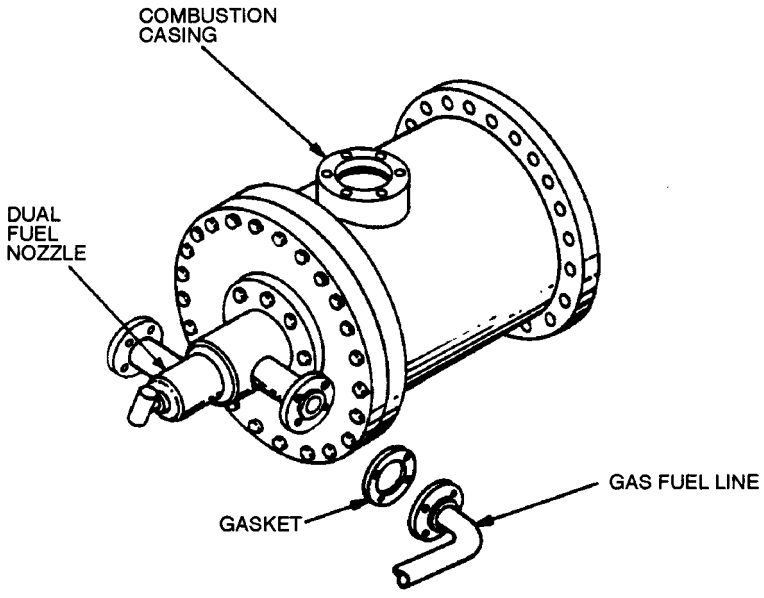



**Figure CI-D5. Removal of Gas Fuel Lines**

**OPERATION 4 - REMOVE ATOMIZING AIR LINES**

1. Identify, tag, and nondestructively matchmark each piping pigtail with the combustion chamber number before removing.

2. Disconnect flange at fuel nozzle assembly. See Fig. CI-D4.

**CAUTION: Support the atomizing air manifold so that as fuel nozzles are disconnected, the weight of the manifold is not on the last fuel nozzle disconnected.**

3. Disconnect flange at atomizing air manifold ring and remove piping. See Fig. CI-D4.

4. Clean all flange faces and cover all pipe openings. Clean all nuts and bolts and throw away old gaskets. Bag bolts, nuts and identify for reassembly.

**CAUTION: Do not use improper size cap or rags to cover pipe openings, they may be accidentally pushed inside the piping.**

**OPERATION 5 - REMOVE GAS FUEL LINES (If Applicable)**

1. Identify, tag, and nondestructively matchmark each piping pigtail for combustion chamber number before removing.

2. Disconnect flanges at fuel nozzle assemblies. See Fig. CI-D5.

**CAUTION: Support the fuel gas manifold so that as fuel nozzles are disconnected, the weight of the manifold is not on the last fuel nozzle disconnected.**

3. Disconnect flanges at gas manifold, and remove piping pigtails. See Fig. CI-D5.

4. Clean all flange faces and cover all pipe openings. Clean all nuts, bolts and discard old gaskets. Bag bolts, nuts and identify for reassembly.

**OPERATION 6 - REMOVE STEAM INJECTION LINES (IF APPLICABLE)**

1. Check to ensure the pigtail and manifold to the corresponding combustion casing is matchmarked. If not matchmarked, do so.

2. Disconnect the flanges at the combustion casing cover and at the manifolds. See Fig. CI-D6.

3. Remove the pigtail and cover all openings on the pigtail, manifold and combustion cover.

4. Clean all nuts, bolts and discard old gaskets. Bag bolts and nuts and identify for reassembly.

**OPERATION 7 - REMOVE WATER INJECTION LINES (IF APPLICABLE)**

1. Identify and tag each set of tubing pigtails to the fuel nozzle body and the water manifold to its corresponding fuel nozzle number.

2. Disconnect the tubing pigtails (1 or 2 per nozzle) from the fuel nozzle and water injection manifold. Hold the fittings steady with a backup wrench to prevent twisting the fitting on the tubing. See Fig. CI-D7.

3. Cap and/or cover all openings on the pigtails, fuel nozzles and manifold.

**OPERATION 8 - REMOVE LIQUID FUEL CHECK VALVES**

**Note:** On dual fuel units, remove purge line check valves also.

1. Identify, and tag, each check valve for later reinstallation to the same fuel nozzle from which it was removed.

2. Remove the check valve by placing a wrench on the hex flats nearest the fuel nozzle. Securely hold the fuel nozzle fitting to prevent twisting the fuel tube when removing the check valve. See Fig. CI-D8.

3. Cap the open ends of both the check valves and the fuel nozzles to prevent entry of foreign material.

**CAUTION: Do not use improper size caps or rags to cover pipe openings, they may be accidentally pushed inside the piping.**

**OPERATION 9 - REMOVE FALSE START DRAIN TUBING**

1. Identify and tag each false start drain tube from chambers 3, 4, 5, 6 and 7.

2. Disconnect the tubing from the combustion outer casing and the false start drain manifold pipe. See Fig. CI-D9.

3. Cap the open ends of the tubing and the open ends on the false start drain manifold.

**CAUTION: Do not use improper size caps or rags to cover pipe openings, they may be accidentally pushed inside the piping.**

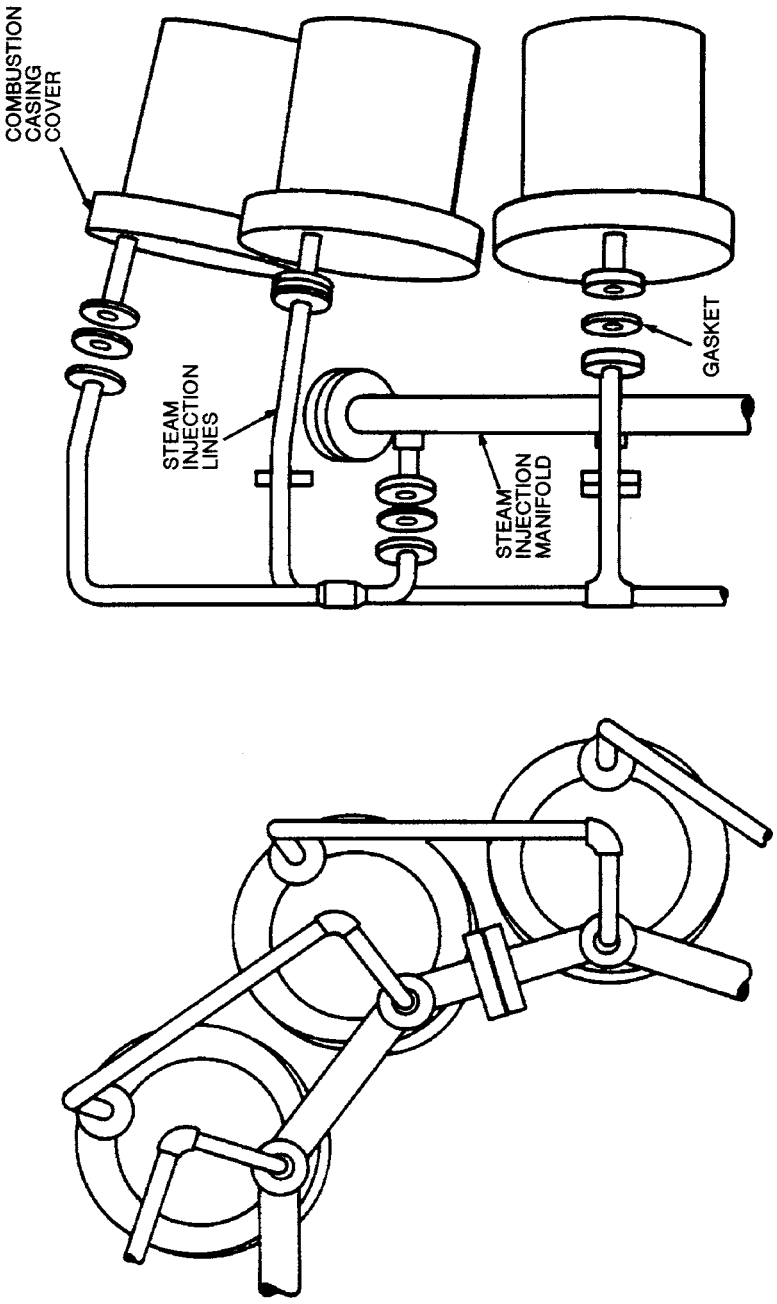



**Figure CI-D6. Removal of Steam Injection Piping**

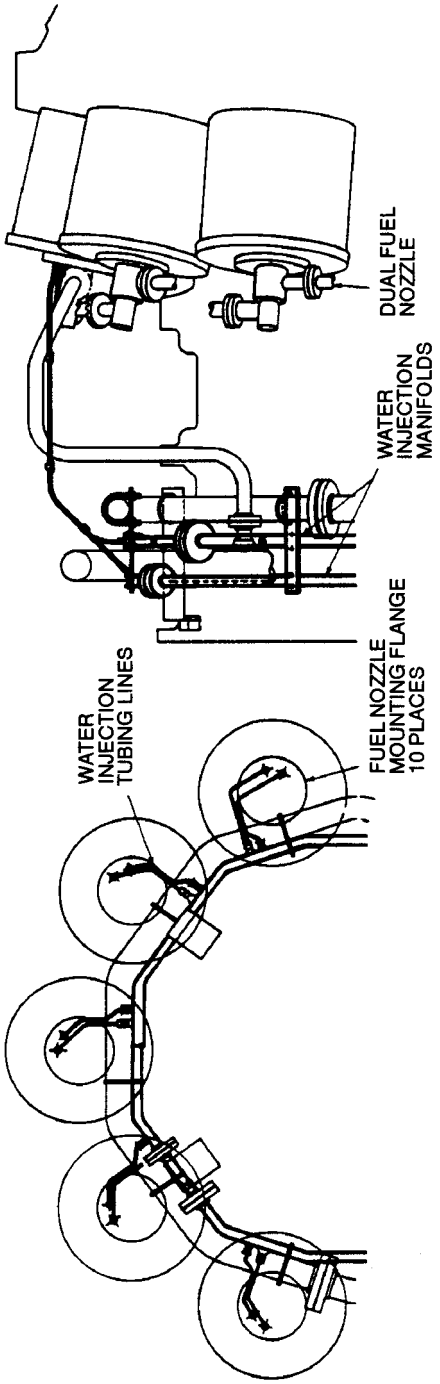



**Figure CI-D7. Removal of Water Injection Piping**

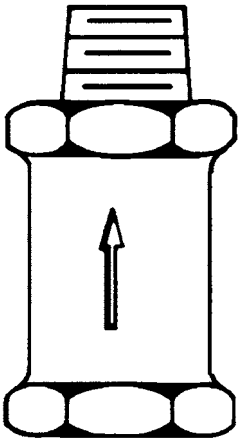


**Figure CI-D8. Fuel Oil Check Valve**

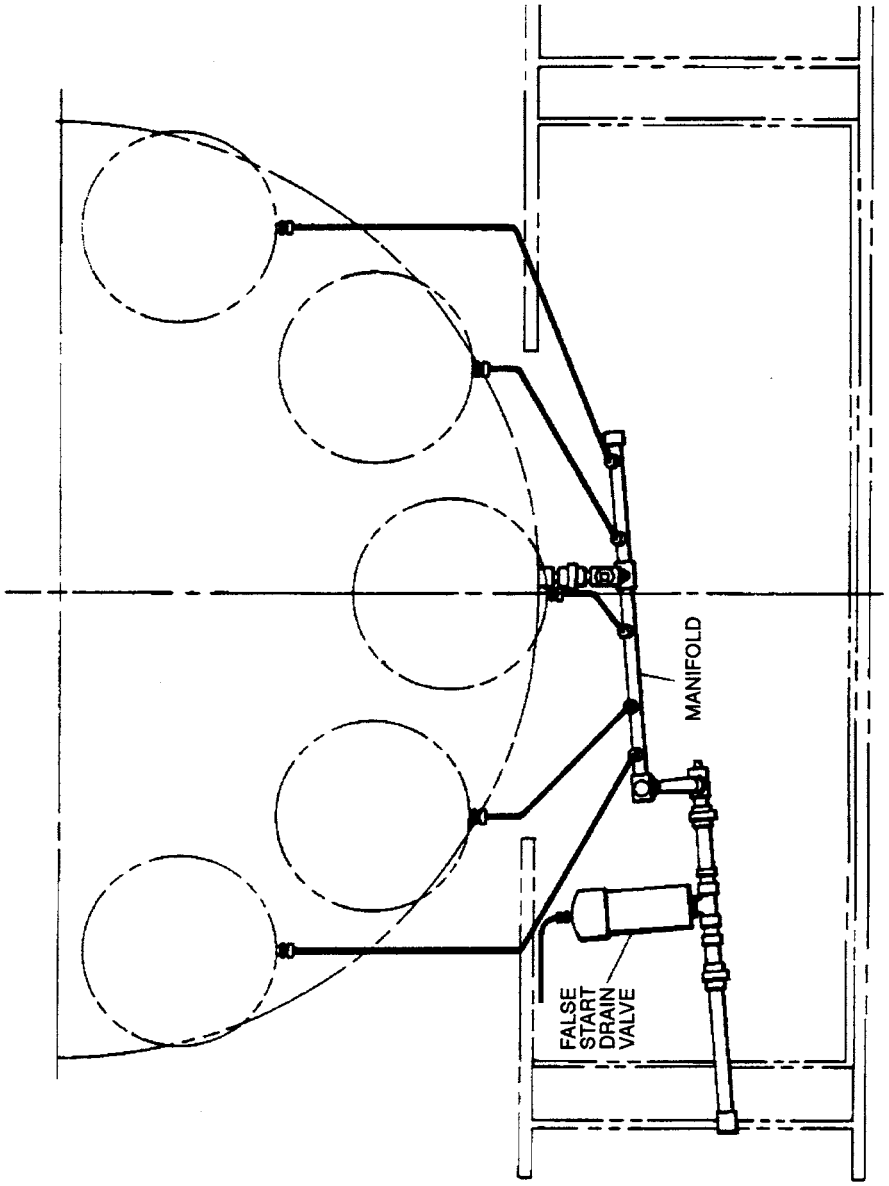



**Figure CI-D9. False Start Drain Piping**

**OPERATION 10 - REMOVE FUEL NOZZLES**

1. Unbolt and remove the fuel nozzle assemblies from combustion can cover. Discard old gaskets. See Fig. CI-D10.

2. Identify and tag nozzles so they can be replaced on same combustion can covers from which they are removed.

3. Remove old anti-seize compound from bolts, and clean all raised face flanges. Bag bolts and identify for reassembly.

4. Be careful that fuel nozzles are not damaged when handling. Place assemblies on wood or heavy card-board, and wrap in plastic bags to keep dirt out.

**OPERATION 11 - REMOVE FLAME DETECTORS AND SPARK PLUGS**

1. Lock out power supply to spark plugs and flame detectors to insure against any possibility of energizing the flame detectors or spark plugs.

2. Remove enclosure screens on flame detectors.

3. Disconnect wiring to flame detector.

4. Unbolt the collar around flame-detector stem which prevents the turbine compartment air from entering the flame-detector enclosure box.

5. Unbolt flame-detector flange, and remove probe from chamber. See Fig. CI-D11.

6. Disconnect spark plug leads at the spark plugs.

7. Unbolt spark plug assembly from casing flange. See Fig. CI-D12.

8. Lift spark plugs out.

**CAUTION: Handle carefully to prevent breaking the internal ceramic parts.**

9. Remove old anti-seize compound on all bolts, bag and identify for reassembly. Clean matching flange surfaces. Do not reuse old gaskets.

**OPERATION 12 - REMOVE ELEVENTH STAGE COOLING AND SEALING AIR LINES**

1. Unbolt the upper half 11th-stage cooling and sealing air piping from the bottom flange on the 5th to 11th stage bearing seal valve (VA14) (if applicable), the compressor casing and the branch off flange of the compressor bleed valve (VA2-2).

2. Remove the piping and cover all openings. Identify and tag any orifices removed for location.

3. Unbolt the lower half 11th stage cooling and sealing air line from the compressor casing lower half and the branch off flange of the piping to the compressor bleed valve (VA2-1).

4. Remove the piping and cover all openings.

**OPERATION 13 - UNBOLT AND OPEN UP COMBUSTION OUTER CASING COVERS**

**CAUTION: Fuel nozzles must always be removed prior to opening combustion casing covers to prevent combustion liner damage.**

1. Remove the cover bolts, clean off old anti-seize compound, bag and identify bolts for reassembly.

2. Open the combustion case cover by pivoting the cover on its hinges. This will permit access to the forward flow sleeve and the combustion liner without the necessity of removing the cover. See Fig. CI-D13.

**CAUTION: If for any reason the hinged covers are removed from the combustion case, the covers weigh approximately 175 pounds. Eyebolts, slings, cables and pulleys should be utilized to remove the covers. Use the turbine compartment enclosure structure or piping as attaching points for the cables, slings, etc.**

3. Remove and scrap the cover ring gasket. DO NOT REUSE.

4. Clean the mating flange surfaces.

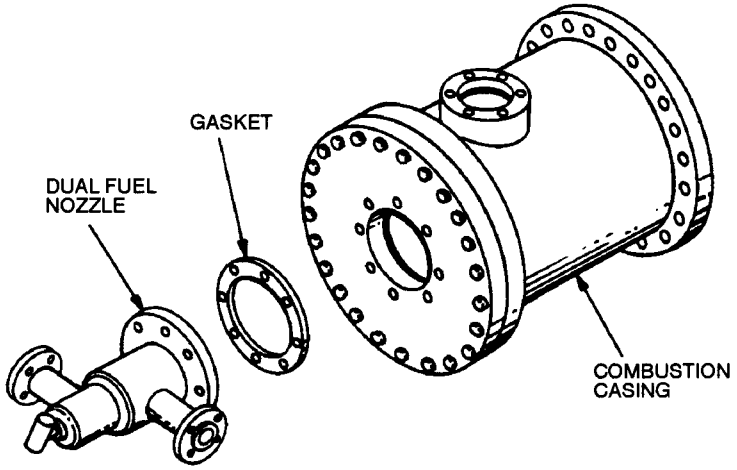



**Figure CI-D10. Removal of Fuel Nozzles**

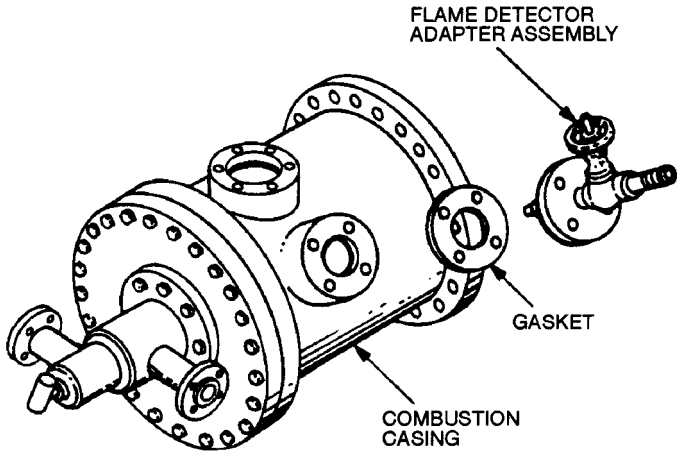



**Figure CI-D11. Removal of Flame Detector**

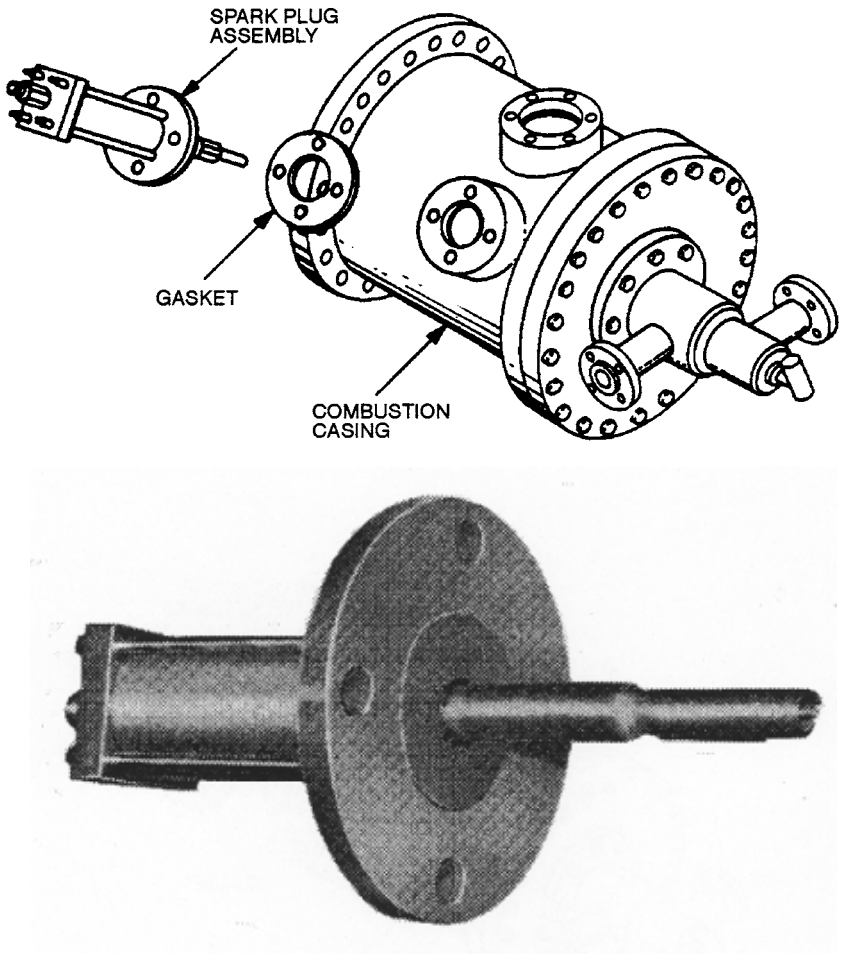



**Figure CI-D12. Removal of Spark Plug**

## OPERATION 14 - REMOVE CROSSFIRE TUBE RETAINERS, COMBUSTION LINERS, CROSSFIRE TUBES AND FORWARD FLOW SLEEVES

1. Remove the crossfire tube spring retainers, (two retainers per combustion casing assembly). Tag and identify all spring retainers according to chamber position. See Figs. CI-D14 and CI-D15.

2. Push the crossfire tube clear of the combustion liner on each side of the combustion case.

3. Using the combustion liner pulling tool and the holes provided in the liner stops, carefully pull the combustion liner straight out to disengage it from the forward flow sleeve liner stops and the transition piece. See Fig. CI-D16.

**CAUTION: Exercise caution when removing combustion liners to ensure that the crossfire tube seals and combustion liner spring seals (hula seals) do not snag the forward flow sleeve liner stops.**

4. Place the combustion liners with the cap end upward on a piece of plywood to prevent damage to the liner. Identify each liner according to chamber position.

**CAUTION: Refer to Section 2, "Standard Practices" for marking of nickel base alloys.**

5. Rotate each inner crossfire tube half section 90° and pull inboard to remove the crossfire tube half sections from each side of the combustion case. Identify and tag each inner crossfire tube half section to its combustion case location. See Fig. CI-D15.

**CAUTION: Flow sleeves can slide out in lower half once bolts are removed.**

6. Remove the four bolts from the forward flow sleeve mounting flange, carefully pull the flow sleeve straight out and place the flow sleeve with the forward end upward on piece of plywood to prevent damage to the flow sleeve . See Fig. CI-D17.

**Note:** Two 0.500-13 jack holes are provided in the flow sleeve flange. Use two 0.500-13 x 1.250 long capscrews to apply axial removal force to the flow sleeve.

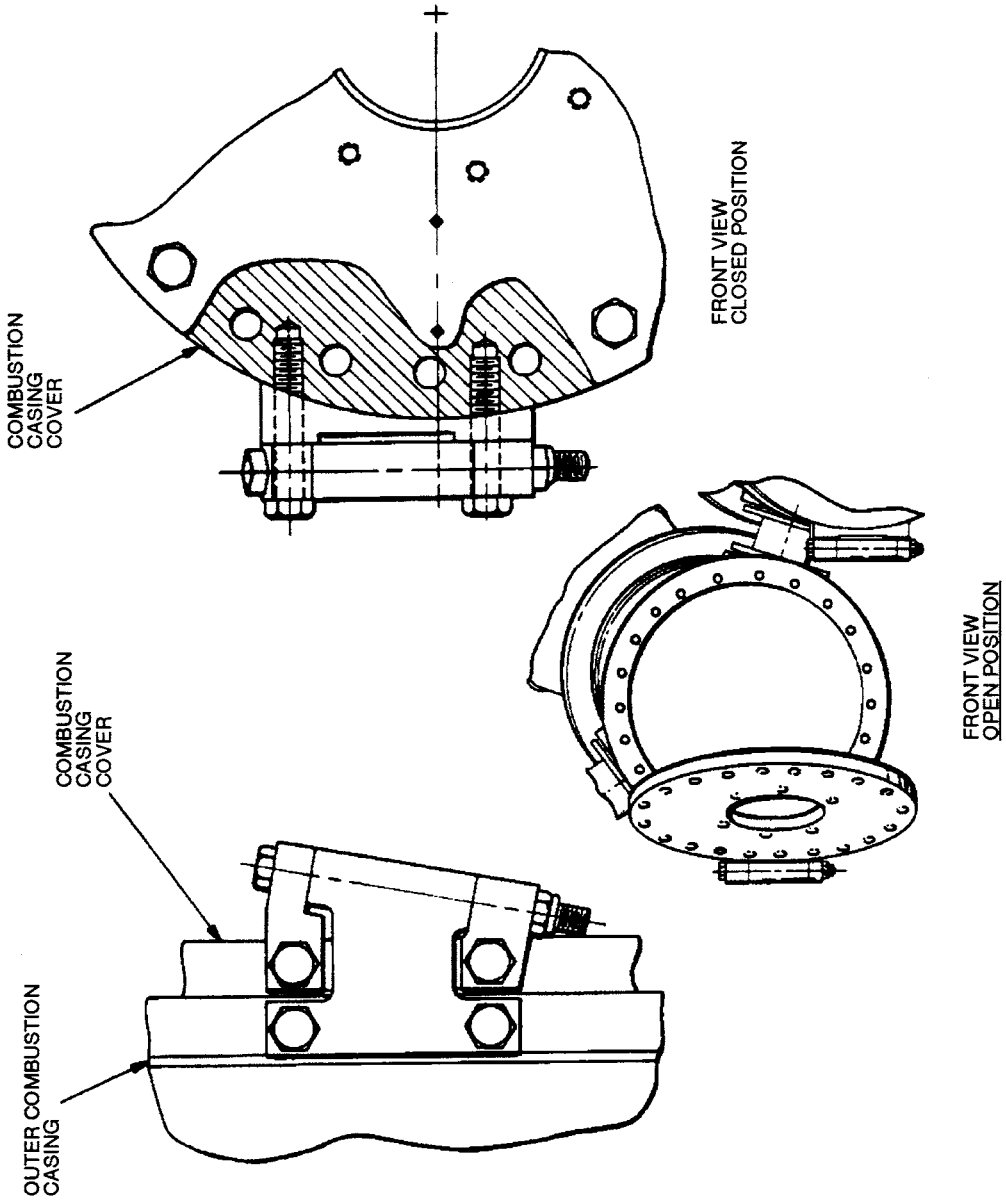



**Figure CI-D13. Combustion Casing Hinged Cover**

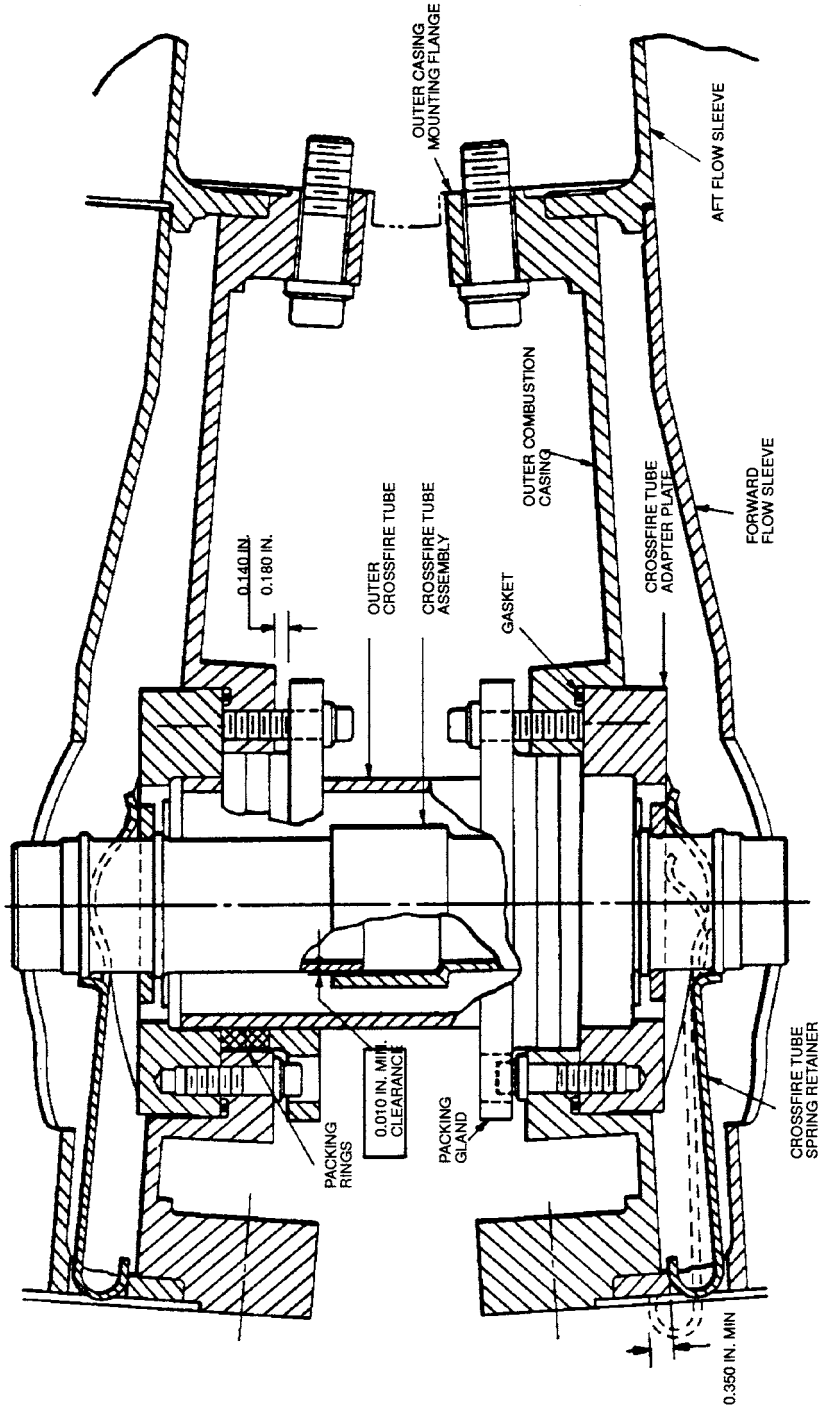



**Figure CI-D14. Crossfire Tube Assembly Cross Section**

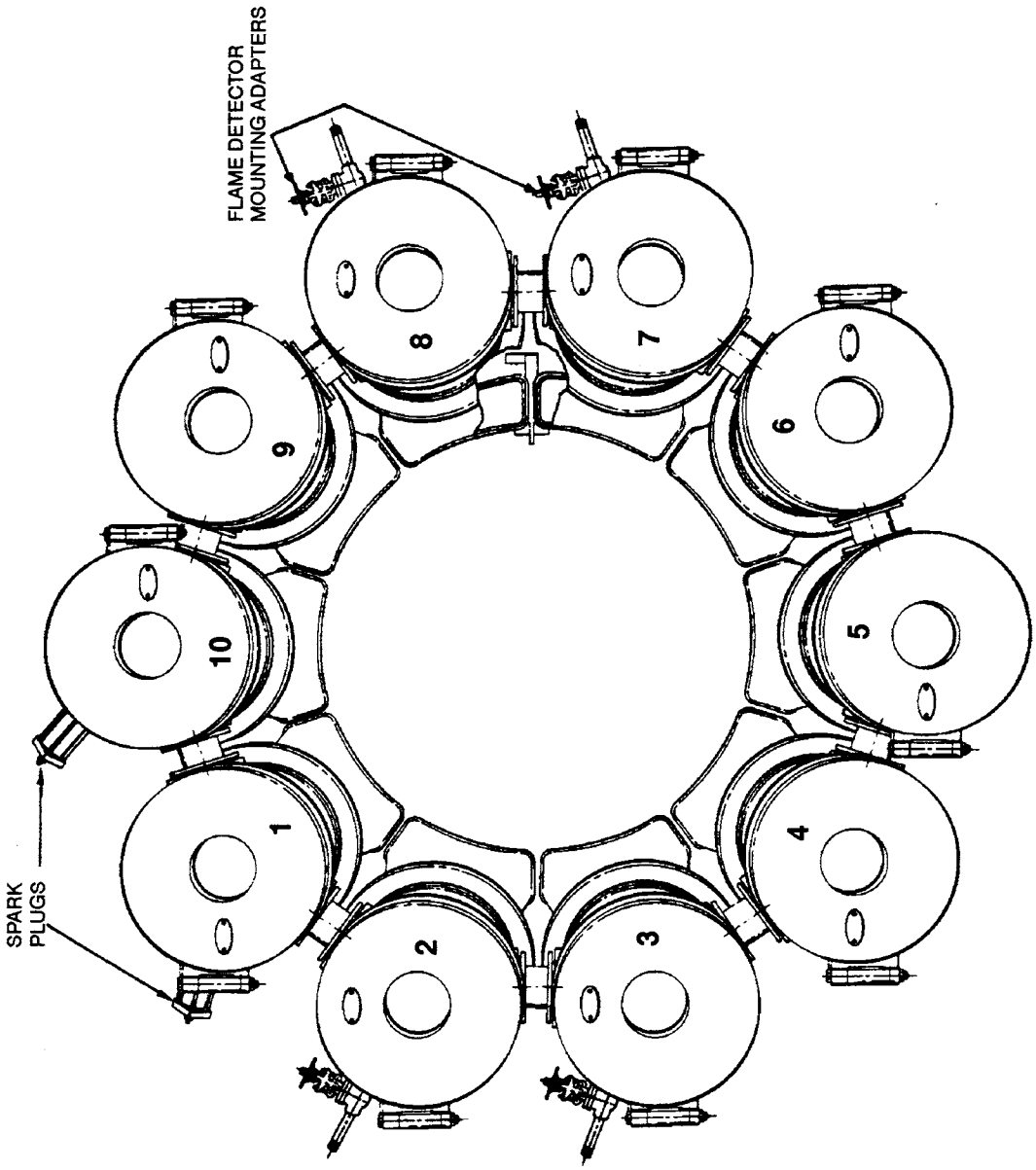



**Figure CI-D15. Combustion Casing Arrangement**

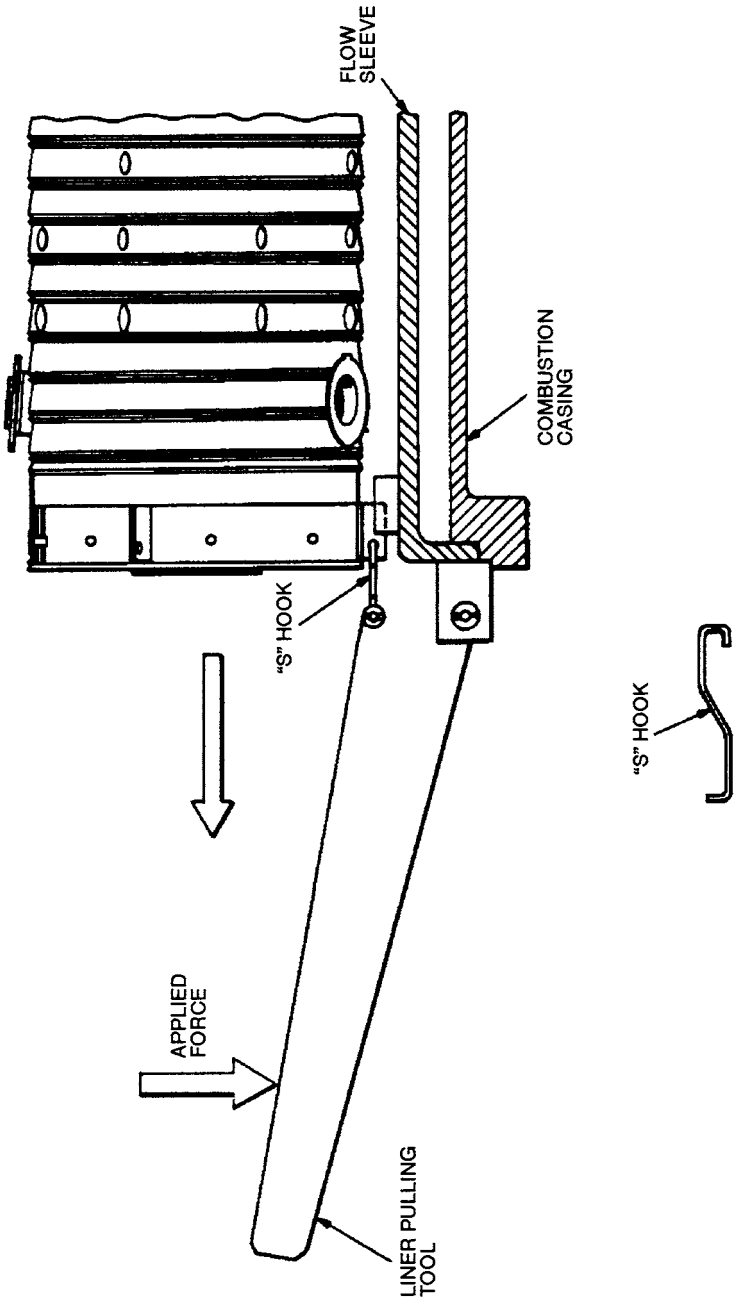
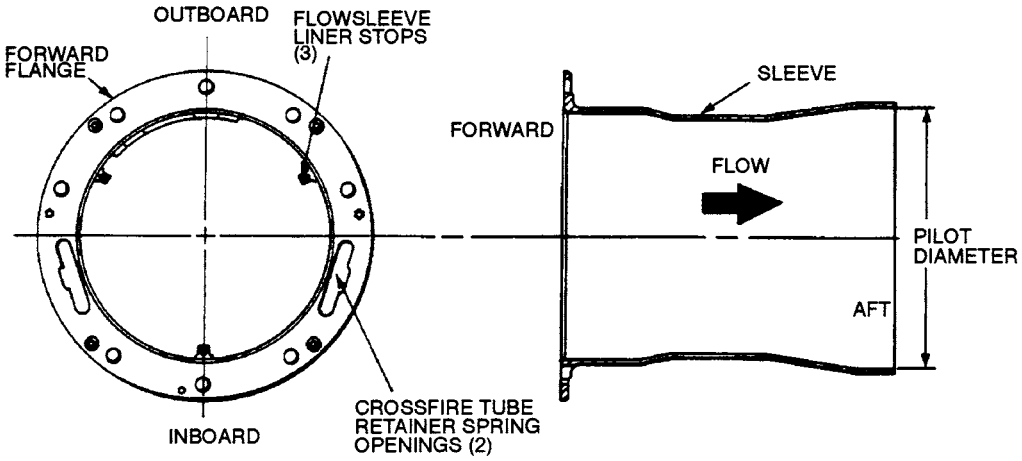



**Figure CI-D16. Combustion Liner Puller Tool**



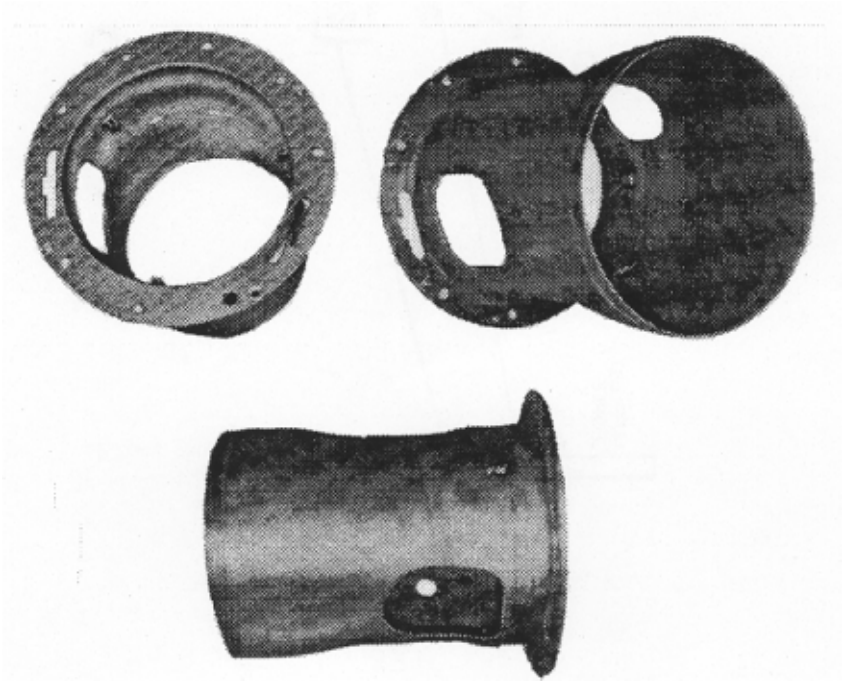



**Figure CI-D17. Combustion Casing Forward Flow Sleeve**

7. Identify the forward flow sleeve according to chamber location.

8. Release the outer crossfire tube packing glands from the combustion casing at each end of the outer crossfire tube.

9. There are two crossfire tube adapter plates assembled in the inside of each combustion casing. Using jack screw holes provided on the combustion casing crossfire tube flanges, remove the adapter plate and gasket from one side only. Discard the gasket. (See Fig. CI-D18.)

10. Carefully work the outer crossfire tube inboard through the combustion casing crossfire tube opening. See Fig. CI-D18.

**Note:** A chain type pipe wrench will help in freeing the outer tube during disassembly.

11. Identify each outer tube, packing gland assembly and bolting according to chamber and location.

12. Discard the ring packings.

13. Remove the remaining combustion liners, inner crossfire tubes, forward flow sleeves and outer tubes by repeating Steps 1 through 12. Identify each piece according to chamber location.

**OPERATION 15 - REMOVE ACCESS PORT BLANK FLANGE FROM UPPER HALF TURBINE CASING OR THE UPPER SECTION OF THE ATOMIZING AIR EXTRACTION MANIFOLD**

**Note:** If unit is designed for inlet deicing piping this blank flange port is used for the deicing piping and the port will not be accessible. The upper half atomizing air extraction manifold must than be removed to gain access to the transition piece aft area. See Fig. CI-D19.

1. Unbolt and remove the access port blank flange (if available) from the upper half turbine casing. If access port is not available proceed to Step 2.

2. Unbolt the atomizing air extraction flange just below the turbine casing horizontal joint and the two upper half flange connection to the turbine casing.

3. Remove the upper extraction pipe, discard gaskets and cover all piping openings. Bag and identify the bolting for reassembly.

## OPERATION 16 - REMOVE OUTER COMBUSTION CASINGS AND AFT FLOW SLEEVES

**CAUTION: Prior to unbolting and removing any combustion outer cases, insure that the location of the outer combustion case mounting flange to the compressor discharge case bulkhead bolting circle is clearly scribed on the combustion case and the compressor discharge casing. This location is required during reassembly in order to maintain the original clearance gap between the two halves of the inner crossfire tubes during reassembly. Refer to the combustion chamber arrangement drawing in the Parts Lists and Drawing Volume of the Service Manual for complete details.**

**Note:** The outer casing flange bolts can best be removed using an impact wrench and a long extension to reach between and behind the combustion casings. Identify each combustion casing for position location.

1. Close each combustion casing cover and secure the cover in place with two bolts allowing it to be removed with the casing.

   **WARNING: Each combustion outer casing and cover weighs approximately 425 pounds and should be rigged accordingly to support this weight prior to start of disassembly of the outer combustion casings.**

2. Starting with the number 10 combustion casing, rig the casing to the lifting means to support the weight. Unbolt the casing from the compressor discharge casing bulkhead.

3. Carefully, using the rigging attached, work the combustion casing outward until the aft flow sleeve clears the bulkhead. Lift the casing clear of the unit.

4. Set and block the outer casing on wood or heavy cardboard so that no damage is done to the aft flow sleeve.

5. Repeat Steps 2, 3 and 4 for combustion casings 1, 2, 3, 7, 8 and 9.

6. Rig number 4 combustion casing to the lifting means. Attach a come-along to the casing to act as a restraint to keep the casing from swinging when it clears the compressor discharge casing bulkhead.

7. Unbolt the casing from the compressor discharge bulkhead and using the rigging attached, carefully work the combustion casing outward until the aft flow sleeve clears the bulkhead. Lift the casing clear of the unit.

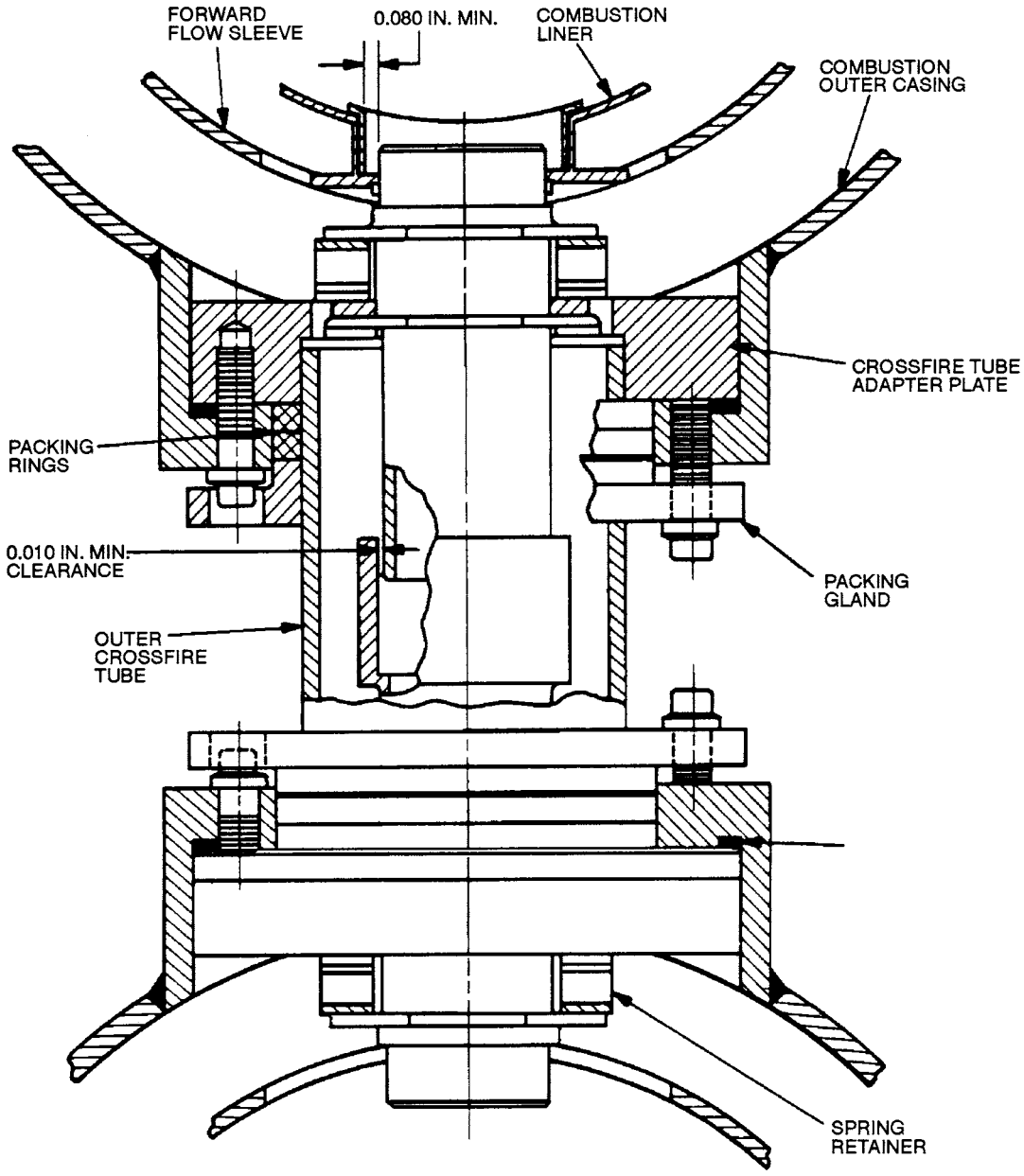



**Figure CI-D18. Crossfire Tube Assembly Cross Section**

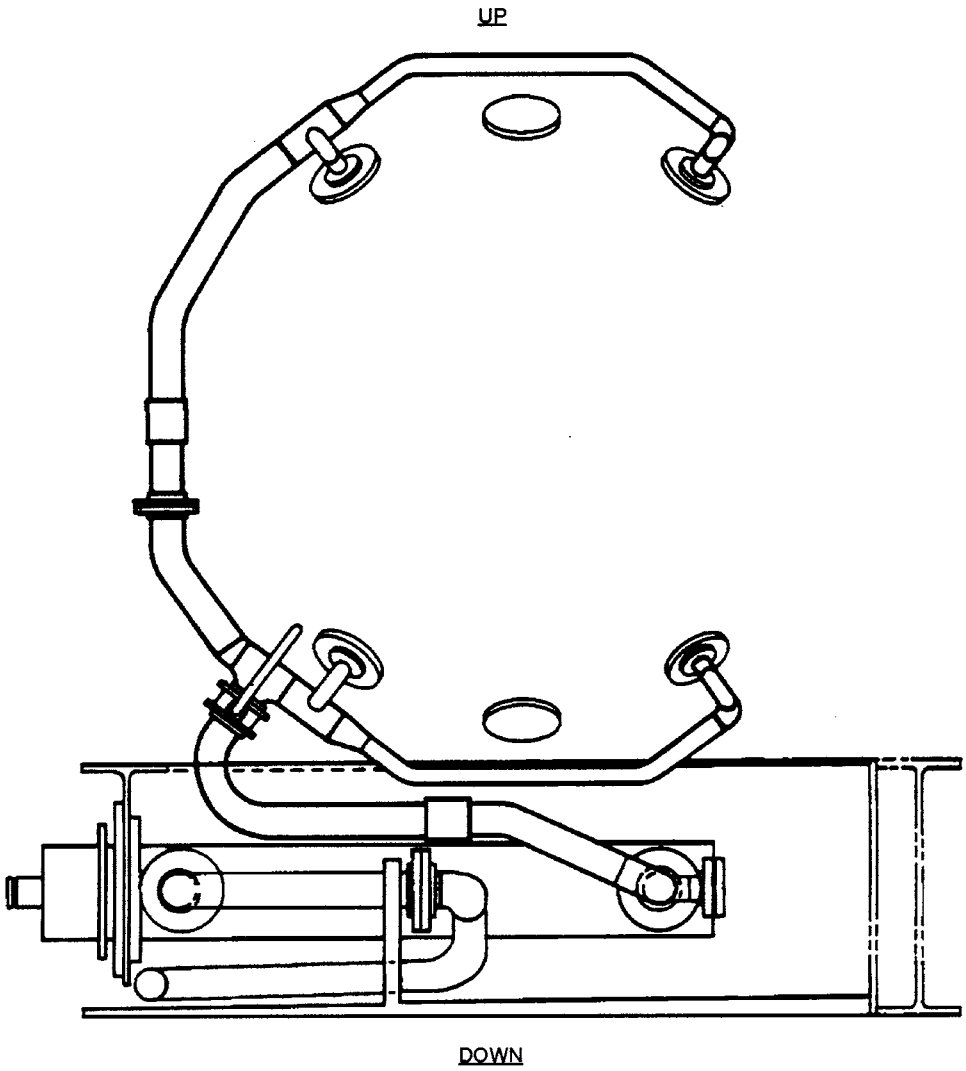



**Figure CI-D19. Atomizing Air Extraction Manifold**

8. Set and block the outer casing on wood or heavy cardboard to avoid damage to the aft flow sleeve.

9. Repeat Steps 6, 7 and 8 for combustion casings number 5 and 6.

10. Remove and discard all used gaskets.

11. Clean all hardware and mating flange surfaces. Bag and identify bolting for reassembly.

**OPERATION 17 - REMOVE TRANSITION PIECES**

**Note:** Depending on what turbine casing access hand hole ports are available, see Operation 15 preceding, the first transition piece to be removed can be number ten (10), one (1) or nine (9) position. Proceed with the following disassembly operation.

1. Remove the shoulder bolt, lockplate and aft side seal stop from between 1 and 10, 10 and 9 transition pieces. Remove the transition piece aft side seals radially outward. Identify the aft side seals and seal stop according to transition piece location. See Fig. CI-D20.

2. Disassemble the forward mounting bracket shoulder bolts, lockplates, support clamp (bullhorn) and forward mounting block. Discard the lockplates and bolts.

**Note:** There are 2 types of transition piece aft brackets. Type "A" has a single aft bracket mounting hole using 1 shoulder bolt and lockplate. Type "B" has 3 mounting holes in the aft bracket using two body bound bolts, two lockplates and a positioning pin.

3. Remove the aft bracket bolt(s) and lockplate(s) that secures the No. 10 transition piece aft bracket to the first-stage nozzle retaining ring. Discard the bolt(s) and lockplate(s). Do not reuse. See Fig. CI-D20.

4. Move the No. 10 transition piece forward and away from the first stage nozzle retaining ring to disengage the floating seals. Remove the transition piece through the compressor discharge casing opening.

5. Identify and tag the transition piece, support clamp and support clamp mounting block shims for chamber location.

**CAUTION: Refer to Section 2, "Standard Practices" for proper marking procedure.**

6. Working through the compressor discharge casing bulkhead ports, repeat Steps 1 through 5 for the remaining transition pieces.

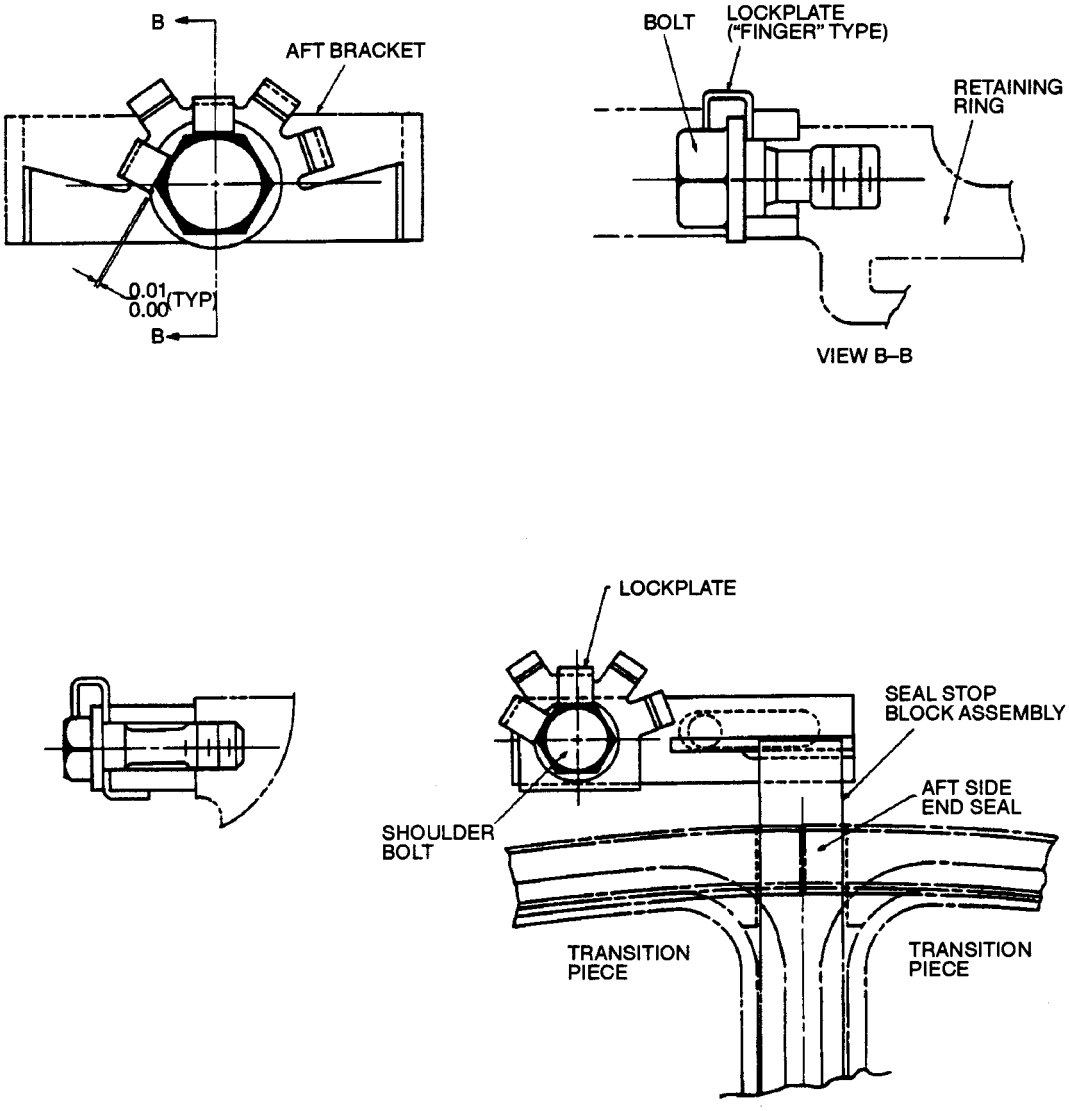



**Figure CI-D20. Transition Piece Aft Bracket and Side Seal Bolts and Lockplates**

## INSPECTION PROCEDURES

### OPERATION 1 - PRESSURE TEST FUEL OIL AND PURGE CHECK VALVES (See Fig. CI-I1)

**WARNING: Pressure testing of the fuel oil check valves requires the use of high pressure oil or air. Use hoses, fittings and associated equipment suitable for high pressure. Do not use faulty equipment since its failure could cause injury. Observe safety precautions applicable to working with high pressure gases.**

1. Using distillate fuel oil, apply 75 psig in the flow direction and check for leakage. There should be zero leakage.

2. Apply 1000 psig oil pressure against the flow direction, to each check valve to ensure against backflow leakage, or use 80 to 100 psi air pressure while submerging the valve under water. There should be zero leakage.

3. If leakage is observed in either direction during the leak tests, the check valve should be replaced or repaired in the following manner:

   a. Valve Disassembly, Inspection and Reassembly

      1. Mount valve vertically in a vise or suitable holding fixture by clamping on the hex of the body portion, (outlet side), taking care not to damage the threaded connection.

      2. Using a proper size box wrench on the hex of the inlet cap, (inlet side), turn the inlet cap counterclockwise to initially loosen. The inlet cap has spring tension on it, so carefully turn out the cap while relieving the spring tension and remove it from the valve body.

      3. Remove the body portion from the vise. Internal parts may now be tapped out by striking the body against hand or other soft surface, or by inserting a small brass or wooden rod into the outlet end and gently pushing on the poppet valve assembly.

      4. Check the "O" ring for cuts or cracks, check the poppet assembly for nicks, burrs etc. and check for dirt or chips that may cause the valve not to seat properly.

      5. Clean the valve body and internal parts with cleaning solvent and allow to dry.

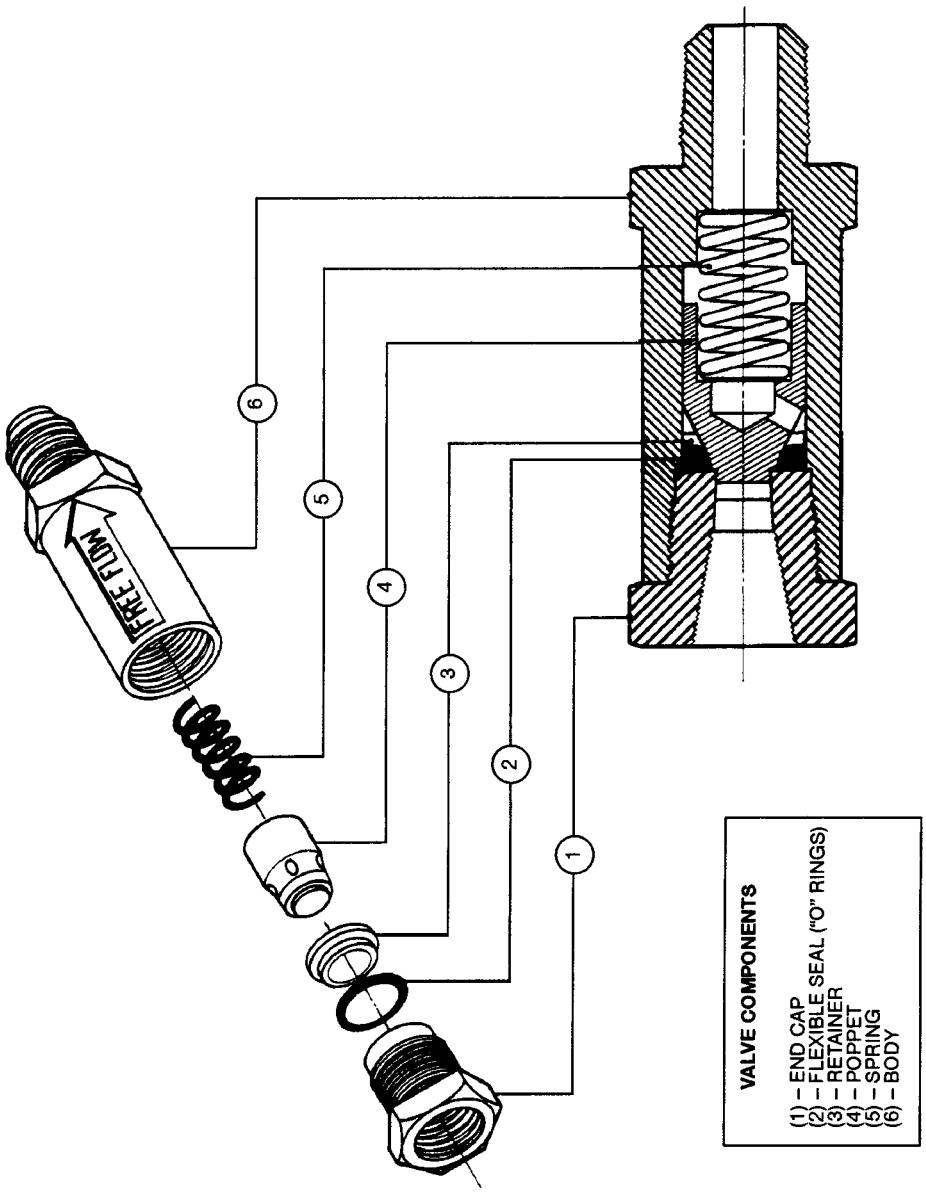



**Figure CI-I1. Fuel Oil Check Valve Assembly**

6. Reassemble the valve using silicone grease or other suitable lubricant on the "O" ring and on the cap threads. Install inlet cap and tighten using a suitable box wrench.

7. Repeat leak checks as described in Steps 1 and 2. If leakage occurs, "O" ring may not be properly seated or was damaged. Disassemble and repeat 4, 5 and 6 above and retest the check valve assembly.

8. If leakage is still present, replace the complete check valve assembly.

**OPERATION 2 - FUEL NOZZLE INSPECTION**

**CAUTION: It is imperative that any fuel nozzle disassembly, cleaning, inspection, reassembly, testing and reinstalling in unit be performed in accordance with the following procedures.**

**GENERAL**

This procedure details the tooling, fixtures, disassembly, inspection, reassembly and the requirements for testing the reassembled fuel nozzles prior to returning to service.

If the customer/user does not have the capability of performing these tests, the nozzles should be returned to Gas Turbine Division or qualified Gas Turbine repair facility for inspection, reassembly and testing.

Your GE Field Service Representative can assist you in the returned material procedures to return the fuel nozzles to the Gas Turbine Division or qualified Gas Turbine repair facility.

Perform fuel nozzle pressure test prior to disassembly. Check for leaking fuel gaskets and/or cracked fuel tube. Bubbles from the atomizing air pipe or from the fuel tube collar interface indicate faulty gasket and/or cracked fuel tube.

The pressure test is recommended prior to disassembly because a visual inspection may not disclose faults in these components.

**Pressure Test Procedure**

1. Install the fuel nozzle assembly in the test fixture as shown in Fig. CI-I2.

**WARNING:** **Use only lines that have been proof tested to 3000 psig and fittings, valves and gauges certified for use with high pressure gases. Observe safety precautions applicable to working with high pressure gases.**

2. Pressurize the fuel nozzle oil fuel passage to a leak test pressure of 300 psig.

3. Close the shutoff valve and submerge the fuel nozzle assembly in a tank of water.

4. Check for leakage at areas shown in Fig. CI-I2.

5. Record findings on inspection form AES/GT-FF-6118.

**Fuel Nozzle Disassembly (After Pressure Test)**

1. Identify, tag and/or mark the outer swirl tip, inner cap and each fuel nozzle body, to prevent mixing similar parts at reassembly. See Fig. CI-I3.

**CAUTION:** **The removal of parts from the fuel nozzle body should be done with extreme care. Small nicks, scratches or other damage to the flow surfaces and orifices can alter the nozzle flow pattern and result in undesirable combustion characteristics. The parts should be placed in clean containers and not placed on steel work benches or surfaces where the parts may be damaged. Wire baskets are acceptable containers. See Fig. CI-I4.**

2. Using an open ended hex wrench and/or crowfoot hex tool, remove the outer swirl tip, inner cap and lockplate from the fuel nozzle body. Discard the lockplate. See Fig. CI-I5.

**Note:** A torque of approximately 275-300 ft. lbs. will be required to release the lockplate stake when removing the outer tip from the fuel nozzle body.

3. Using an open ended hex wrench and/or crowfoot hex tool, remove the transition piece from the fuel nozzle body. Identify the transition piece assembly to the nozzle body from which it was removed. See Fig. CI-I6.

4. Remove and discard the gasket which seals between the nozzle body and the transition piece assembly. Use care in removing the gasket so as not to nick or scrape the gasket sealing surface.

5. Record any abnormalities or defects found during disassembly on AES/GT-FF-6118.

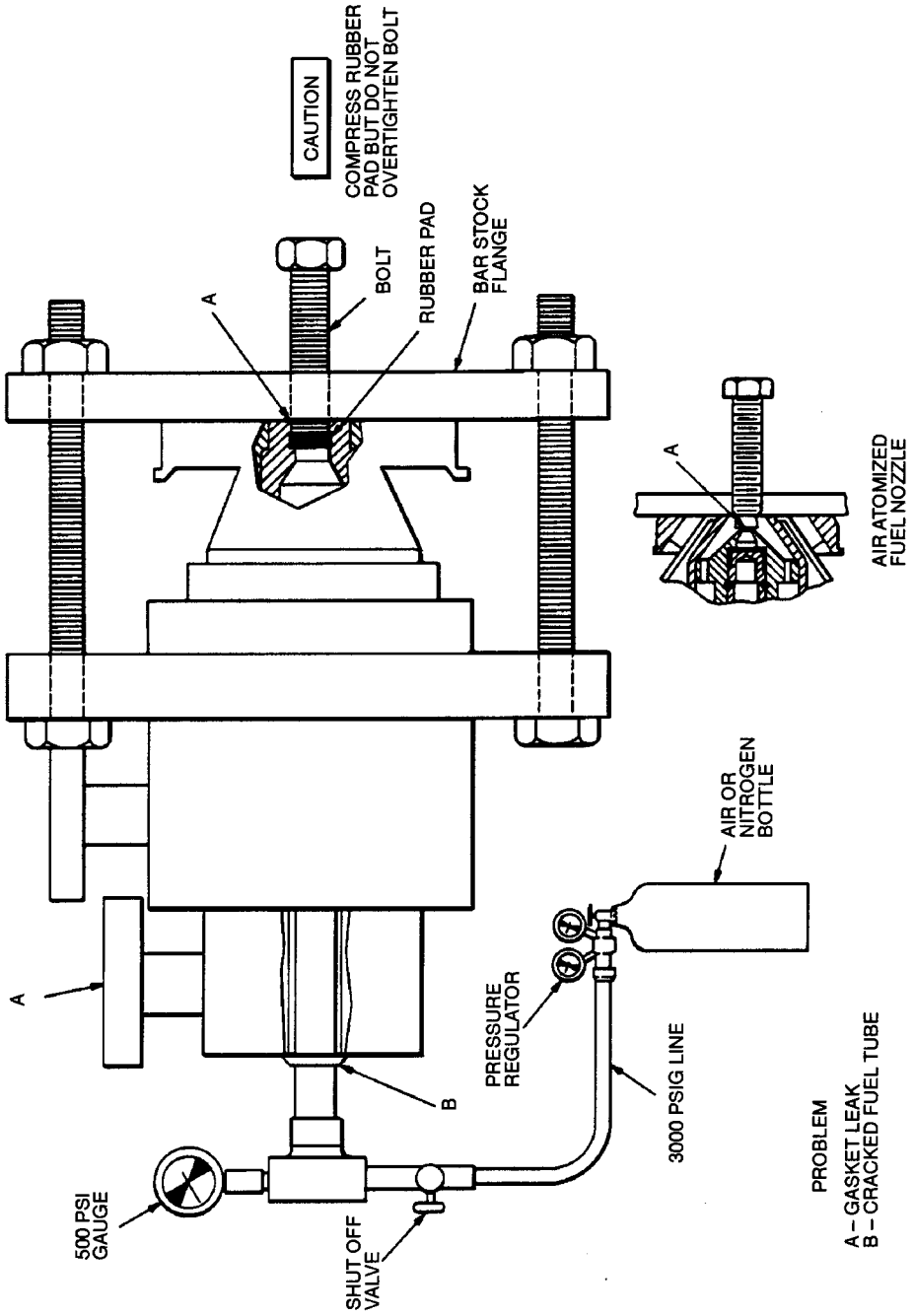



**Figure CI-I2. Fuel Nozzle Assembly Test Fixture**

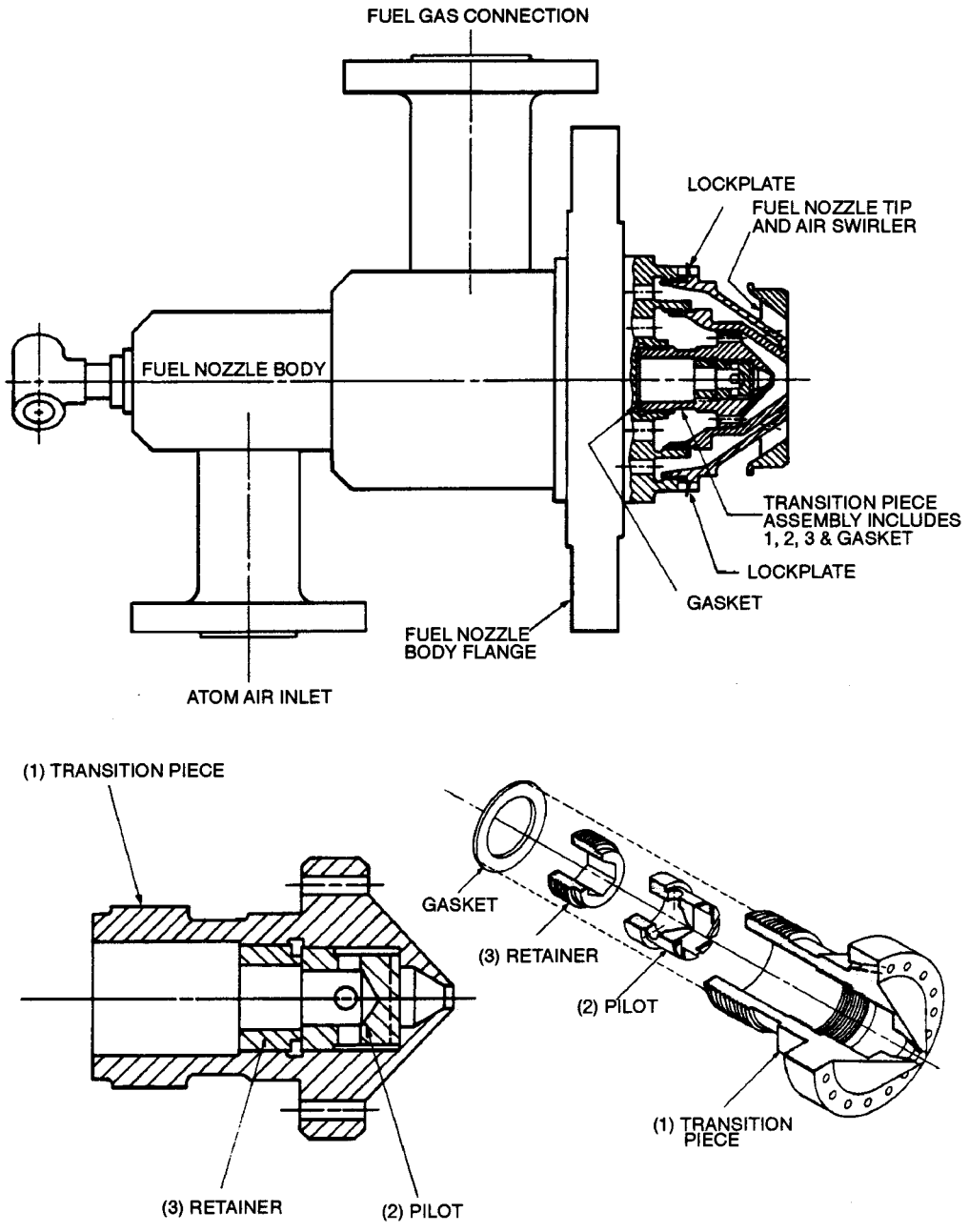



**Figure CI-I3. Fuel Nozzle Assembly Detail – Dual Fuel**

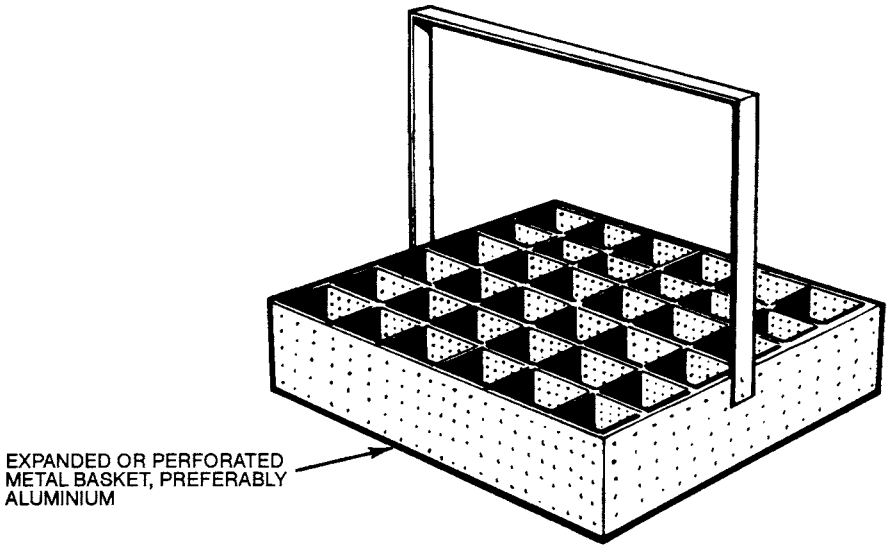



**Figure CI-I4. Fuel Nozzle Cleaning Parts Basket**

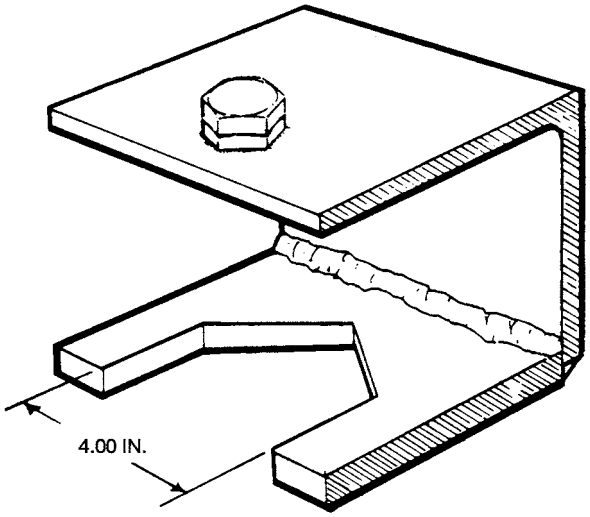



**Figure CI-I5. Fuel Nozzle Crowfoot Wrench Outer Swirl Tip**

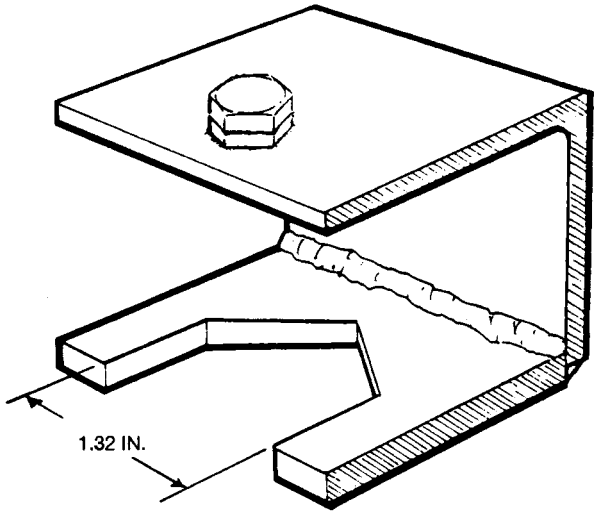



**Figure CI-I6. Fuel Nozzle Crowfoot Wrench – Transition Piece**

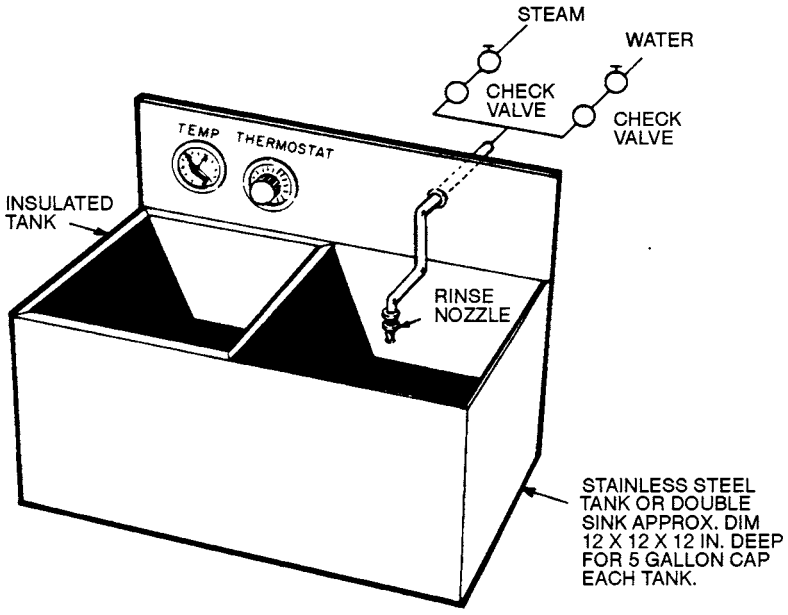



**Figure CI-I7. Typical Fuel Nozzle Cleaning and Rinsing Tank**

**Note:** If the transition piece assemblies are to be returned to a General Electric Company facility for cleaning, inspection, reconditioning and calibration, make sure that each assembly is wrapped separately in suitable packing material to ensure protection against damage during shipping, and identification for turbine number and customer location is marked on each package (use AES/GT-FF-6118 forms).

6. Remove the retainer from the transition piece using a 0.500 inch straight hex wrench. Use caution in handling the transition piece because the exit orifice can be easily damaged.

7. Using a 0.156 dia. wooden dowel and a lightweight mallet, gently tap the fuel nozzle pilot out of the back of the transition piece, exercising caution to prevent damaging the swirl slots of the pilot.

8. Record any abnormalities or defects found during disassembly of the transition piece on form AES/GT-FF-6118.

**Fuel Nozzle Inspection And Cleaning (After Disassembly)**

**Note:** The length of time required to thoroughly clean the fuel nozzles will largely depend upon the fuel used, turbine running time, and the type of cleaner used. The following method is the recommended approach for cleaning fuel nozzle components.

1. Inspect parts for major carbon buildup, and for plugged slots or orifices prior to cleaning. If there has been a fuel oil leak into the atomizing air passages, it will be necessary to remove the oil and carbon deposits from the body. Such a condition must be corrected before further turbine operation. Record findings on AES/GT-FF-6118.

2. Large pieces of face carbon which can be picked or rubbed off with the fingers may be removed. This will shorten the time required in the cleaning tank.

**CAUTION: Do not use metallic scrapers, pen knives, or wire brushes to remove carbon from fuel metering passages.**

3. A typical cleaning and rinsing facility is shown in Fig. CI-I7. Immerse the parts in a clean solution like those suggested in Table CI-1, and allow the parts to soak at the solution temperature recommended by the supplier until clean. Follow the manufacturer's instruction for use, the temperature required and the safety precautions to be observed.

   Ultrasonic cleaning equipment may also be used to clean fuel nozzle parts. Consult ultrasonic equipment manufacturer for cleaning equipment recommendations, cleaning solution to be used, temperatures required and the safety precautions to be observed.

**TABLE CI-1**
**RECOMMENDED CLEANING COMPOUNDS**

**CAUTION: Follow the manufacturer's instruction for use, the temperature required and the safety precautions to be observed.**

| COMPOUND | MANUFACTURER |
|---|---|
| Lix Diesel Klean, Heavy | Lix Corporation<br>716 East 85th Street<br>Kansas City, Missouri |
| *Turco 4008 | Turco Products Inc.<br>6135 South Central Avenue<br>Los Angeles, California |
| Bendix Metal Clene | Bendix Products<br>South Bend, Indiana |
| Oakite Solution | Oakite Products Inc.<br>50 Valley Road<br>Berkeley Heights, New Jersey |

*When extremely tenacious deposits are encountered, a 1/2 hour presoak in Turco Super Carb or equivalent is recommended.

**WARNING: Cleaning solutions are extremely caustic. Wear protective clothing, gloves and face shield to prevent contact with skin or eyes. If solution contacts skin or eyes, irrigate thoroughly with clean water and immediately obtain medical aid.**

4. Remove the parts from the cleaning solution, and spray with water and steam to knock off loosened carbon.

5. Rinse the parts, and coat them with either clean fuel oil or light machine oil.

6. Visually inspect all parts, keeping in mind that the contact surfaces of the fuel nozzle parts are machined to a fine finish so the flow rates of fuel oil or gas and atomizing air through the nozzle assembly will be within specified limits. When inspecting parts, ascertain that there are no dents, burrs and nicks on any surfaces which will result in an interference on reassembly, or affect fuel or air flow. Special attention should be given to the condition of the swirl slots in the fuel nozzle pilot, and the surfaces with which the nozzle pilot are in contact. If the swirl slots show any indication of crushing, galling or erosion, or if the surfaces with which they are in contact are not flat and smooth, the parts must be replaced. Record findings on inspection form AES/GT-FF-6118.

**CAUTION: Do not attempt to remachine damaged parts. This can cause irregular fuel flow, erratic combustion system operation, and subsequent turbine damage.**

7. Inspect the atomizing air cone, transition piece components, outer tip and air swirler for nicks, scratches, dents and unusual wear. Record findings on inspection form AES/GT-FF-6118.

8. Inspect the atomizing air cone and transition piece to determine if parts have been adversely affected by erosion. The depth of the erosion and remaining thickness of the metal may be measured with a deep throat caliper. Figure CI-I8 shows a cut-away of the atomizing air cone showing the normal thickness of the wall. Figure CI-I9 shows a cut-away of the transition piece showing the normal thickness of the wall.

**Note:** When measuring always try to position the caliper tips perpendicularly to the area being measured.

9. Complete additional data requested on inspection forms AES/GT-FF-6118 and AES/GT-FF-6000 and forward to your General Electric Company Field Service Representative.

10. Erosion may be projected based on the inspection measurements and operating time as follows:

    a. Establish current erosion depth. Original wall thickness less remaining wall thickness.

    b. Extrapolate future erosion depth using simple linear relationship

       Present Depth (1) = x

       Current Interval + Next Interval
       Fired Hour Fired Hour
       Current Interval
       Fired Hour

    c. **Example:** Assume inspection interval of 7850 fired hours. Where current erosion depth was 0.015 inch and the next planned interval to be 8000 fired hours.

       0.015 inch x 7850 8000
       Fired Hours + Fired Hours = 0.030 Inch
       7850 Fired Hours

       0.030 inch is the total expected erosion depth at next inspection interval.

    d. For projected erosion allow a minimum thickness of remaining material to be 0.030

inch on the atomizing air cone and 0.010 inch on the transition piece.

**Fuel Nozzle Reassembly**

1. New parts are frequently plastic coated, be sure that all plastic material is removed before reassembling the nozzle. Ascertain that no plastic material remains lodged on inside surfaces.

2. Do not use Teflon tape or liquid sealant anywhere in the fuel nozzle assembly.

3. Ascertain that all gaskets are new. **Never reuse an old gasket.**

4. Use FEL-PRO C-102 anti-seize compound on all threaded parts.

5. During reassembly of the fuel nozzle, use the torque values specified on the fuel nozzle assembly drawing in the Parts Lists and Drawings volume of the Service Manual.

6. Use a torque wrench with a range from zero to approximately 20 percent above the maximum torque required for assembly. The accuracy should be within ±5 percent.

7. Hold the transition piece lightly, but firmly, in a vise, using two soft metal blocks of sufficient width and depth to grip the hex of the transition piece housing in the vise jaws. See Fig. CI-I3.

8. Insert the fuel nozzle pilot into the transition piece body, making sure all parts bottom properly against their seats. See Fig. CI-I3.

9. Wipe anti seize compound lightly on the internal transition piece threads and the retainer piece threads.

10. Screw the retainer into the transition piece. Using a straight 0.500 inch hex wrench, socket, and torque wrench, torque the retainer to 40 ft. lbs. See Fig. CI-I3.

11. Using a suitable holding fixture, secure the fuel nozzle body in a vertical position, bore side up.

12. Wipe anti-seize compound lightly on the external threads of the transition piece assembly and on the corresponding threads in the fuel nozzle body.

13. Carefully install a new gasket into the fuel nozzle body, making sure that it is seated against the retaining seat in the fuel nozzle body.

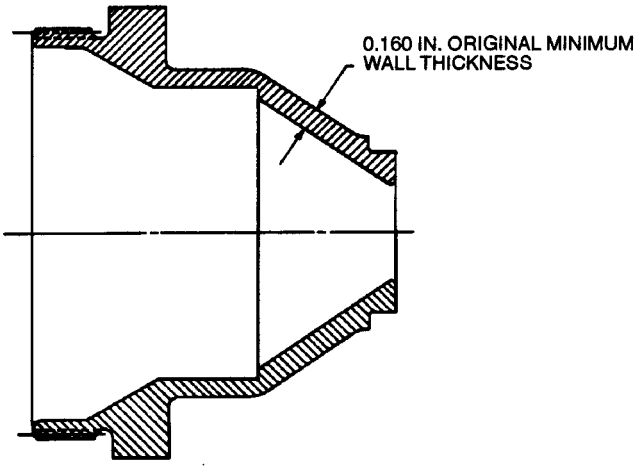



**Figure CI-I8. Fuel Nozzle Cut-Away of Atomizing Air Cone showing Normal Wall Thickness**

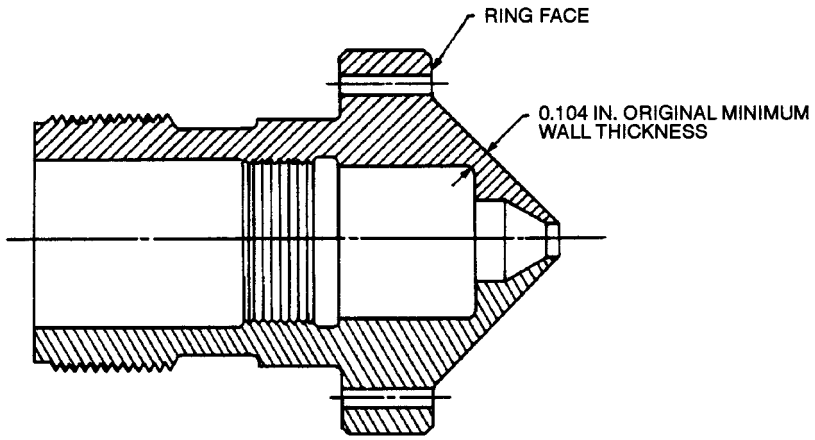



**Figure CI-I9. Fuel Nozzle Cut-Away of Transition Piece showing Normal Wall Thickness**

14. Screw the transition piece assembly into the fuel nozzle body finger tight. Using the special crowfoot wrench and a torque wrench, torque the transition piece assembly from 125 to 135 ft. lbs. Torque the assembly only once.

**CAUTION: Do not loosen and retorque the assembly without replacing the gasket.**

15. Check the integrity of the transition assembly installation by pressure testing to 300 psig pressure. Leaks in the transition piece assembly indicate a faulty gasket. See Fig. CI-I2.

**CAUTION: Exercise extreme care when assembling pressure test fixture to the transition piece assembly. Damage to the tip of the transition piece can occur if bar stock flange is not pulled down squarely and/or if the rubber pad has been damaged or cut. If a transition piece is damaged during test, do not attempt any repairs, replace with new transition piece.**

16. Reinstall the fuel nozzle body in the holding fixture.

17. Wipe anti-seize compound lightly on the threads of the atomizing air cone and on the corresponding threads in the fuel nozzle body. Screw the atomizing air cone into the fuel nozzle body. Using the special crowfoot wrench and a torque wrench, torque the atomizing air cone from 35 to 40 ft. lbs.

18. Install the fuel nozzle lockplate on the fuel nozzle body. See Fig. CI-I10.

19. Wipe anti-seize compound lightly on the threads of the outer swirl tip and on the corresponding threads in the fuel nozzle body. Screw the outer swirl tip into the fuel nozzle body. Using the special crowfoot wrench and torque wrench, torque the outer swirl tip from 60 to 70 ft. lbs.

20. Check the clearance at the front (downstream) face of the fuel nozzle. The recess of the inner cone inside the outer swirl tip should be 0.010 inch or less. Check the lockplate, it should offer some resistance to rotation around the swirl tip shoulder, but should still be capable of being rotated. The lockplate should not be clamped tight or completely flattened at assembly. See Fig. CI-I10.

21. If during either of these checks problems are found, the fuel nozzle should be disassembled and the drawing dimensions should be checked. Contact your GE Field Service Representative for assistance, if needed.

**Note:** If after assembly and final torquing of the outer swirl tip to the fuel nozzle body, if any of the lockplate slots lineup across from each other, lock the lockplate to the outer swirl tip slot.

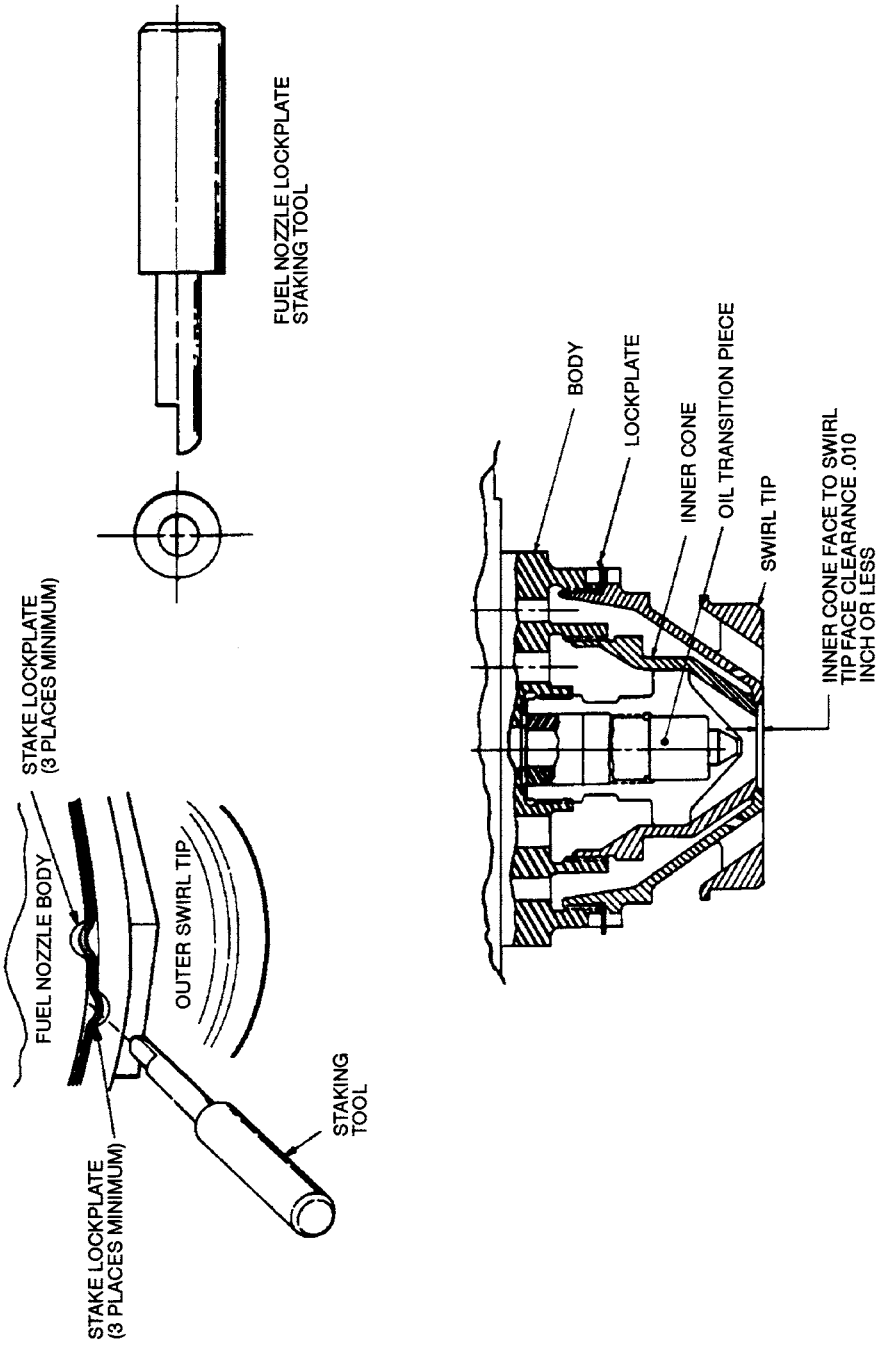



**Figure CI-I10. Fuel Nozzle Lockplate and Assembly Clearance**

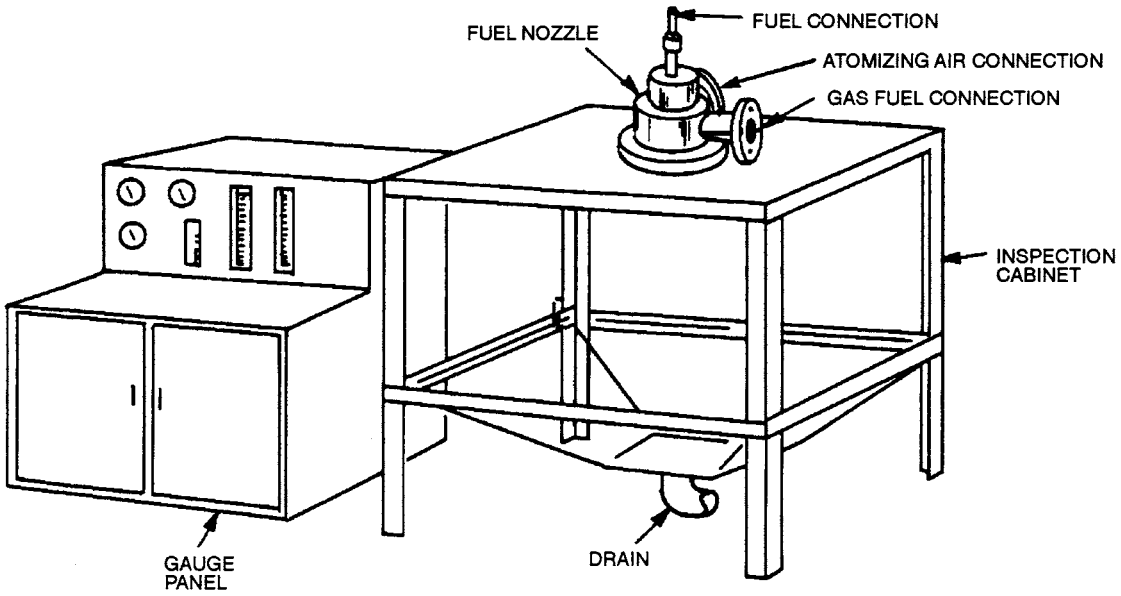



**Figure CI-I11. Typical Fuel Nozzle Flow and Spray Pattern Set Up**

22. Lock the outer swirl tip to the fuel nozzle body with the fuel nozzle lockplate. Lock the lockplate using the locking tool in a minimum of three (3) places on the nozzle body and a minimum of three (3) places on the outer swirl tip. See Fig. CI-I10.

**Fuel Nozzle Testing**

**CAUTION: The fuel nozzle must pass the following air flow, and fuel flow tests to insure integrity of assembly and function before being returned to service or storage.**

1. Fuel Nozzle Flow Check

   a. The flow through the nozzle cannot be adjusted. Burrs, nicks, or dents on mating surfaces will adversely affect flow.

   b. Test for flow in accordance with the requirements on the fuel nozzle assembly drawing in the Parts Lists and Drawings Volume of the Service Manual.

2. Fuel Nozzle Liquid Fuel Flow and Spray Pattern Check

   a. Install the fuel nozzle assembly in a test fixture as illustrated in Fig. CI-I11.

   b. Check for a symmetrical spray pattern of the fuel nozzle assembly.

   c. Check the fuel flow through the fuel nozzle in accordance with the requirements specified on the fuel nozzle assembly drawing in the Parts Lists and Drawings Volume of the Service Manual.

   d. Excess (rich) fuel flow indicates that the exit orifice and/or the pilot slots and holes may be too large, or improper pilot assembly, allowing too much oil to pass by the pilot. Low (lean) fuel flow indicates that the holes and slots in the nozzle may be plugged or contain burrs which restrict flow.

**Fuel Nozzle Storage**

1. Serviceable fuel nozzles should be stored as complete assemblies in separate containers designed to prevent damage to nozzles during storage. If the fuel nozzles are to be stored for an indefinite time, it is recommended that they be stored in a corrosion-free atmosphere.

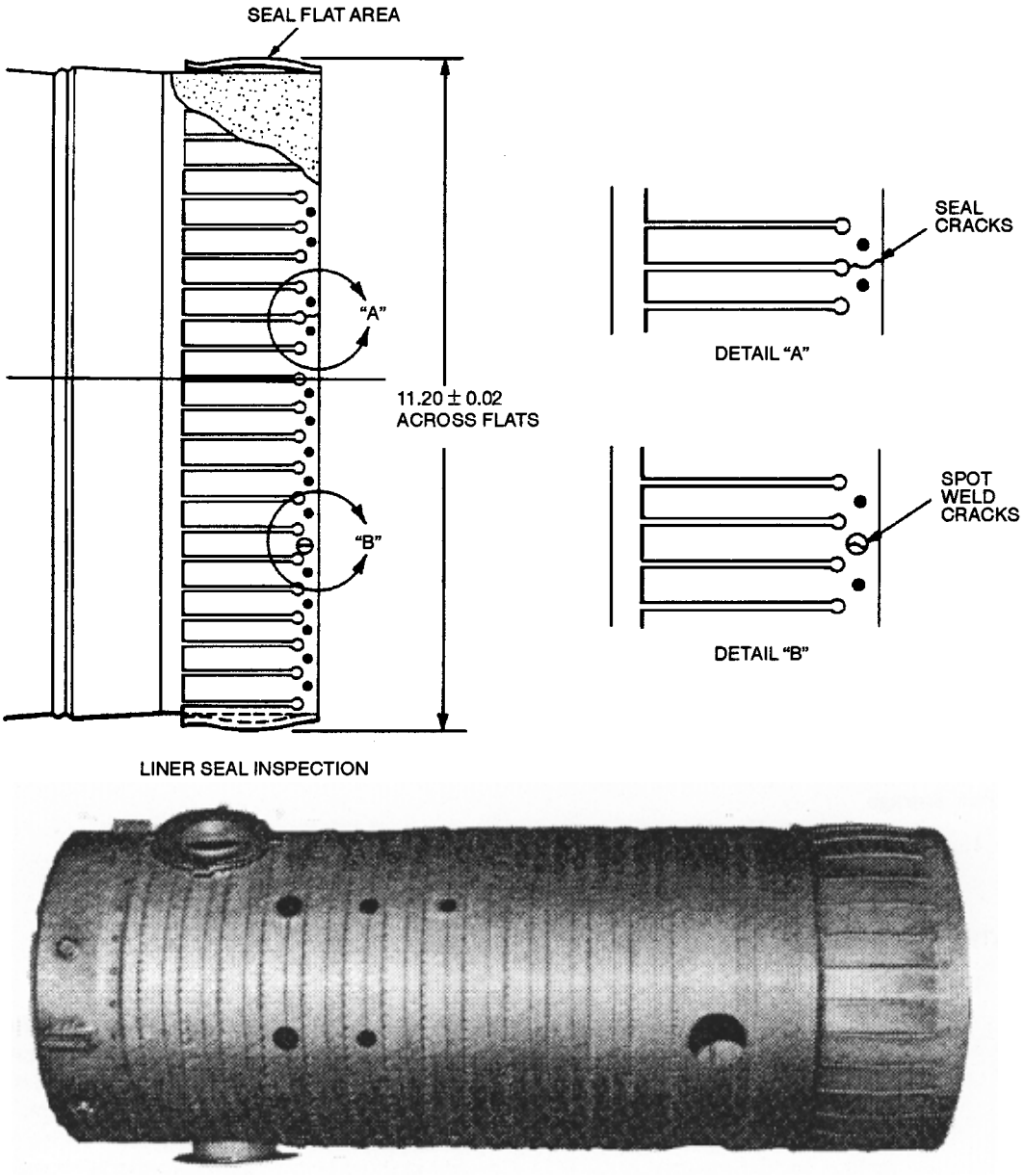



**Figure CI-I12. Combustion Liner Spring Seal (Hula Skirt) Inspection Areas**

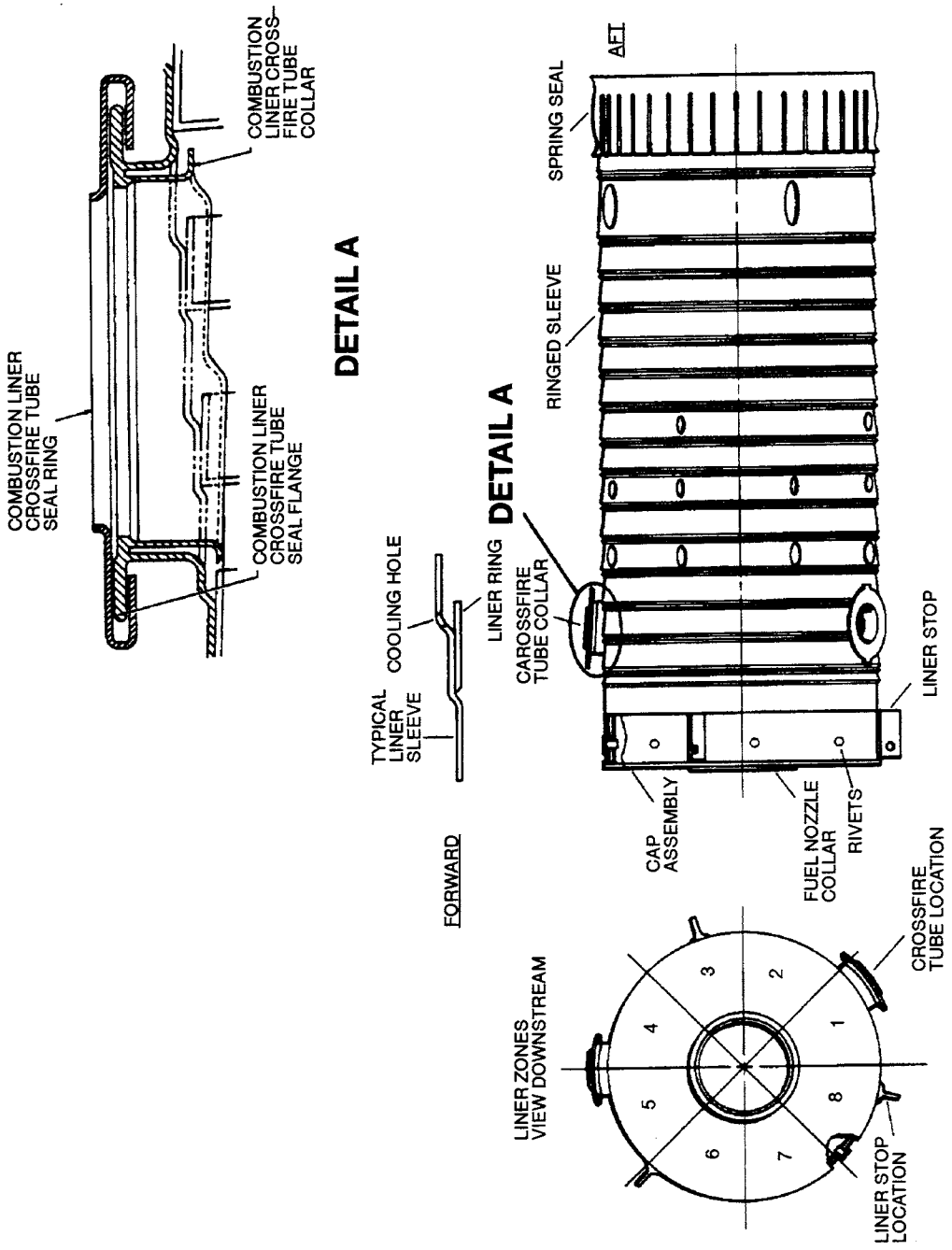
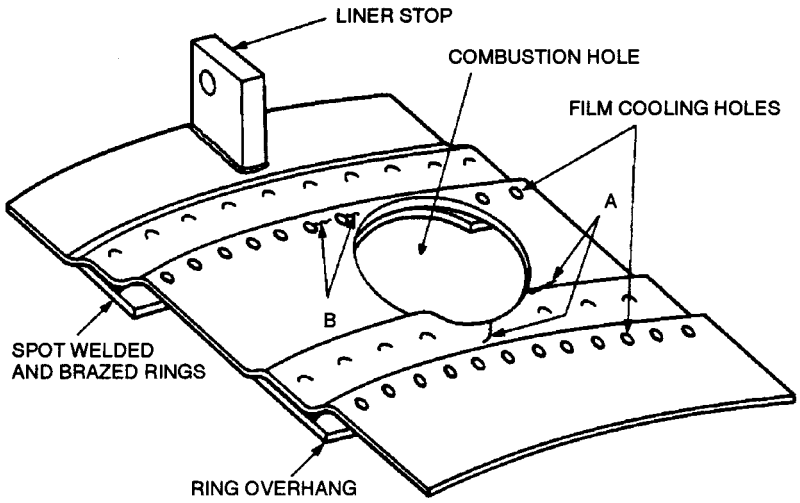



**Figure CI-I13. Typical Slot Cooled Combustion Liner**



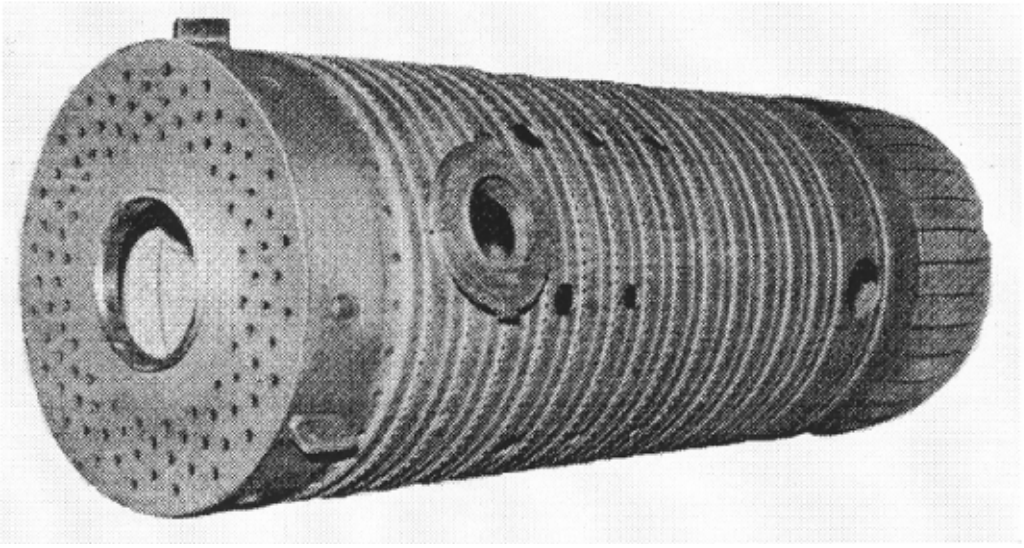



**Figure CI-I14. Typical Slot Cooled Combustion Liner Rolled Ring Section**

## OPERATION 3 - INSPECT COMBUSTION LINERS

1. Report Combustion Liner Inspection on AES/GT-FF-6121 Inspection Form.

**NOTE:** Combustion casing liners will not be dealt with by any specific drawing number. Repair welding techniques will not be described except to note that GTAW (Gas Tungsten Arc Welding) is the only weld repair procedure allowed on MS6001 combustion liners. It is generally recommended that combustion liners requiring repairs, replacement of caps, liner sleeves, and spring seals be sent to a qualified General Electric Company Gas Turbine repair facility, and not repaired in the field. Coated rod should not be used. Deterioration less than maximum allowable limits indicates that reuse may be permissible provided consideration has been given to past operating hours and firing conditions in relation to the future operations expected.

### TABLE CI-2
### INSPECTION LIMITS, SLOT COOLED LINERS

**CAUTION: The inspection criteria in this Table CI-2 applies to General Electric Company supplied slot cooled combustion liners only.**

| Inspect | Maximum Acceptance Limits | Disposition (For Parts Beyond Max. Limits) |
|---|---|---|
| 1. Liner Sleeve (Outer Shell) | | |
| a. Cracks (Except at Film Cooling Holes) | Any number, 3/16 inch long. See Fig. CI-I14 | * Replace |
| b. Cracks (At Film Cooling Holes) | None | * Replace |
| c. Burn through or missing metal | None | * Replace |
| d. Out-of-roundness | 1/8 inch from original contour | Reform using hand tools |
| e. Local distortion | 3/32 inch from original contour with no cracks | * Replace |
| f. Liner wear beneath spring seals | 20% material thickness beneath seal | * Replace |
| 2. Liner Rings | 0.060 inch min. gap | * Replace |
| a. Ring slot height | 0.060 inch min. gap | * Replace |
| b. Axial cracks | None | * Replace |
| c. Circ. cracks | None | * Replace |
| d. Burn through (missing metal) | None | * Replace |

| | | |
|---|---|---|
| e. Visual braze flow (overtemperature) | None | * Replace |
| 3. Cap and Cowl | | |
| a. Axial cracks between louvers. | 1/8 inch long, one per ligament. Allow up to 3 per cap. | * Replace. |
| b. Louver end cracks. | 1/8 inch long, 50%of louvers. | * Replace. |
| c. Louver-to-louver cracks | 3 per row each separated by 5 ligaments | * Replace. |
| d. Row to adjacent row cracks | None | * Replace. |
| e. Distortion | 1/8 inch from original contour. | * Replace. |
| f. Cap to liner looseness | 0.030 inch Max. axial. | * Replace. |
| g. Cowl plate weld cracks | None | * Replace |
| h. Cowl plate cracks | None | * Replace. |
| i. Fuel nozzle collar wear | 0.040 inch metal removed thru 90° arc | Repair weld. |
| j. Burn through or missing metal | None | * Replace |
| k. Warping or local distortion of cowl plate at fuel nozzle collar | 0.200 inch from original location | * Replace |
| 4. Liner Spring Seal | | |
| a. Axial cracks | None | Replace seal |
| b. Circ. Cracks | None | Replace seal |
| c. Distortion (flattening) Pi-tape OD | 0.200-0.230 inch<br>11.18/11.22 inch dia. | Replace seal<br>Replace seal |
| d. Wear | 30% mat'l removal at slots | Replace seal |
| e. Missing leaves | None | Replace seal |
| f. Spot weld cracks | Three per liner, No more than two adjacent. | Replace seal |
| g. Bent leaves | Any number | Straighten and penetrant check |
| h. Loss of elasticity | Seal leaves must be in contact with liner, no clearance | Replace seal |
| 5. Crossfire Tube Collars | | |
| a. Cracks | 0.060 inch long 2 places | * Replace |

| | | |
|---|---|---|
| b. Wear | 2.52 inch maximum I.D. | * Replace |
| c. Distortion | 2.48 inch minimum I.D. | Reform with hand tools |
| d. Burning at crossfire tube collar | None | Replace collar |
| 6. Crossfire Tube Collar Air Seal | | |
| a. Wear | 0.025 inch minimum thickness | Replace |
| b. Distortion | 0.020 inch out of flat | Replace |
| 7. Liner Stops | | |
| a. Side wear | 0.140 inch minimum thickness | Repair |
| b. End wear | 0.020 inch metal removed | Repair |
| c. Cracks (weld or parent metal) | None | Repair weld |
| 8. Liner Rivets | | |
| a. Weld cracks | None | Repair |
| b. Missing rivets | None missing | Replace rivets |

* "Replace" does not mean liner must be scrapped; part can be repaired by qualified GE personnel.

2. Inspect Slot Cooled Liners

   a. Table CI-2 is a guide to aid in decision making on reuse, repair or discarding of slot cooled combustion liners.

   b. Figures CI-I 12, CI-I 13 and CI-I 14 should be used to identify areas of the slot cooled combustion liner called for in Table CI-2.

   c. Weld cracks shall be detected using fluorescent penetrant. Zyglo ZL2, ZL22 and ZL54 are acceptable.

   d. Cracks emanating from cooling holes or combustion holes should be detected using a bright light and ten (10) power glass. The use of penetrants over the liner surface is discouraged except where confirmation is required .

   e. Figure CI-I 14 shows a typical slot cooled combustion hole and portion of two bands of cooling holes. Areas that should be looked at closely are identified as follows.

      1. It is important to verify that there are no cracks emanating from the combustion holes (A) and no cracks between the cooling holes, or from the cooling holes toward the combustion holes (B).

2. The cooling ring overhang should be inspected to detect cracks or warpage. Particularly inspect the overhang in the cooling panel behind the crossfire tube ports and combustion holes.

3. Check the liner stops for wear, cracks or deformation due to improper handling.

4. Liner stop welds should be inspected for cracks.

f. Inspect the liner spring seals (Hula seals) for axial cracks, circumferential cracks, distortion (flattening), wear, missing leaves, bent leaves or spotweld cracks. See Table CI-2 for inspection criteria.

g. If any spring seals are reshaped, ensure the leaf curvature height is similar to adjacent leaves and "PI" tape dimension (11.20 + 0.02 inches).

h. Liquid penetrant check any spring seal leaves that have been reshaped.

i. Inspect liner rivets for weld cracks and missing rivet ends.

j. Inspect the combustion liner fuel nozzle collar for wear.

k. Inspect crossfire tube collar and air seal for cracks, wear and distortion.

l. Inspect the cap assembly, cowl plate to ring, for weld cracks.

m. Inspect the cap assembly to determine if it is loose in liner (0.030 inch allowed).

n. Report inspection findings on Inspection Form AES/GT-FF-6121.

**OPERATION 4 - INSPECT CROSSFIRE TUBES AND SPRING RETAINERS**

1. Inspect crossfire tubes for evidence of distortion, cracks, missing metal, burn-through and wear. Inspect the flanges for cracks. Use Table CI-3 for inspection criteria.

2. Inspect crossfire tube retainers for wear or spalling of hard facing and evidence of distortion. Use Table CI-3 for inspection criteria.

3. Report inspection findings on inspection form AES/GT-FF-6120.

**OPERATION 5 - INSPECT TRANSITION PIECES**

1. Clean and fluorescent penetrant inspect each of the following areas on transition pieces for cracking. See Fig. CI-I15 and CI-I16 and Table CI-4 for inspection criteria.

   a. Aft bracket to body for cracks.

   b. Aft brackets for cracks and wear.

   c. Aft frame internal corner cracks.

   d. Aft frame external corner cracks.

   e. Forward brackets for weld cracks right and left sides.

   f. Cooling holes for cracks.

   g. Transition piece body for cracks.

2. Use the transition piece inspection criteria Table CI-4 to completely inspect the transition pieces after fluorescent penetrant test.

3. Record all inspection findings on inspection report form AES/GT-FF-6122.

4. Repair welding procedures.

**Note:** Repair welding procedures will not be described except to note that GTAW (Gas Tungsten Arc Welding) is the only weld repair procedure allowed on MS6001 transition pieces. It is generally recommended that transition pieces requiring repairs be sent to a qualified General Electric Company Gas Turbine repair facility.

**OPERATION 6 - INSPECT COMBUSTION FORWARD AND AFT FLOW SLEEVES**

1. Using a bright light and ten (10) power magnifying glass, inspect the combustion casing flow sleeves.

2. Use Figs. CI-I 17 and CI-I 18 for identification of parts and Table CI-5 for inspection limits. Record all inspection findings on Inspection Report Form AES/GT-FF-6123.

   a. Inspect the sleeve for body or weld cracks, out-of-roundness, burn-through, missing metal and wear.

   b. Inspect forward flange to body weld for cracks or cracks in the flange.

   c. Inspect liner stops and rivets for wear, and liner stops for weld cracks.

**OPERATION 7 - INSPECT COMBUSTION CASINGS**

1. Inspect inside and outside of combustion casings for cracks, bulging, buckling signs of overheating and corrosion.

2. Inspect casings 3, 4, 5, 6 and 7 false start drain holes for plugging or blockage.

3. Inspect each combustion casing cover and hinge for cracking, freedom of hinges, loose hardware and/or hinge mounts.

4. Inspect the combustion casing covers sight port glass for cracks. Replace all cracked sight port glass with a new assembly.

**Note:** After inspection and/or replacement of sight port glasses, reinstall the pipe plug in the assembly. Do not stake pipe plugs at this time. After cleanup, visual inspection, pre-start and startup checks are completed, stake the pipe plug in two places.

5. Report condition and findings on inspection report form AES/GT-FF-6119.

**OPERATION 8 - INSPECT FIRST STAGE NOZZLE**

1. Use a borescope through the inspection ports, if applicable. Inspect the first stage nozzle for corrosion, erosion cracks, deposits and foreign object damage. See Section 4 Scheduled Turbine Maintenance, "Special Inspections" of the Service Manual.

2. If borescope inspection cannot be performed make a visual inspection of the first stage nozzle.

3. Report condition and findings on inspection report forms AES/GT-FF-6095 and 6096.

**OPERATION 9 - INSPECT AND TEST SPARK PLUGS**

**WARNING:** **If this gas turbine installation is in a hazardous area location as defined by National Protection Association (NFPA)-70, and the site requires a Class I, Division 1 or 2, Group D certified ignition system, any parts which are replaced as part of maintenance or servicing procedures must be identical to those as originally installed, per General Electric Company in order for the system certification to continue to be in effect.**

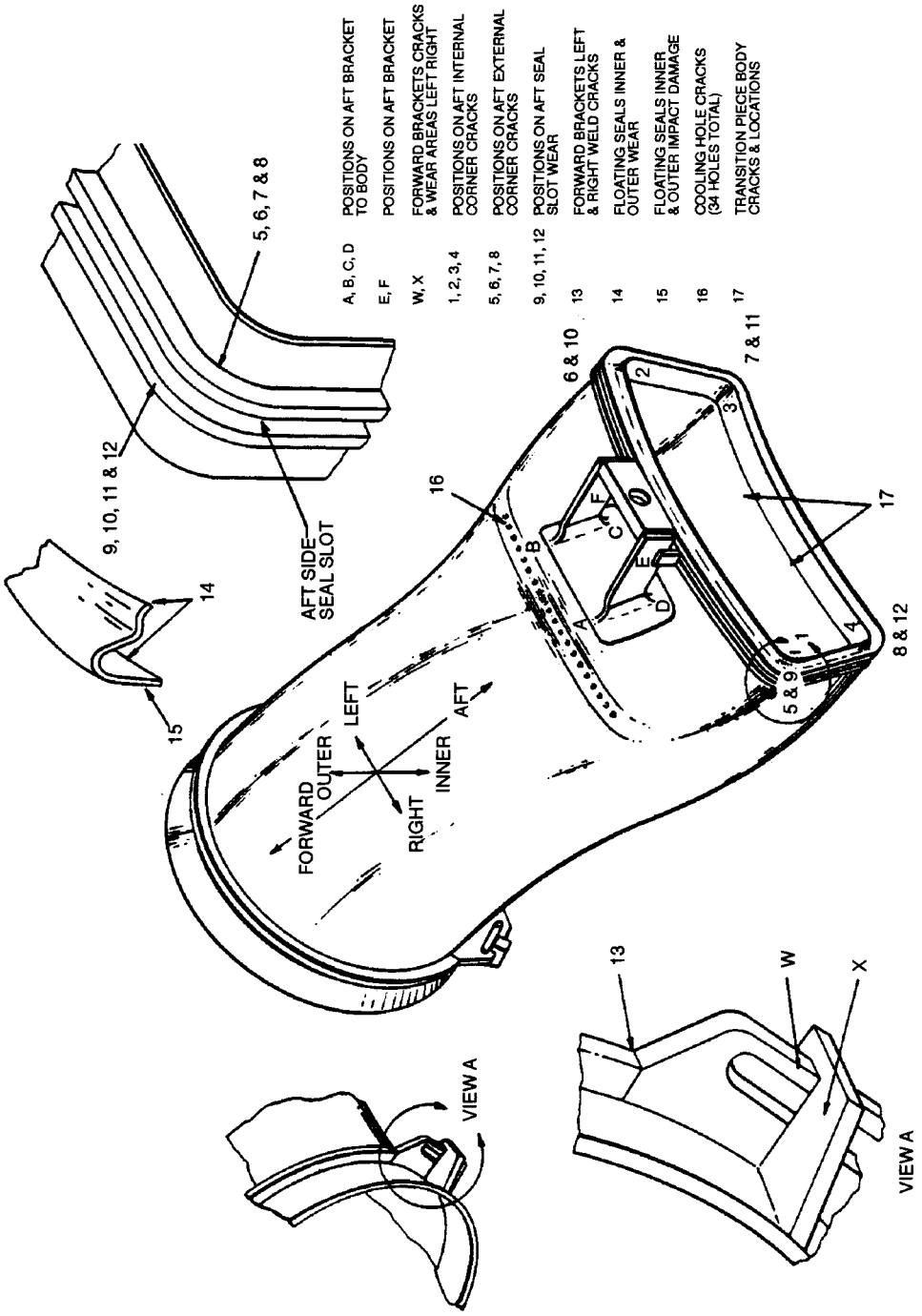



**Figure CI-I15. Transition Piece ("A" Style) Inspection Area Locations**

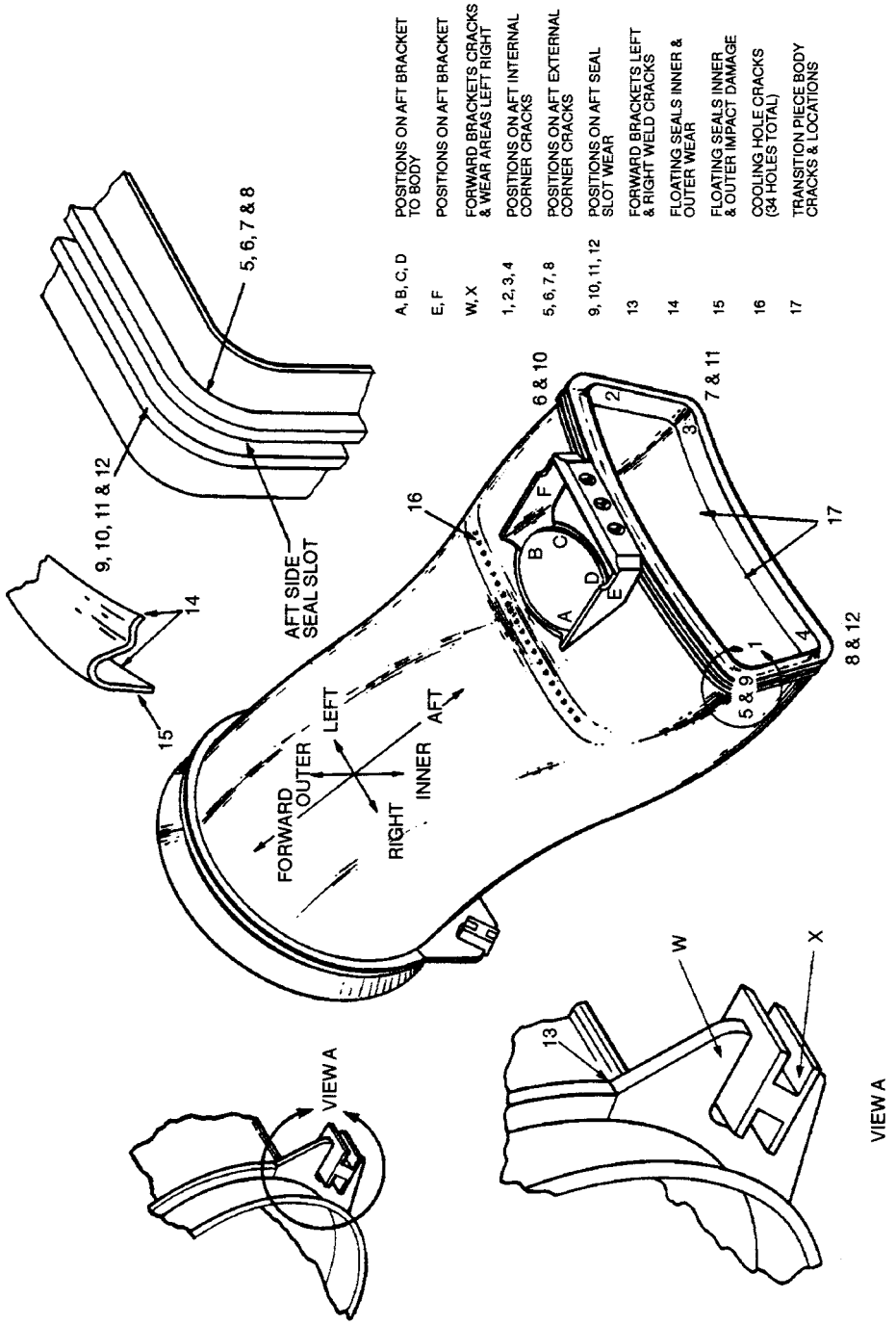



**Figure CI-I16. Transition Piece ("B" Style) Inspection Area Locations**

## TABLE CI-3
## INSPECTION LIMITS CROSSFIRE TUBES AND RETAINERS

**CAUTION: The inspection criteria in this Table CI-3 applies to General Electric Company supplied crossfire tubes and retainers only.**

| Inspect | Maximum Acceptance Limits | Disposition (For Parts Beyond Max. Limits) |
|---|---|---|
| 1. Crossfire Tubes | | |
| a. Burn through; missing metal | None | Replace |
| b. Distortion | None | Replace |
| c. Cracks | None | Replace |
| d. Flange wear | 0.14 inch minimum thickness | Replace |
| e. Tube wear | 0.11 inch minimum thickness | Replace |
| 2. Crossfire Tube Spring Retainers | | |
| a. Wear | 20% of original material | Replace |
| b. Evidence of distortion | None | * Replace |

## Table CI-4
## TRANSITION PIECE ASSEMBLY INSPECTION LIMITS

**CAUTION: The inspection criteria in this Table CI-4 applies to General Electric Company supplied transition pieces only.**

| Inspect | Maximum Acceptance Limits | Disposition (For Parts Beyond Max. Limits) |
|---|---|---|
| 1. Aft Side Seal Slots | | |
| a. Wear | 0.070 inch metal removal | * Replace |
| b. Cracks | None | * Replace |
| 2. Body | | |
| a. Body or weld cracks | 0.1875 inch length | * Replace |
| b. Distortion | 0.0625 inch from original contour | * Replace |
| c. Corrosion | 0.030 inch thick | |
| d. Deposits | 0.030 inch reduction from original material thickness | Remove Deposit |

| | | |
|---|---|---|
| 3. Aft Bracket | | |
| a. Weld or body cracks | None Inside and outside surface | * Replace |
| 4. Aft Picture Frame/Body Corners | | |
| a. Weld or body cracks | None | * Replace |
| b. Distortion | 0.200 inch | * Replace |
| c. Wear | 0.070 inch metal removed | * Replace |
| 5. Forward Support Rings | | |
| a. Weld or body cracks | 0.1875 inch long. Inside and outside surface | * Replace |
| b. Wear (forward support engagement surface) | 0.030 inch from original thickness | * Replace |
| 6. Floating Seal Retainers | | |
| a. Weld or retainer cracks | None | * Replace |
| b. Wear | 50% of original material | * Replace |
| 7. Floating Seals | | |
| a. Wear | 50% of original material thickness | * Replace |
| b. Cracks | None | * Replace |
| c. Radial movement | 0.080 inch minimum from nominal diameter | * Replace |
| 8. Side Seals | | |
| a. Surface Wear | 0.050 inch metal removal | * Replace |
| b. Cracks | None | * Replace |
| c. Distortion | 0.020 inch out of flat | * Replace |
| 9. Forward Support (Bullhorn) | | |
| a. Cracks | None | * Replace |
| b. Slot wear | 0.040/0.060 inch from original dimension | * Replace |

* "Replace" does not mean liner must be scrapped; part can be repaired by qualified GE personnel.

## TABLE CI-5
## INSPECTION LIMITS FOR FLOW SLEEVES

**CAUTION: The inspection criteria in this Table CI-5 applies to General Electric Company supplied flow sleeves only.**

| Inspect | Maximum Acceptance Limits | Disposition (For Parts Beyond Max. Limits) |
|---|---|---|
| 1. Sleeve | | |
| a. Cracks | None | * Replace |
| b. Distortion | 0.125 inch out of round | Reform |
| c. Wear (Pilot Diamter) | 0.010 inch per side | * Replace |
| 2. Forward Flange | | |
| a. Distortion | 0.010 inch out of round | Replace |
| b. Drain hole blockage | None | Clean |
| 3. Liner Stops | | |
| a. Rivet wear | 0.030 inch from original diameter | * Replace |
| b. Mount sidewall wear | 0.015 per side | * Replace |

* "Replace" does not mean liner must be scrapped; part can be repaired by qualified GE personnel.

## Table CI-6
## SPARK PLUG INSPECTION LIMITS

**CAUTION: The inspection criteria in this Table CI-6 applies to General Electric Company supplied spark plugs only.**

**WARNING: If this gas turbine installation is in a hazardous area as defined by National Protection Association (NFPA) -70, and the site requires a Class I, Division 1 or 2, Group D certified ignition system, any parts which are replaced as part of maintenance or servicing procedures must be identical to those as originally installed, per General Electric Company in order for the system certification to continue to be in effect.**

**Table CI-6**
**SPARK PLUG INSPECTION LIMITS**

| Inspect | Maximum Acceptance Limits | Disposition (For Parts Beyond Max. Limits) |
|---|---|---|
| 1. Spark Plug | | |
| a. Binding | None Allowed | Disassemble and clean |
| b. Galling | None Allowed | Polish with fine abrasive cloth |
| c. Terminal extension cracks | None Allowed | Replace |
| e. Ceramic insulator cracks | None | Replace |
| f. Core assembly cracks | None Allowed | Replace |
| g. Bad threads on electrical connection | None | Replace |
| h. Test firing | No internal arcing allowed | Replace insulator |

1. Inspect the spark plug assembly for binding, galling, arcing, cracks, ceramic component cracks and damage threads on electrical lead connection. See Table CI-6 for inspection criteria. See Fig. CI-I 19.

2. Disconnect the spark plug lead and protect the end termination.

3. Check the movement of spark plug piston by pulling on the lead end of the piston.

**CAUTION: Do not pull on the spark plug lead at any time.**

If any binding exists, the cylinder assembly must be disassembled and the binding eliminated.

To disassemble the cylinder assembly, remove the four nuts and lockwashers from the top of the cylinder; then remove the top of the cylinder.

**CAUTION: Take care not to lose or damage the nickel sealing rings at the top and bottom of the cylinder.**

Remove the spring and pull the piston assembly out.

Disassemble the piston assembly by grinding off the welds on the locking tab and/or releasing the lockplate tabs, if applicable, that secures the retaining nut. Unscrew the retaining nut, remove the lockplate, if applicable. Discard the lockplate, do not reuse. Remove the inner assembly containing the ceramic insulator sleeve, terminal extension, retainer, core assembly and two nickel washers. Carefully separate the inner assembly components.

**CAUTION: Care should be taken when removing the terminal extension (ceramic rod) from the core assembly to avoid breakage of the core assembly's ceramic shell.**

Inspect all components for cracks, wear, damage and arcing. Replace parts as required. Remove all foreign material, such as rust, dirt or chips from the cylinder and piston assemblies. Remove all oxidation and galling marks. (See Fig. CI-I 19.)

4. Reassemble the spark plug as follows: install new nickel washers on either side of the porcelain core assembly. This washer must conform to the contours of the core. This assembly should be carefully inspected to insure correct washer orientation and that two washers are not doubled up in either case. Insert the spring seal end of the terminal extension into the core assembly.

**CAUTION: Exercise care in inserting the terminal extension to prevent breakage of the core assembly.**

Insert the core/extension assembly into the retainer. Check positioning of the nickel washers on each side of the core for correct positioning. To facilitate assembly, hold core/extension/retainer assembly in the vertical position, with core upward and insert the into piston assembly. Hold spark plug cylinder vertically, mounting flange down and install piston assembly and spring into the cylinder, and using the nickel sealing ring, install the top cover plate and bolts.

Install outer insulator, lockplate if applicable, and retaining nut hand tight. Centralize the center electrode between the two ground electrodes and adjust the tips to insure that gaps of 0.090 ± 0.005 inch exist between the center and side electrodes. Torque the retaining nut to 30-50 ft. lbs., but do not bend lockplate if applicable, at this time. To insure good electrical contact between the core and the terminal extension, tap lightly on the ceramic rod with a small plastic rod through the open end of the piston rod where the electrical cable makes it's connection.

Upon completion of assembly, check the spark plug for freedom of movement (binding) by pulling on the lead end of the piston and working the assembly in and out several times. The total stroke is 2.50 inches.

**CAUTION: Exercise care in making this check. Do not allow the piston rod assembly to slam back from the full retracted position.**

5. Prior to installing the spark plug assembly to the unit, test the spark plug as follows: reconnect the spark plug lead, with the plug out of the unit but with its body resting against the turbine shell (or other grounded metal). Make sure the center electrode is not grounded.

**WARNING: Do not test spark plugs (ignitors) in or near an area with explosive atmosphere.**

Be careful of high voltage when testing. Energize the ignition circuit and check the spark of the plug for approximately two (2) minutes. If heating occurs at a point about 4 inches up from the tip, this indicates arcing between the core and ceramic rod contact. This can be remedied by disassembly of the spark plug and cleaning of the core electrical contact. If arcing does not occur at the tip of the electrodes, recheck for clean electrodes and recheck the electrode gaps at 0.085 to 0.095 inch between center and side electrodes. Gaps on either side of the center electrode must be equal within 0.010 inch.

If the plug still does not fire properly, check voltage to the ignition transformer, and check ignitor lead voltage by disconnecting the lead from the ignitor and arcing the lead to ground.

**Note:** Be sure that the tip of the lead is within one-half inch of ground before ignition voltage is applied.

If no arcing occurs between the tip of the lead and ground, the transformer and/or lead are defective and should be replaced. If arcing occurs, the spark plug is faulty and should be inspected for an open circuit, short etc.

After the spark plug assembly has been successfully test fired, disconnect the ignition lead and form lockplate or weld a tab between the retaining nut and the piston assembly as shown in Fig. CI-I19. The tab material should be AISI 304 chrome nickel steel, bright annealed, its dimensions should be 0.060 x 0.250 x 0.500. Use GTAW welding process with filler ER308. In situations where GTAW welding is not available, stick weld with weld electrode E308-15, E308-16. Upon completion of welding the tab on the spark plug, check to insure there is no restriction in the retraction or extending of the spark plug. For those spark plug assemblies using a lockplate to secure the retaining nut, using a pair of pliers, squeeze the lockplate tabs over two diagonally opposite flats on the retaining nut. DO NOT HAMMER. See Fig. CI-I19.

**OPERATION 10 - INSPECT AND TEST FLAME DETECTORS**

**Note:** The flame detector tube is contained in the flame sensor subassembly. Combined with the body and window subassembly, the two subassemblies form the ultraviolet flame sensor. If during inspection and testing the sensor is found to be faulty, it should be replaced as a complete assembly.

1. Clean the scanner lenses. Reconnect flame detector wiring and energize the control panel. With power on the control panel, check that FL-1 and FL-2 indicating lights are on.

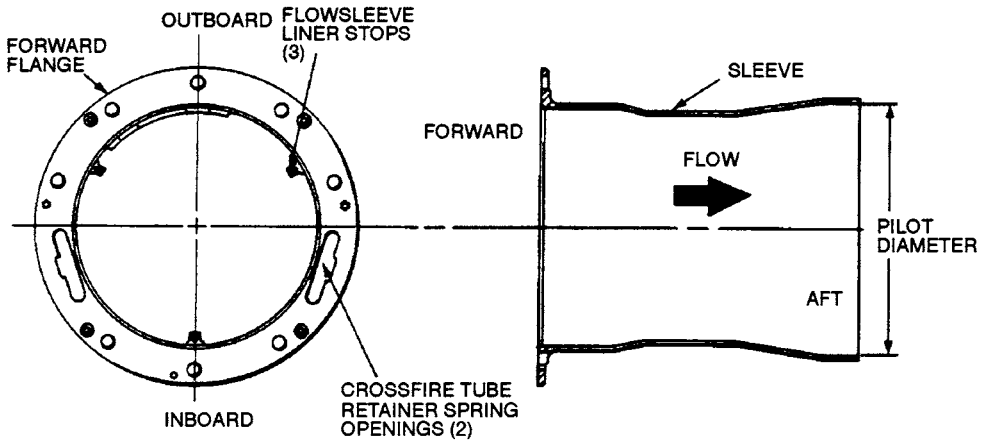
**CAUTION: Reverse polarity will destroy the detector.**

2. With the detector amplifier power on, verify base wiring polarity at the sensor junction box by reading 290 VDC (minimum) across leak pair.

3. Turn power off and connect black detector lead to plus (+) and the white lead to minus (-).

4. Using a ultraviolet source at the sensor, assure that each flame detector picks up and drops out when ultraviolet source is placed in front of the sensor and removed.

**Note:** A high level of ultraviolet is defined as a match or candle at a distance of less than 2 inches. The unit, however, should be capable of detecting a match or candle at approximately 18 inches.



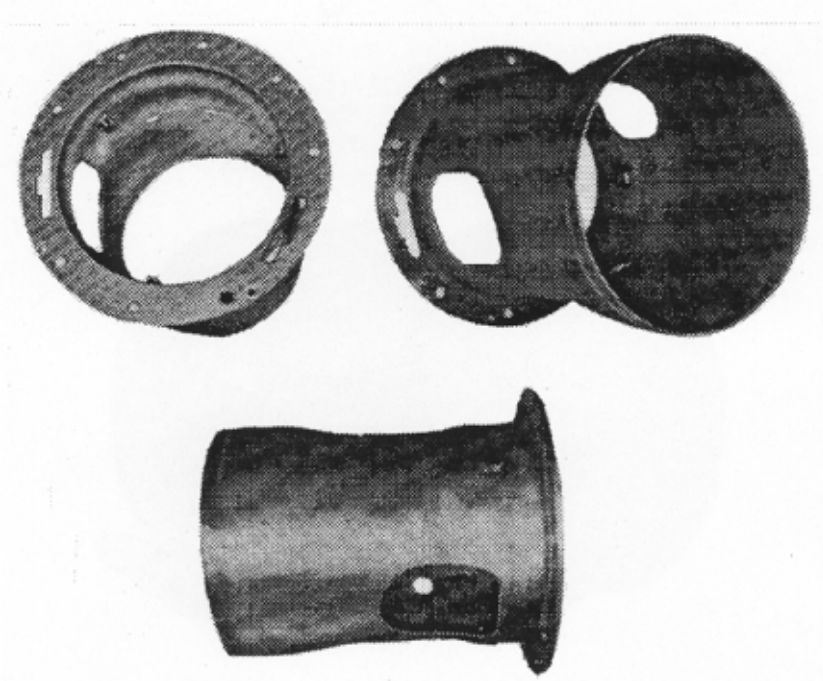



**Figure CI-I17. Combustion Casing Forward Flow Sleeve**

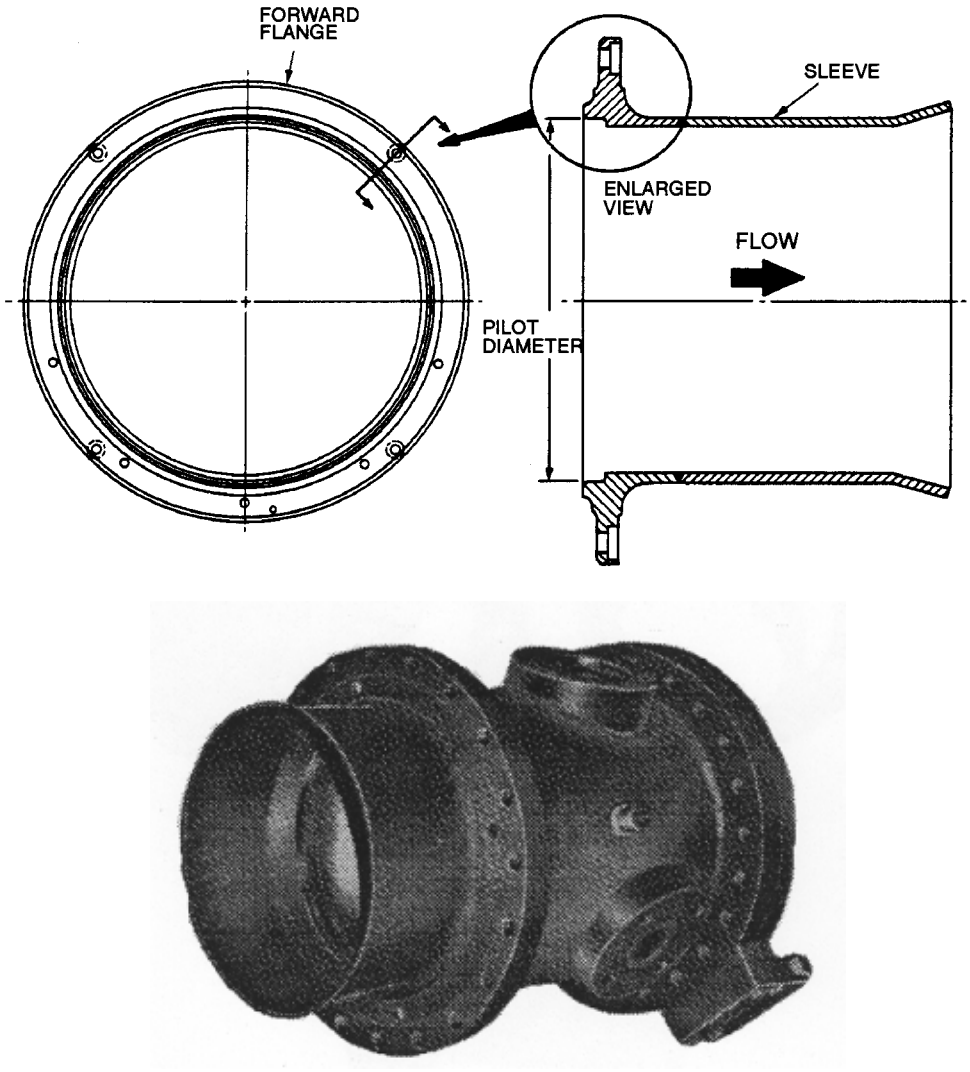



**Figure CI-I18. Combustion Casing Aft Flow Sleeve**

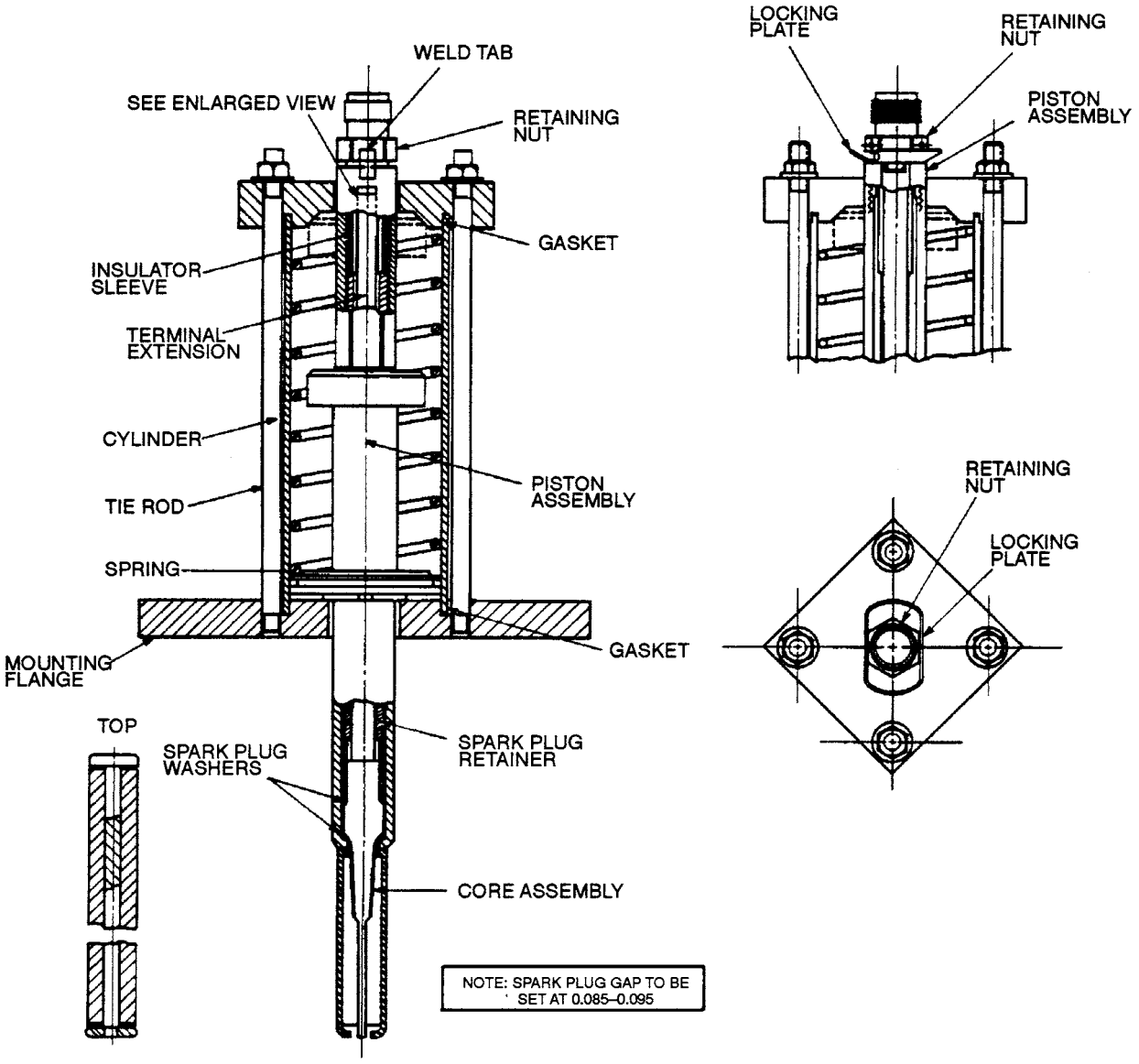



**Figure CI-I19. Spark Plug Assembly with Tab Weld or Lockplate**

## REASSEMBLY

### OPERATION 1 - TRANSITION PIECE PREASSEMBLY INSPECTION AND CHECKS

1. Prior to installation, inspect transition pieces for obvious handling or transportation damage; such as deformed floating seals and pinched end seal slots and damaged inner brackets. Trial fit end seals into transition piece end seal slots before installing transition piece into the unit.

2. Trial fit each support clamp (bullhorn) to its respective transition piece prior to installing into the unit.

3. Check first stage nozzle segments for axial displacement (Steps) between adjoining segments. Displacement greater than 0.030 inch should be reseated with a five (5) pound rawhide or nylon mallet. Do Not Use Metal Hammer.

4. Check to insure that the first stage nozzle transition piece seal slots are free from damage such as nicks, burrs or any other faults which would interfere with the proper assembly of the transition piece floating seals to the first stage nozzle.

5. Check all helicoils for cross threading, elongation, stripping or insert backing out. Remove and replace all damaged helicoils.

### OPERATION 2 - INSTALL TRANSITION PIECES ("A" STYLE DESIGN)

**Note:** This operation is applicable to the "A" style design transition piece only. The "A" style transition piece has a fin aft bracket with only one (1) mounting hole. See Fig. CI-R1.

**CAUTION: Use new bolts and lockplates. Do not reuse hardware previously removed.**

1. Install guide pin, hand tight only, into aft bracket bolt hole in the first stage nozzle retaining ring in the number five (5) position. See Fig. CI-R1.

**Note:** With the turbine casing in place the last transition piece to be installed can be in the number 1, 9 or 10 location, dependent on which turbine casing access port was used at disassembly. Install the transition pieces starting with number five (5) position; then install 4, 6, 7, 3, 2, 8, 1, 9 and 10 last or as stated above number 1 or 9 position last. With the turbine casing removed install number five (5) first then 4, 6, 7, 3, 2, 8, 1, 9 and 10 last.

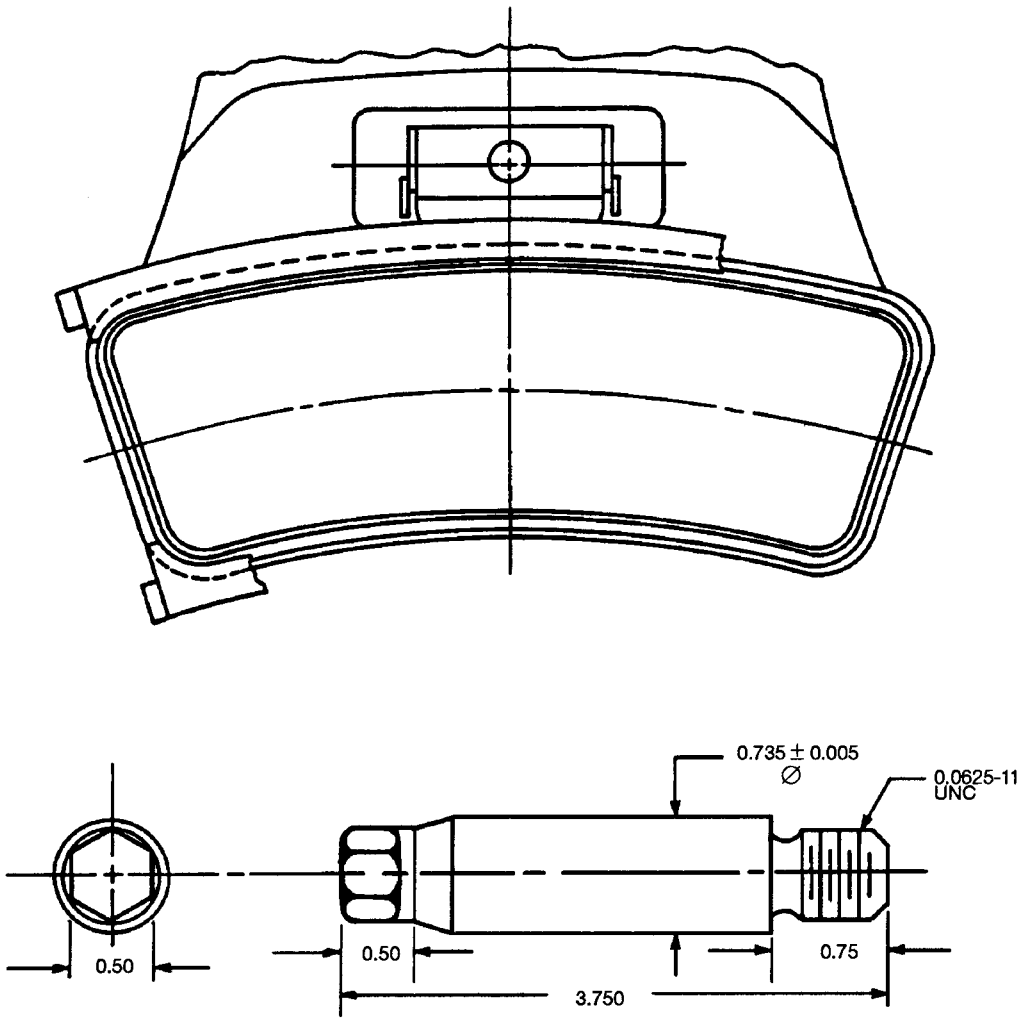



**Figure CI-R1. Guide Pin for "A" Style Transition Piece Installation**

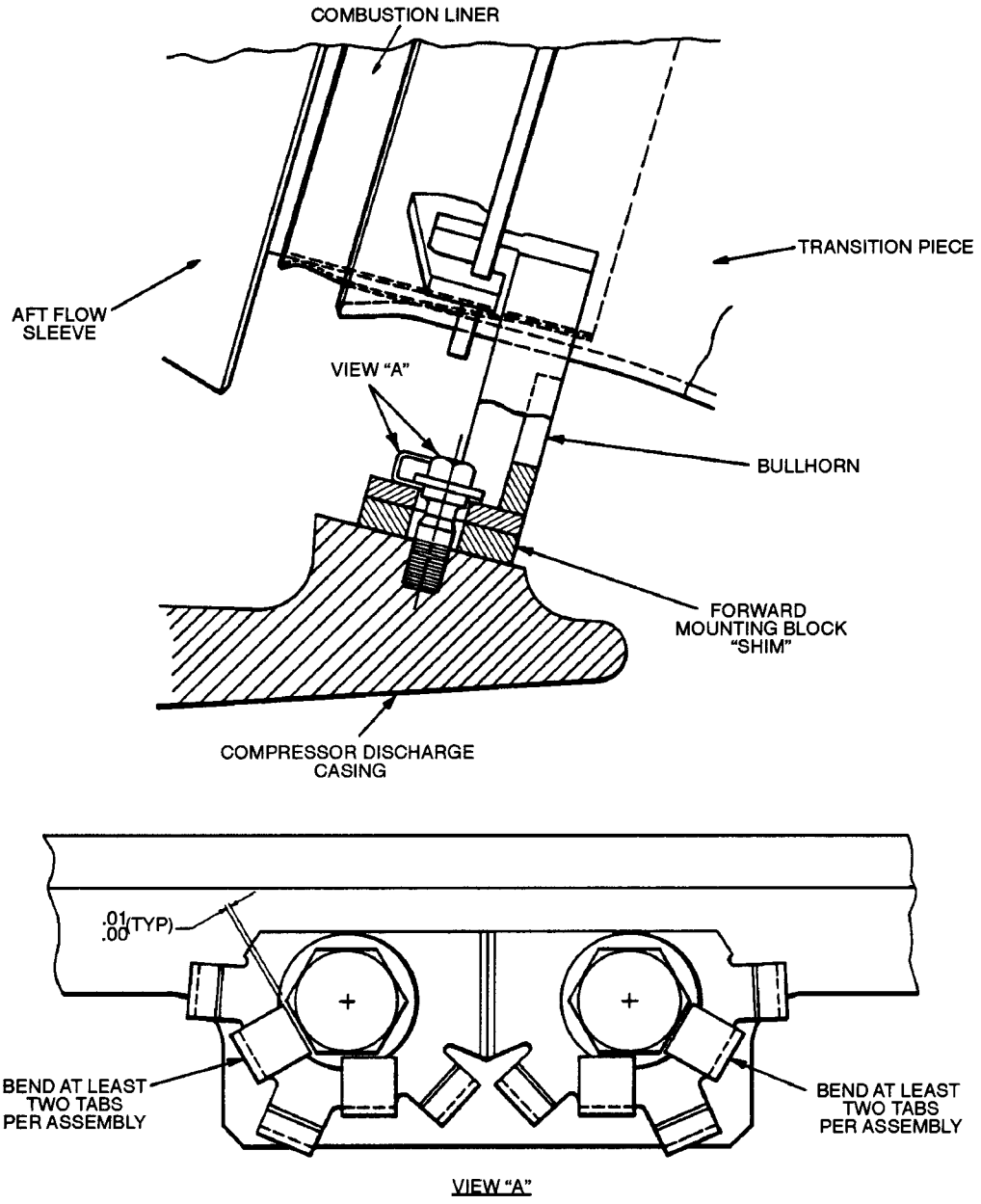



**Figure CI-R2. Transition Piece "A" Style Support Clamp (Bullhorn) Assembly**

2. Slide number five (5) transition piece over the guide pin to support the transition piece. Insert the floating seals into their respective slots in the first stage nozzle retaining ring. Prior to installation of all bolts, lubricate threads with Fel Pro C-102. Assemble the support clamp (bullhorn) to the transition piece forward mounting bracket slots. Install the support clamp (bullhorn) forward mounting shims, lockplates and bolts. Bolt the support clamp (bullhorn) to the compressor discharge casing. Do not tighten bolts or bend lockplates at this time. See Fig. CI-R2.

3. Remove the guide pin and install the aft bracket bolt and lockplate and tighten to the first stage nozzle retaining ring. Prior to torquing the bolt, check for proper floating seal engagement to the first stage nozzle retaining ring slots. Torquing aft bracket bolt with floating seal out of its slot will result in damage to seal. Torque aft bracket bolt to 60-67 lb. ft. Do not bend lockplate tabs at this time.

**Note:** Torquing procedure; to minimize variations in clamping force of bolt for given torque values, bolts must be torqued backed off, then retorqued.

4. Torque the support clamp (bullhorn) bolts to 33-37 lb. ft. then loosen and remove bolts and lockplates from the support clamp.

5. Install guide pin into number four (4) position and slide number four (4) transition piece over the guide pin to support the transition piece. Insert the floating seals into their respective slots in the first stage nozzle retaining ring. Assemble the support clamp (bullhorn) to the transition piece forward mounting bracket slots. Install the support clamp (bullhorn) lockplates, and bolts. Bolt the support clamp (bullhorn) to the compressor discharge casing. Do not tighten or bend lockplate tabs at this time.

6. Remove the guide pin and install the aft bracket bolt and lockplate and tighten to the first stage nozzle retaining ring. Prior to torquing the bolt, check for proper floating seal engagement to the first stage nozzle retaining ring slots. Torquing aft bracket bolts with floating seals out of their slots will result in damage to seal. Torque aft bracket bolts to 60-67 lb. ft. Do not bend lockplate tabs at this time.

7. Torque the support clamp bolts to 33-37 lb. ft. then loosen and remove bolts and lockplates from the support clamp.

8. Repeat Steps 5, 6 and 7 for number six (6) transition piece.

9. Starting with number five (5) transition piece, check the spacing between four and five, five and six transition pieces at the transition piece side seal slot. The spacing should be between 0.36-0.53 inch. To align the number five (5) transition piece the aft bracket bolts on number four (4), five (5) and six (6) transition pieces may have to be loosened. After alignment is achieved on number five (5) transition piece, retorque the aft bracket bolts to 60-67 lb. ft. See Fig. CI-R3.

10. Measure the radial and circumferential displacement of the transition piece with respect to the compressor discharge casing. Using a carpenters square or two scales (rulers), measure the concentricity of the transition piece to the compressor discharge casing at four (4) points (radial outside position and every 90° from that point). If displacement does not exceed 0.100 inch radially or circumferentially, proceed to Step 11. See Fig. CI-R4.

**CAUTION: If measured displacement exceeds 0.100 inch further assembly should not be attempted. Contact your General Electric Company Field Service Representative for corrective action.**

11. With the number five (5) transition piece aligned per Step 10 above, determine the required thickness of the support clamp (bullhorn) forward mounting block shims. Shim thickness measurements must be made so that when the support clamp bolts are torqued to 33-37 lb. ft. a contact should exist between the transition piece and the support clamp. See Fig. CI-R5.

**CAUTION: If the spacing between adjacent transition pieces or the contact condition cannot be achieved further assembly should not be attempted until the cause is resolved. It is recommended that your General Electric Company Field Service Representative be consulted for recommended action to resolve the problem.**

12. Reinstall forward support clamp (bullhorn) bolts and lockplates. Torque the forward support clamp (bullhorn) bolts on number five (5) transition piece to 33-37 lb. ft.

13. Check to assure that the gap between number five (5), four (4) and six (6) transition pieces and face of the first stage nozzle is a minimum of 0.090 inch and not to exceed 0.160 inch. Make checks at six (6) points. Contact your General Electric Company Field Service Representative if these clearances cannot be obtained. See Fig. CI-R6.

14. Bend the lockplate tabs on the transition piece aft bracket mounting bolt and the forward support clamp (bullhorn) bolts to secure the bolts in place. See Figs. CI-R2 and R7.

15. Repeat Steps 5 through 7 and 9 through 14 for all other transition piece positions in the sequence specified.

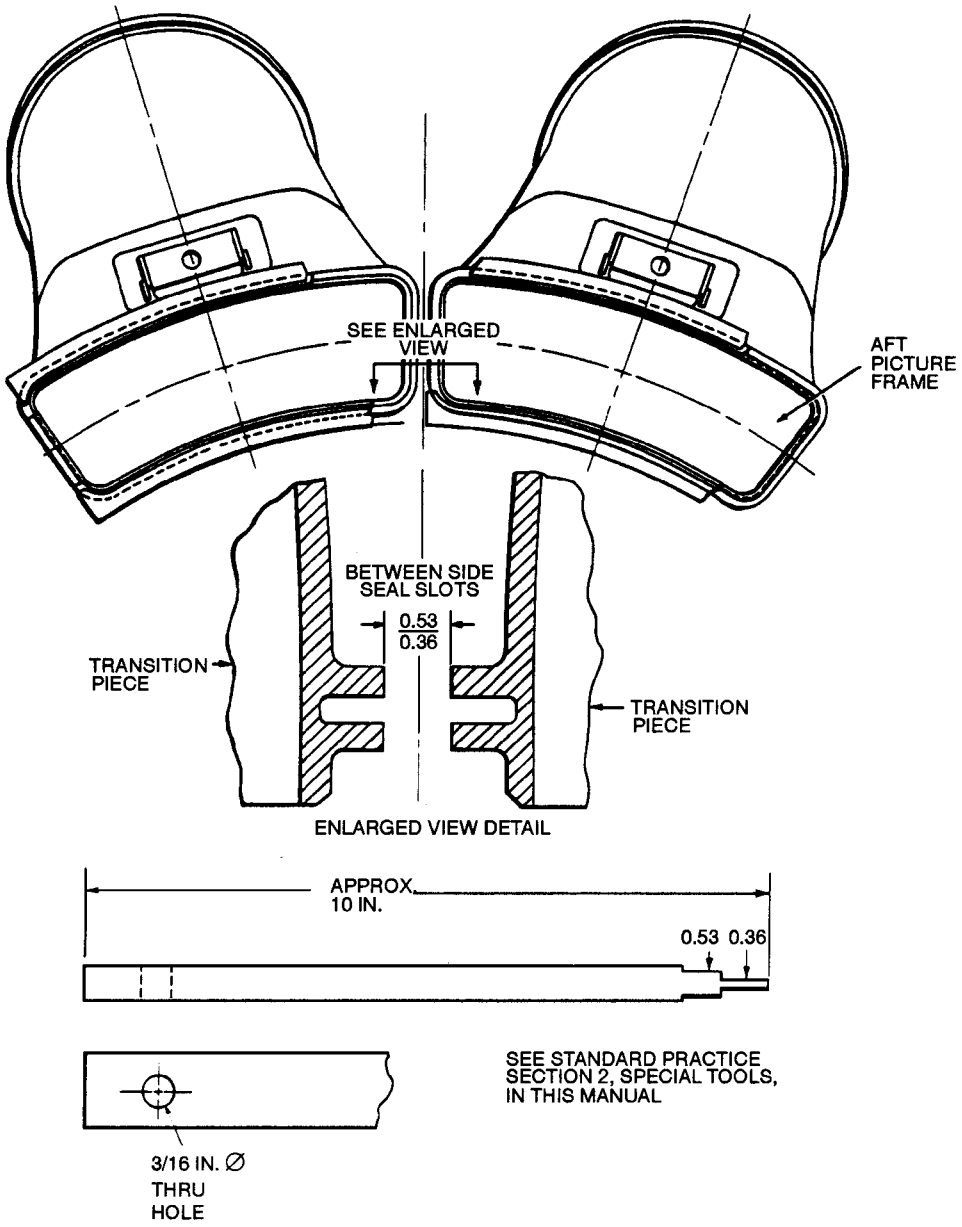



**Figure CI-R3. Transition Piece "A" Style Side Seal Clearance Checks**

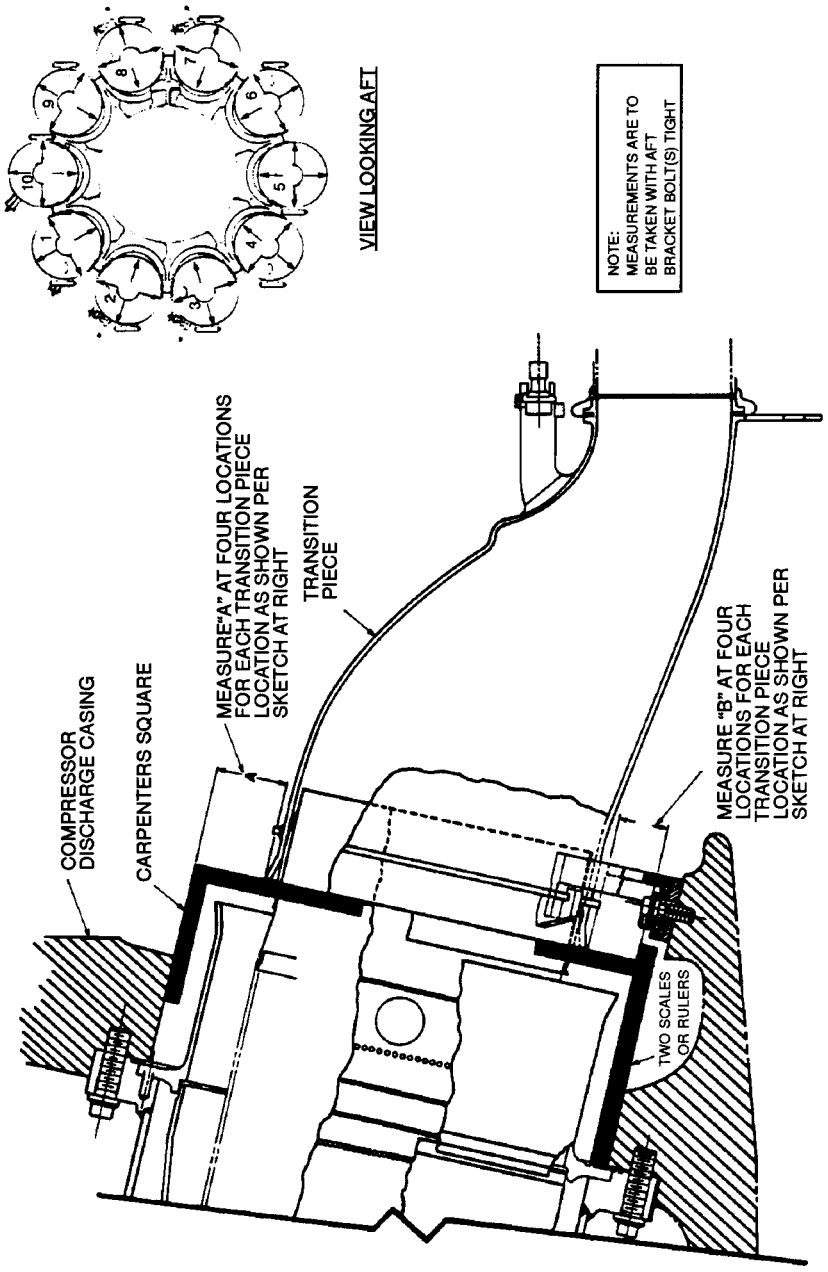



**Figure CI-R4. Transition Piece Centralization Checks Set Up**

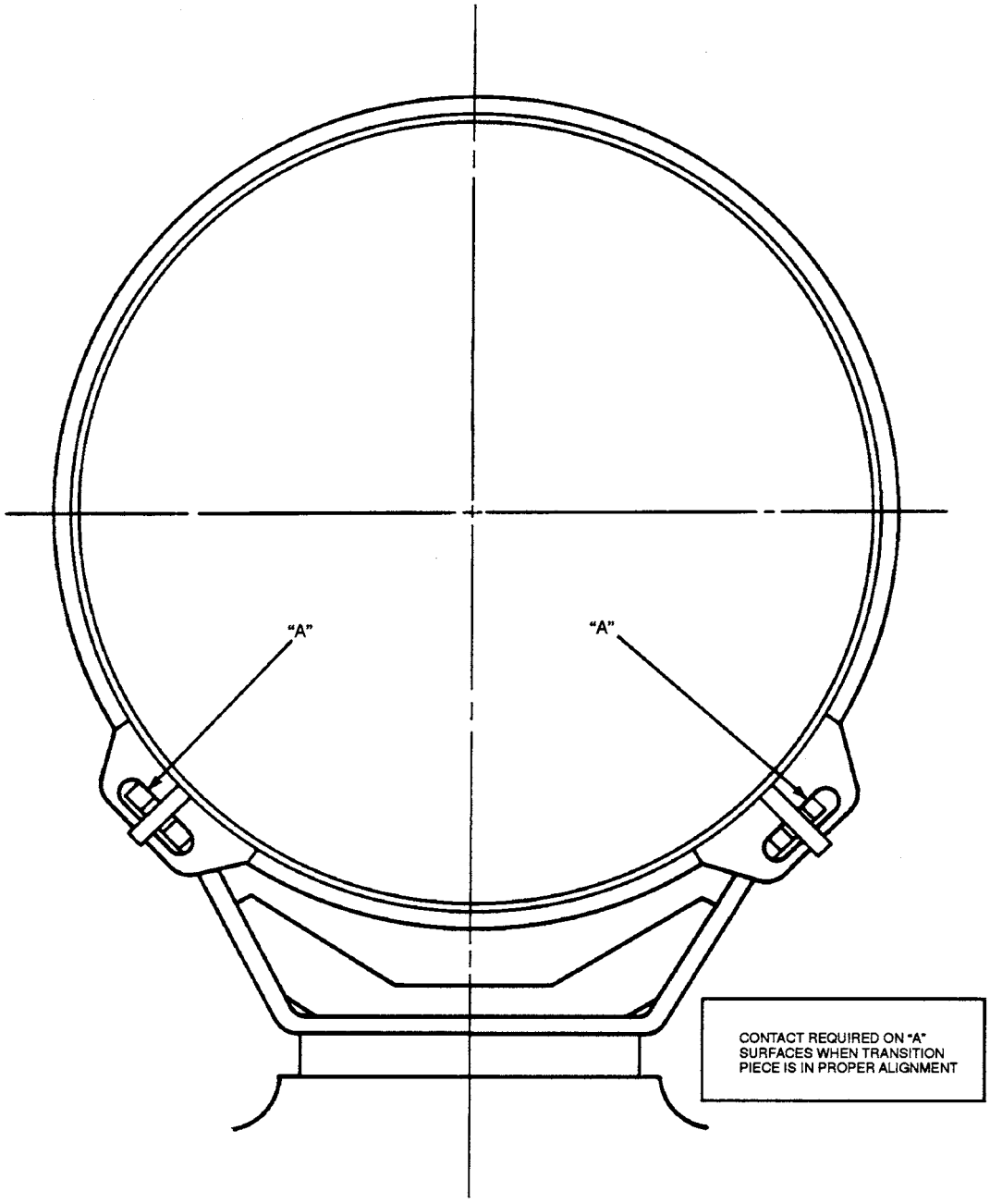



**Figure CI-R5. Transition Piece "A" Style Support Clamp (Bullhorn) Assembly Limits**

16. Install aft side seals between adjacent transition pieces, e.g.; the aft side seals between position five (5) and four (4) transition pieces is installed after position five (5) and four (4) transition pieces are correctly assembled and between five (5) and six (6) transition piece six (6) is properly assembled, etc.

**CAUTION: Aft side seals should be installed by hand force only. Do not attempt further assembly if greater than hand force is required.**

17. Install aft side seal and stop seal assembly, bolt and lockplate. Torque aft side seal stop assembly bolt to 41-46 lb. ft. and bend lockplate tabs to secure stop seal bolt in place. See Fig. CI-R8.

**OPERATION 2 - INSTALL TRANSITION PIECES ("B" STYLE DESIGN)**

NOTE: This operation is applicable to the "B" style design transition piece only. The "B" style transition piece has a circular aft bracket with three mounting holes. See Fig. CI-R9.

**CAUTION: Use new bolts and lockplates. Do not reuse hardware previously removed.**

1. Install the transition piece guide pin, if removed, into the center mounting hole of the first-stage nozzle retaining ring in the number five (5) position. See Fig. CI-R9.

**Note:** The last transition piece to be installed can be in the 1, 9 or 10 location dependent on which turbine casing access port was used at disassembly. Install the transition pieces starting with number 5; then install 4, 6, 7, 3, 2, 8, 1, 9 and 10 last or as stated above number 1 or 9 last.

2. Slide number five (5) transition piece over the guide pin to support the transition piece. Insert the floating seals into their respective slots in the first stage nozzle retaining ring. Prior to installation of all bolts, lubricate threads with Fel Pro C-102. Assemble the support clamp (bullhorn) to the transition piece forward mounting bracket slots. Install the support clamp (bullhorn) lockplates and bolts. Bolt the support clamp (bullhorn) to the compressor discharge casing. Do not tighten bolts or bend lockplates at this time. See Fig. CI-R10.

3. Install the aft bracket bolts and lockplates and tighten to the first stage nozzle retaining ring. Prior to torquing the bolts, check for proper floating seal engagement to the first stage nozzle retaining ring slots. Torquing aft bracket bolts with floating seal out of its slot will result in damage to seal. Torque aft bracket bolts to 75-85 lb. ft. Do not bend lockplate tabs at this time.

**Note:** Torquing procedure; to minimize variations in clamping force of bolt for given torque values, bolts must be torqued backed off, then retorqued.

4. Torque the support clamp (bullhorn) bolts to 33-37 lb. ft. then loosen and remove bolts and lockplates from the support clamp.

5. Slide number four (4) transition piece over the positioning pin to support the transition piece. Insert the floating seals into their respective slots in the first stage nozzle retaining ring. Assemble the support clamp (bullhorn) to the transition piece forward mounting bracket slots. Install the support clamp (bullhorn) lockplates, and bolts. Bolt the support clamp (bullhorn) to the compressor discharge casing. Do not tighten bolts or bend lockplates at this time.

6. Install the aft bracket bolts and lockplates and tighten to the first stage nozzle retaining ring. Prior to torquing the bolts, check for proper floating seal engagement to the first stage nozzle retaining ring slots. Torquing aft bracket bolts with floating seals out of their slots will result in damage to seal. Torque aft bracket bolts to 75-85 lb. ft. Do not bend lockplate tabs at this time.

7. Torque the support clamp bolts to 33-37 lb. ft. then loosen and remove bolts and lockplates from the support clamp.

8. Repeat Steps 5, 6 and 7 for number six (6) transition piece.

9. Starting with number five (5) transition piece, check the spacing between four and five, five and six transition pieces at the transition piece side seal slot. The spacing should be between 0.36-0.53 inch. To align the number five (5) transition piece the aft bracket bolts on number four (4), five (5) and six (6) transition pieces may have to be loosened. After alignment is achieved on number five (5) transition piece, retorque the aft bracket bolts to 75-85 lb. ft. See Fig. CI-R11.

10. Measure the radial and circumferential displacement of the transition piece with respect to the compressor discharge casing. Using a carpenters square or two scales (rulers), measure the concentricity of the transition piece to the compressor discharge casing at four (4) points (radial outside position and every 90° from that point). If displacement does not exceed 0.100 inch radially or circumferentially, proceed to Step 11. See Fig. CI-R12.

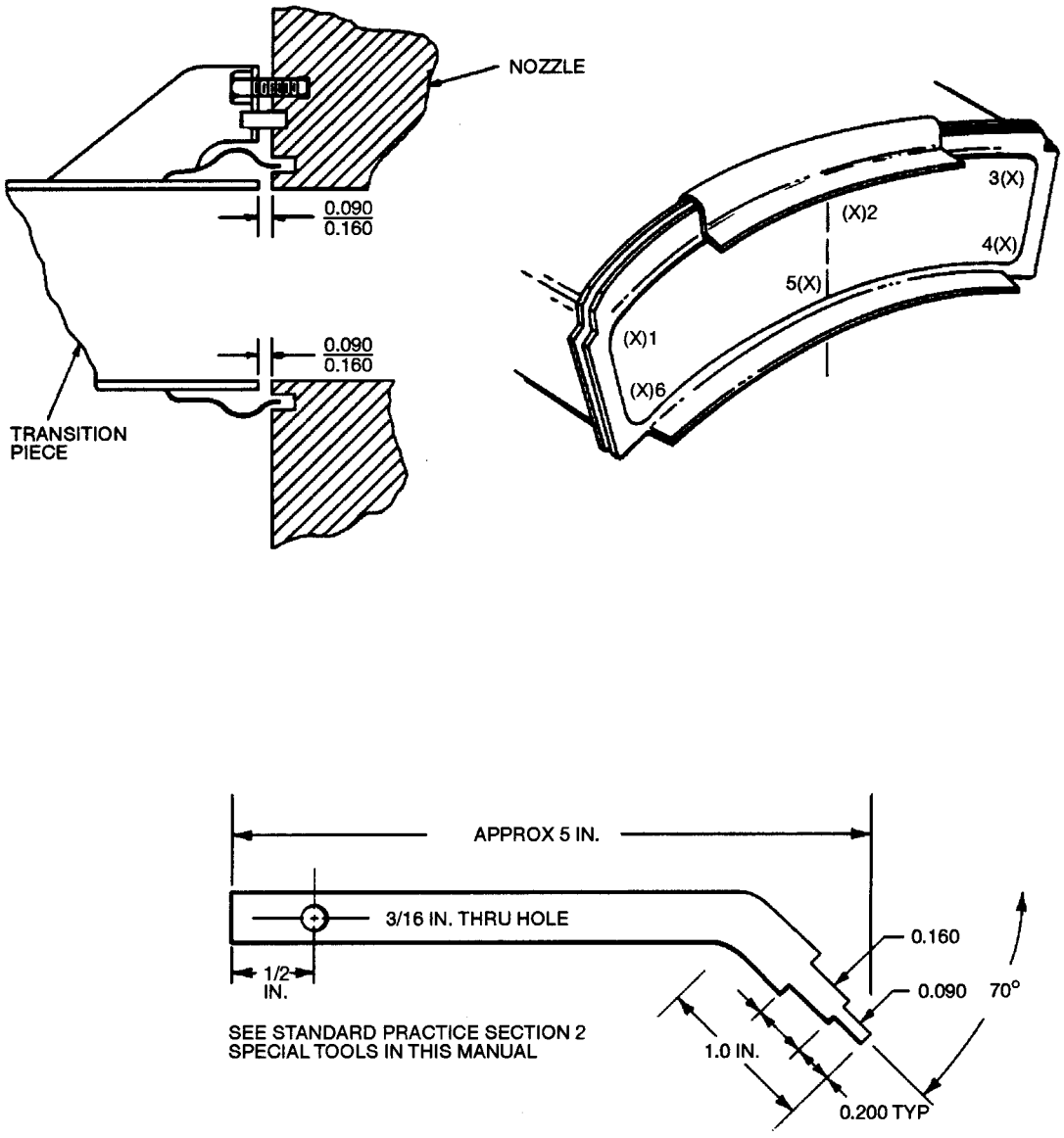



**Figure CI-R6. Transition Piece "A" Style First Stage Nozzle to Retaining Ring Clearance Checks**

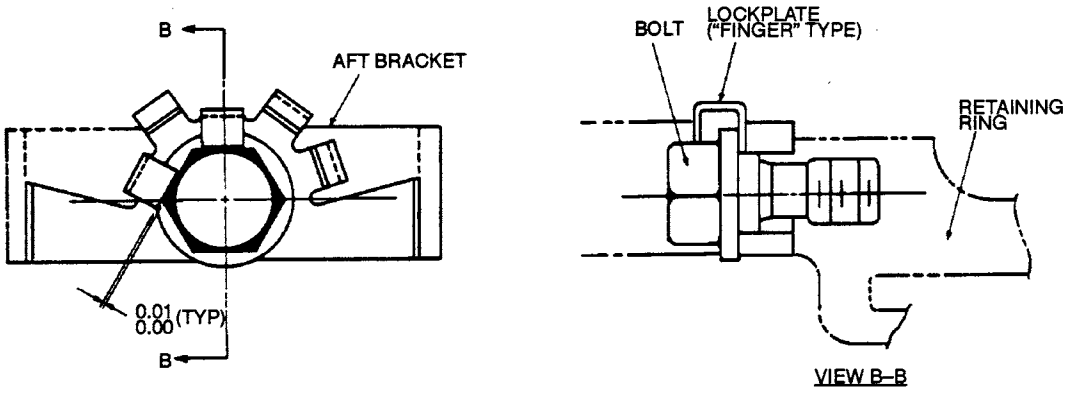



**Figure CI-R7. Transition Piece "A" Style Lockplating of Aft Bracket Bolt**

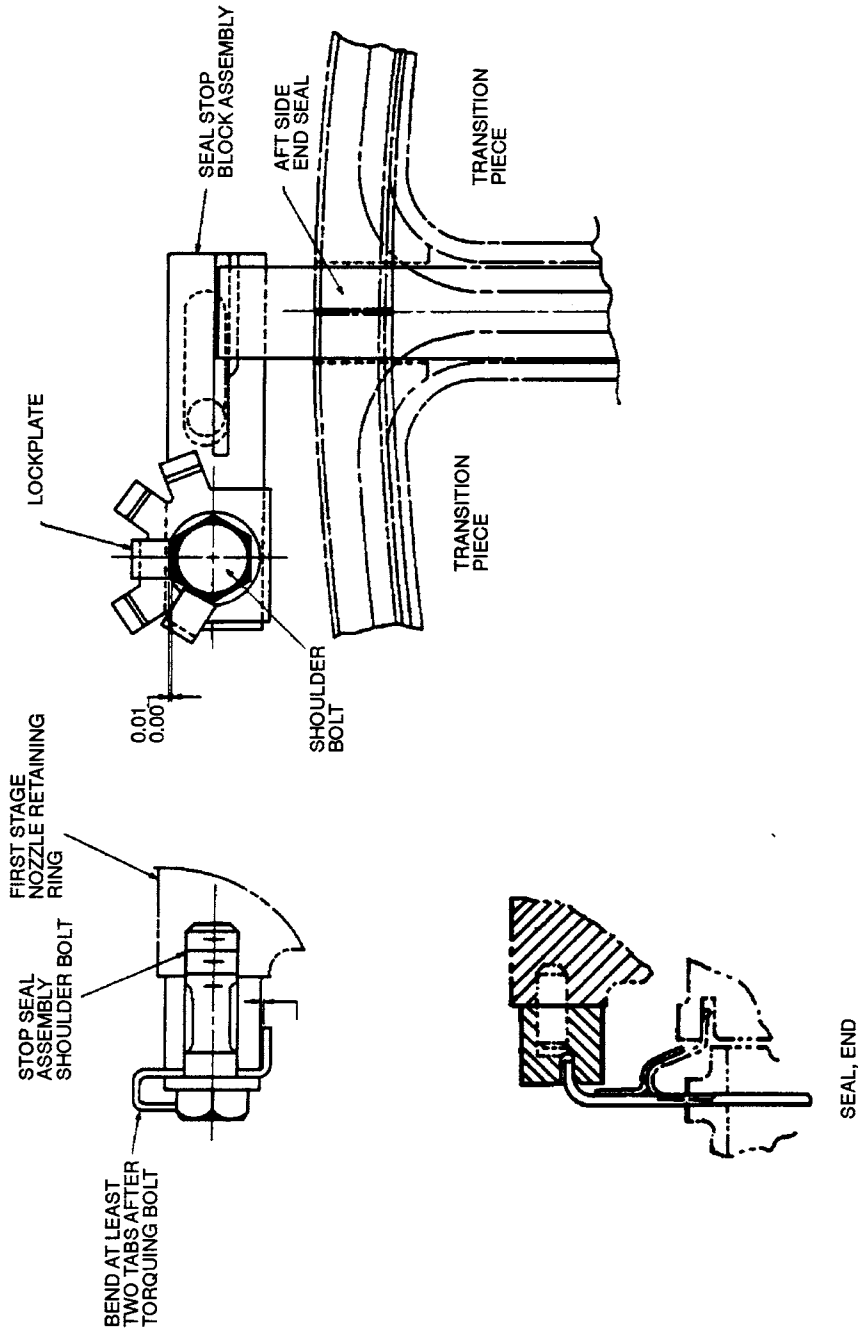



**Figure CI-R8. Assembly of End Seal and Stop Block to "A" Style Transition Piece**

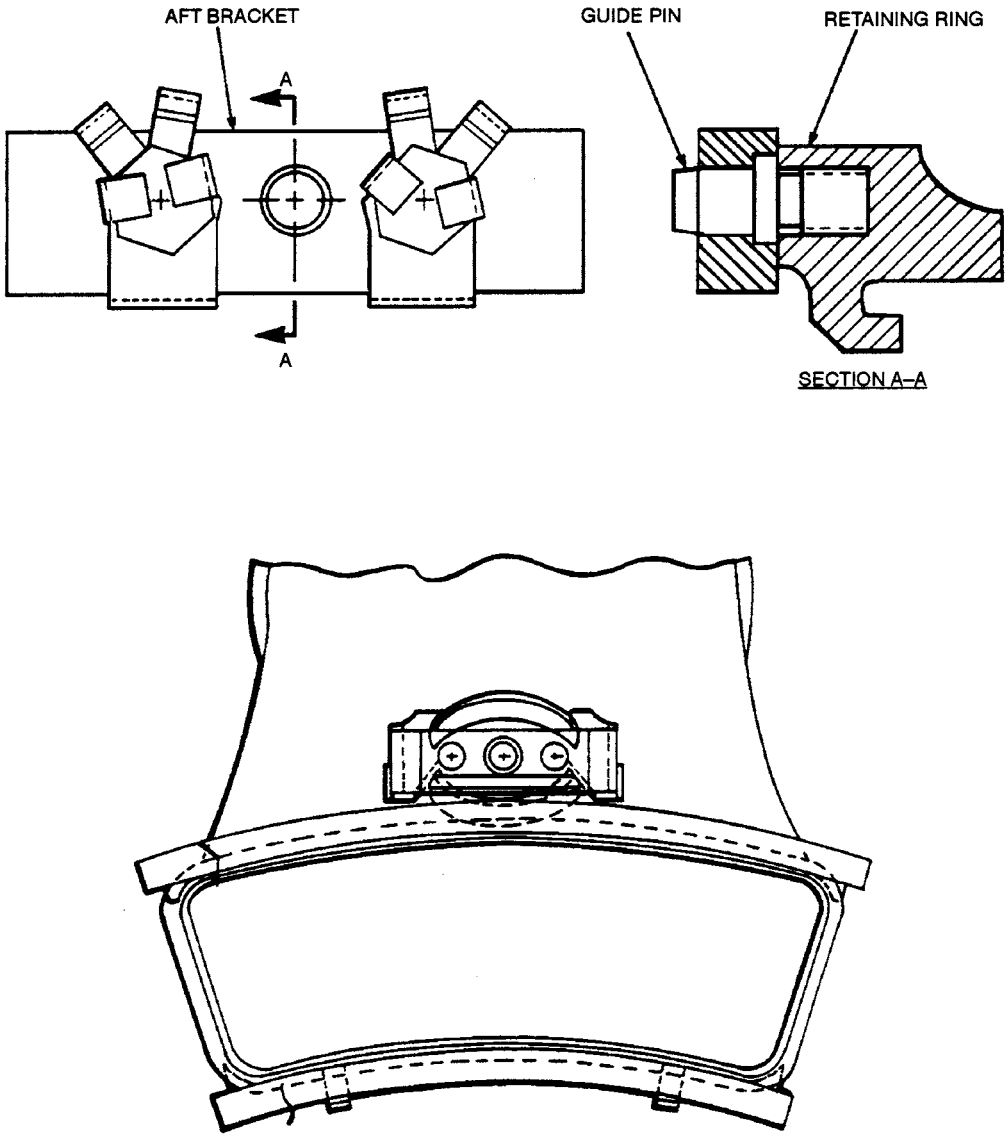



**Figure CI-R9. Transition Piece "B" Style Aft Bracket Guide Pin Assembly**

**CAUTION: If measured displacement exceeds 0.100 inch further assembly should not be attempted. Contact your GE Field Service Representative for corrective action.**

11. With the number five (5) transition piece aligned per Step 10 above and the loading is achieved, a contact condition must exist when the support clamp (bullhorn) bolts are final torqued. See Fig. CI-R13.

**Note:** Nominal shim thickness for the "B" style support clamp is 0.0 inch. A maximum of three (3) shims for a thickness of 0.100 inch may be used if required to achieve proper loading of the support clamp.

**CAUTION: If the spacing between adjacent transition pieces or the support clamp loading requirements cannot be achieved further assembly should not be attempted until the cause is resolved. It is recommended that your General Electric Company Field Service Representative be consulted for recommended action to resolve the problem.**

12. Reinstall forward support clamp (bull-horn) bolts and lockplates. Torque the forward support clamp (bullhorn) bolts on number five (5) transition piece to 33-37 lb. ft.

13. Check to assure that the gap between number five (5), four (4) and six (6) transition pieces and face of the first stage nozzle is a minimum of 0.090 inch and not to exceed 0.160 inch. Make checks at six (6) points. Contact your General Electric Company Field Service Representative if these clearances cannot be obtained. See Fig. CI-R14.

14. Bend the lockplate tabs on the transition piece aft bracket mounting bolt and the forward support clamp (bullhorn) bolts to secure the bolts in place. See Fig. CI-R10 and R15.

15. Repeat Steps 5 through 7 and 9 through 14 for all other transition piece positions in the sequence specified.

16. Install aft side seals between adjacent transition pieces, e.g.; the aft side seal between position five (5) and four (4) transition pieces is installed after position five (5) and four (4) transition pieces are correctly assembled and between five (5) and six (6) transition piece six (6) is properly assembled, etc.

**CAUTION: Aft side seals should be installed by hand force only. Do not attempt further assembly if greater than hand force is required.**

17. Install aft side seal and stop seal assembly, bolt and lockplate. Torque aft side seal stop assembly bolt to 45-55 lb. ft. and bend lockplate tabs to secure stop seal bolt in place. See Fig. CI-R16.

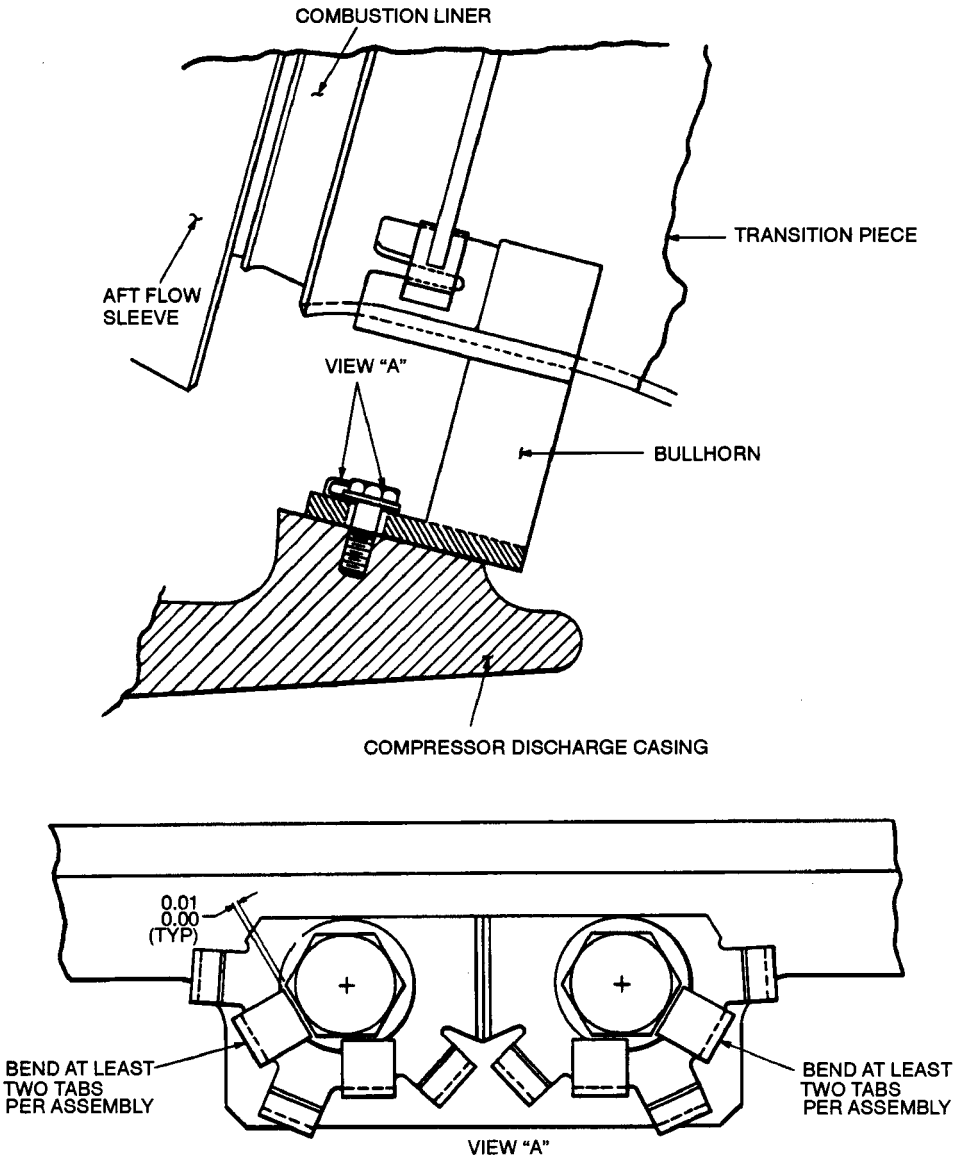



**Figure CI-R10. Transition Piece "B" Style Support Clamp (Bullhorn) Assembly**

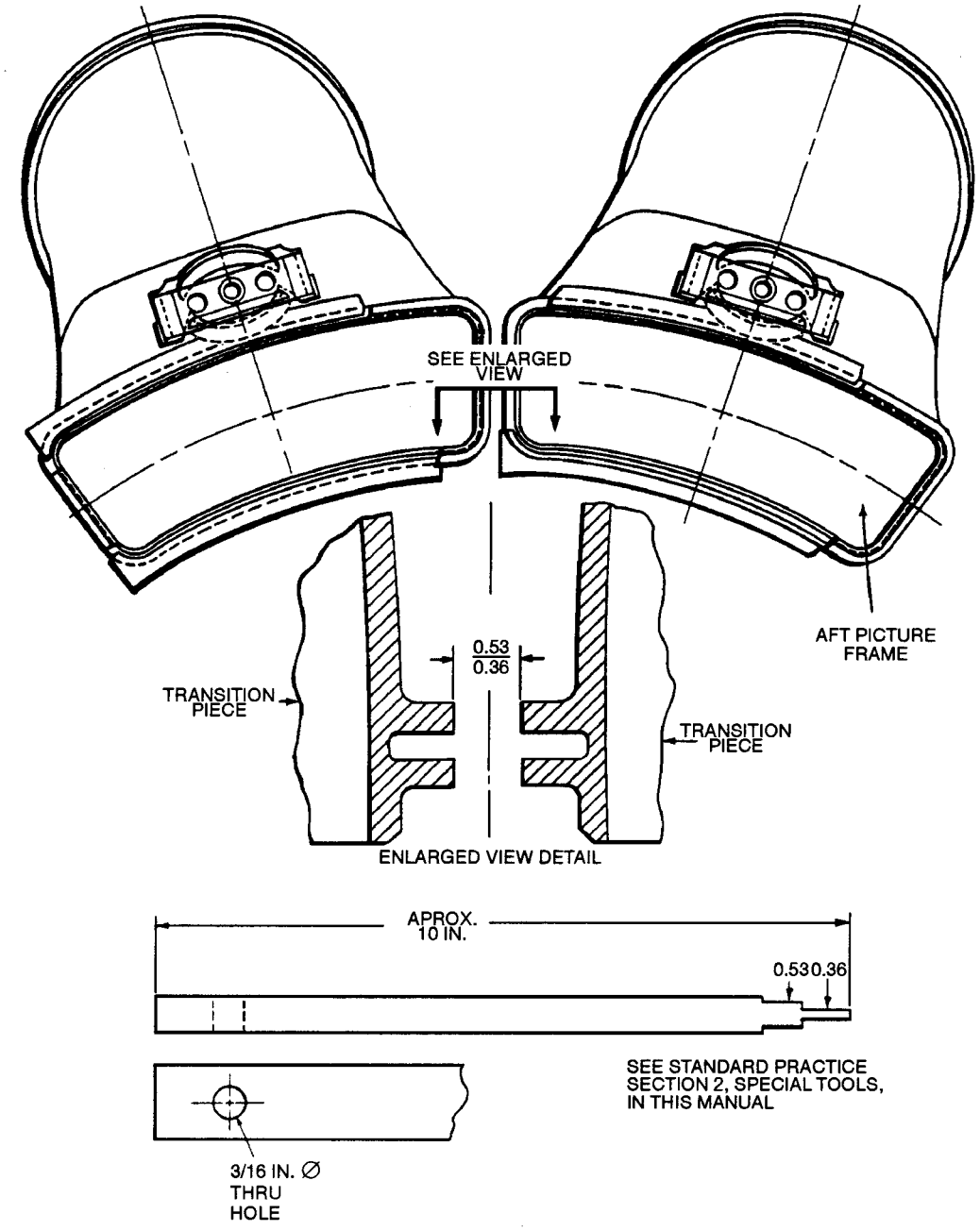



**Figure CI-R11. Transition Piece "B" Style Side Seal Clearances**

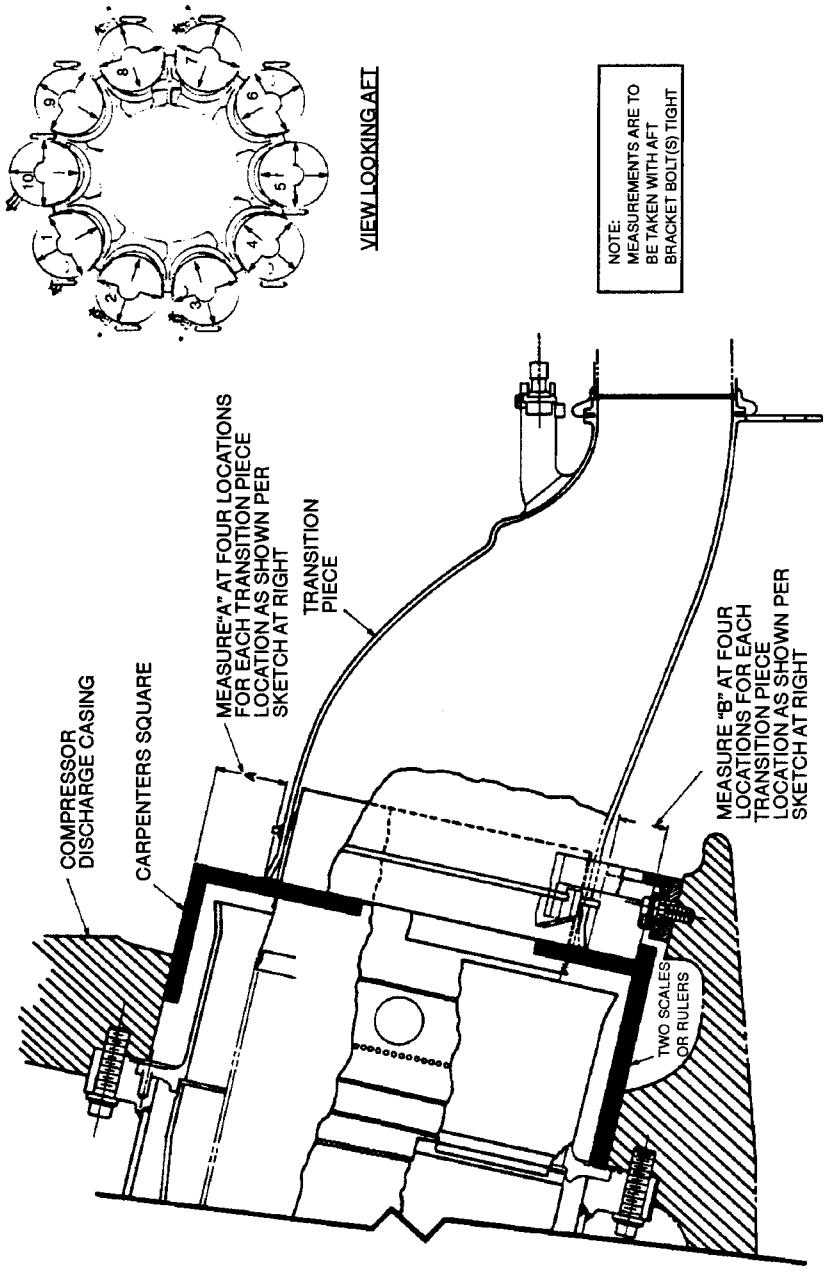



**Figure CI-R12. Transition Piece Centralization Checks Set Up**

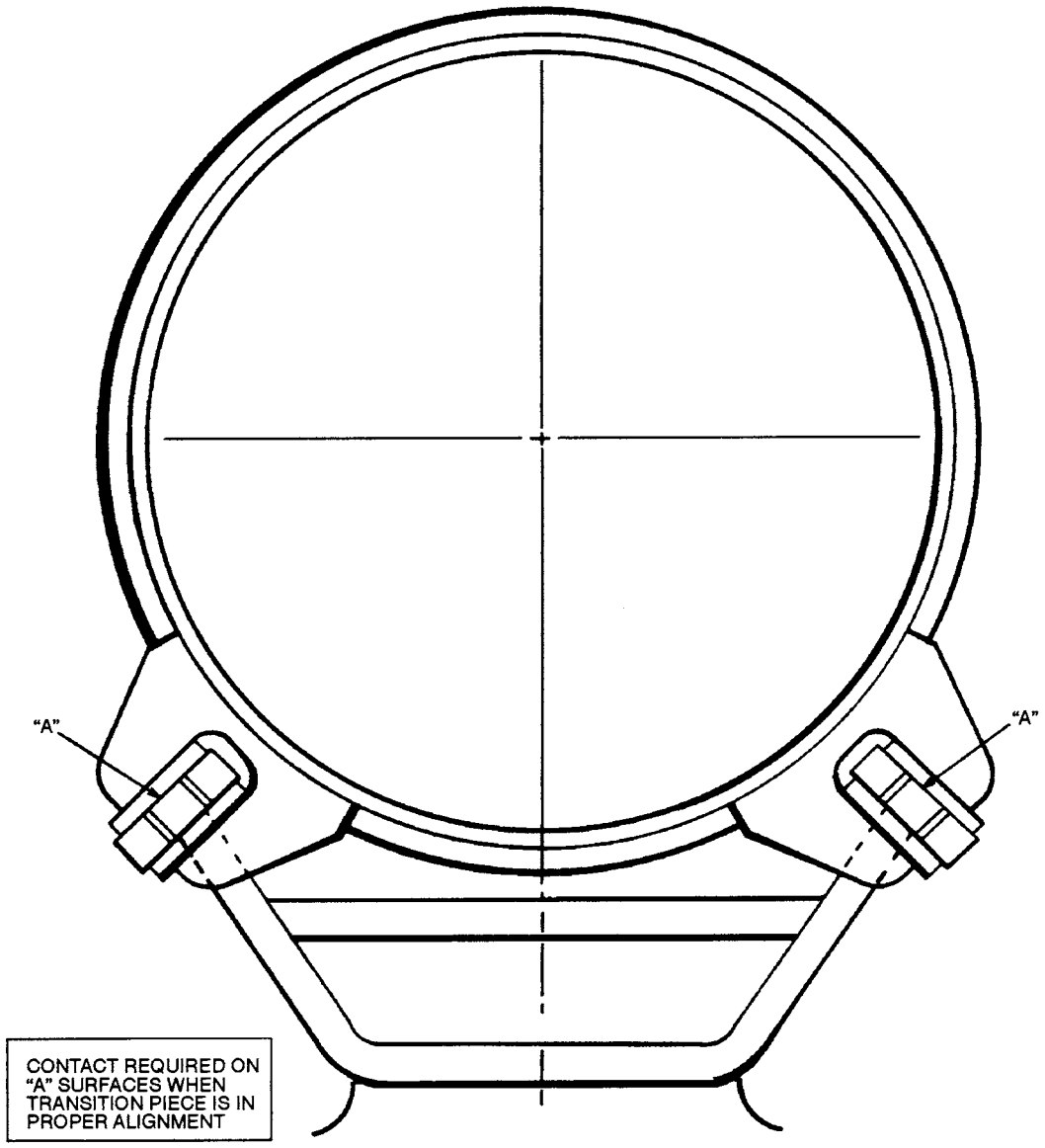



**Figure CI-R13. Transition Piece "B" Style Support Clamp (Bullhorn) Assembly Checks**

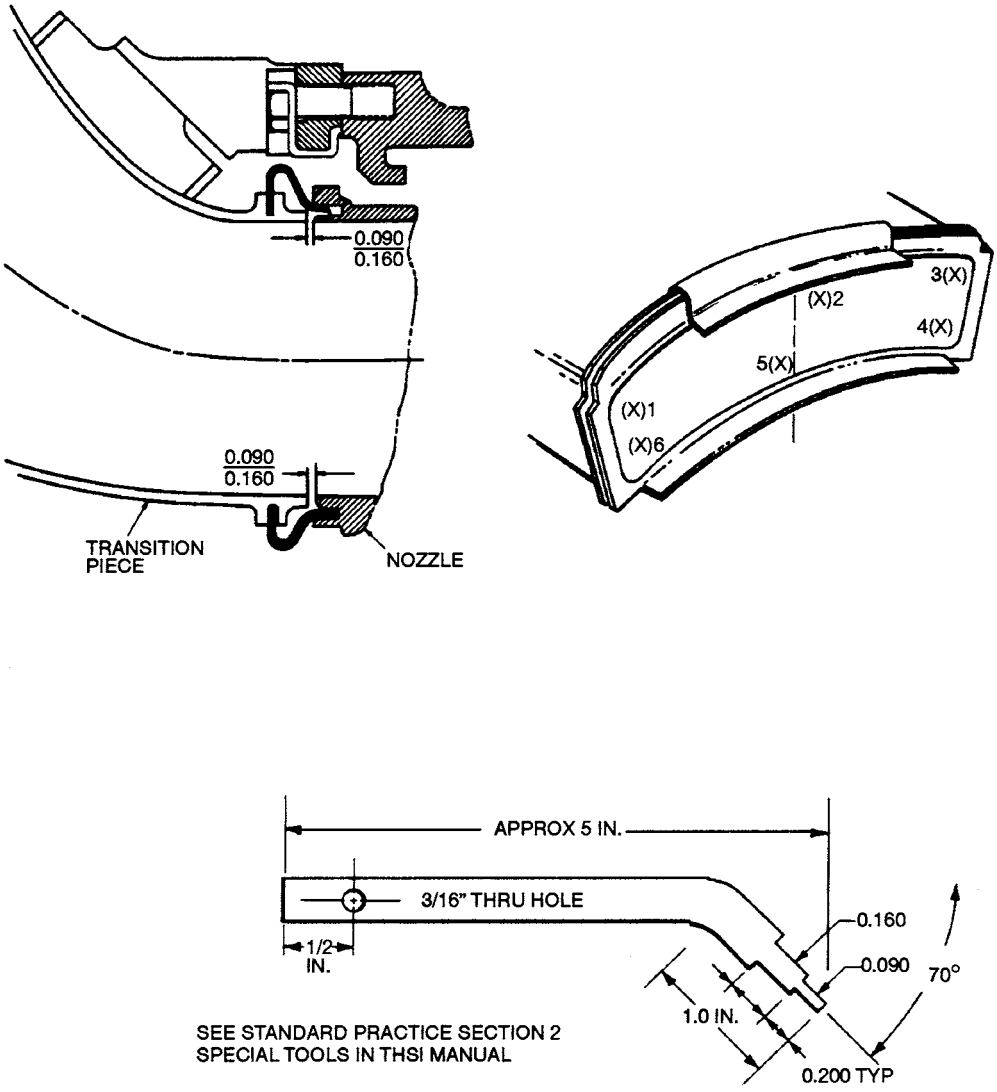



**Figure CI-R14. Transition Piece "B" Style First Stage Nozzle to Retaining Ring Clearance Checks**

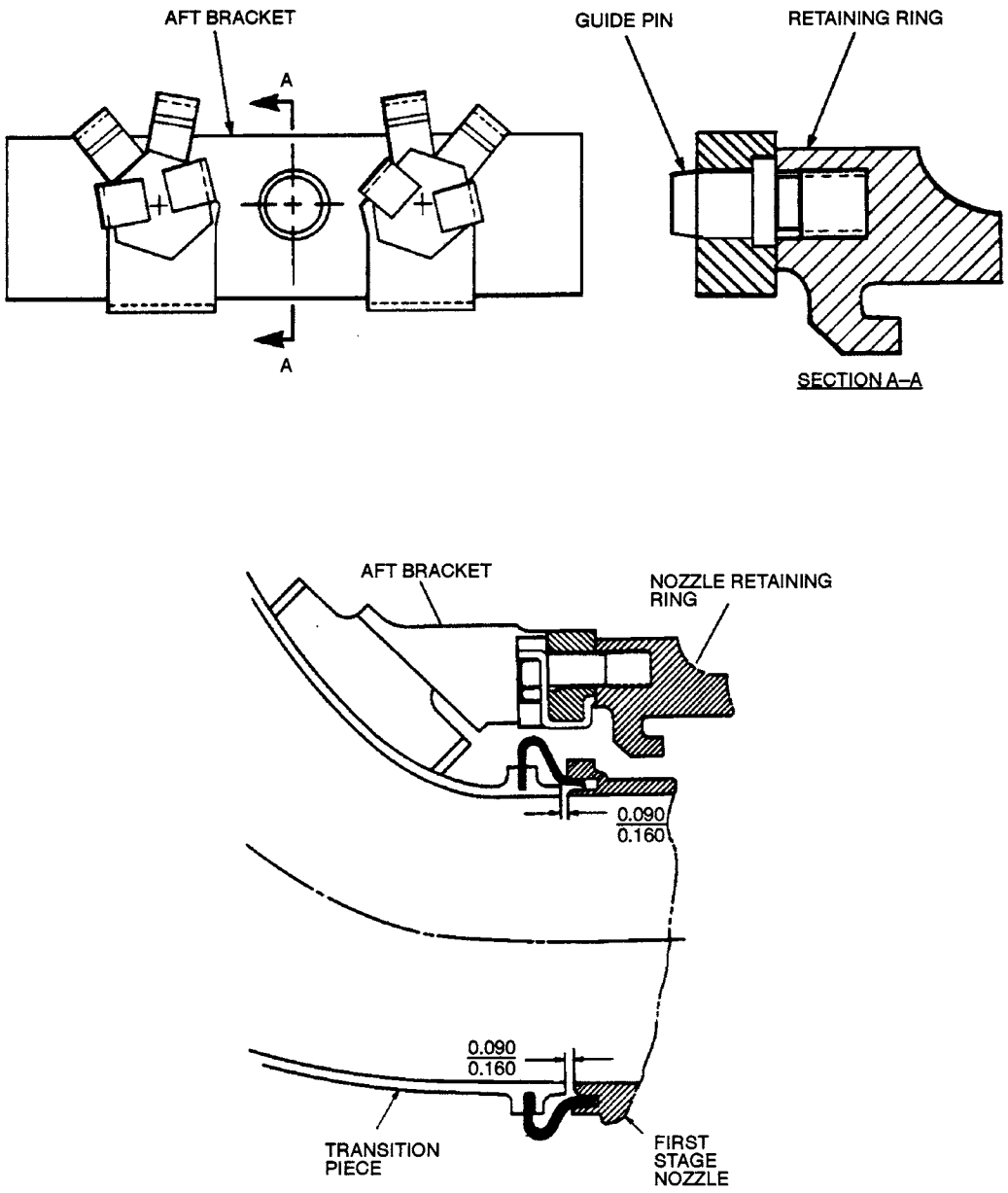



**Figure CI-R15. Transition Piece "B" Style Lockplating of Aft Bracket Bolts**

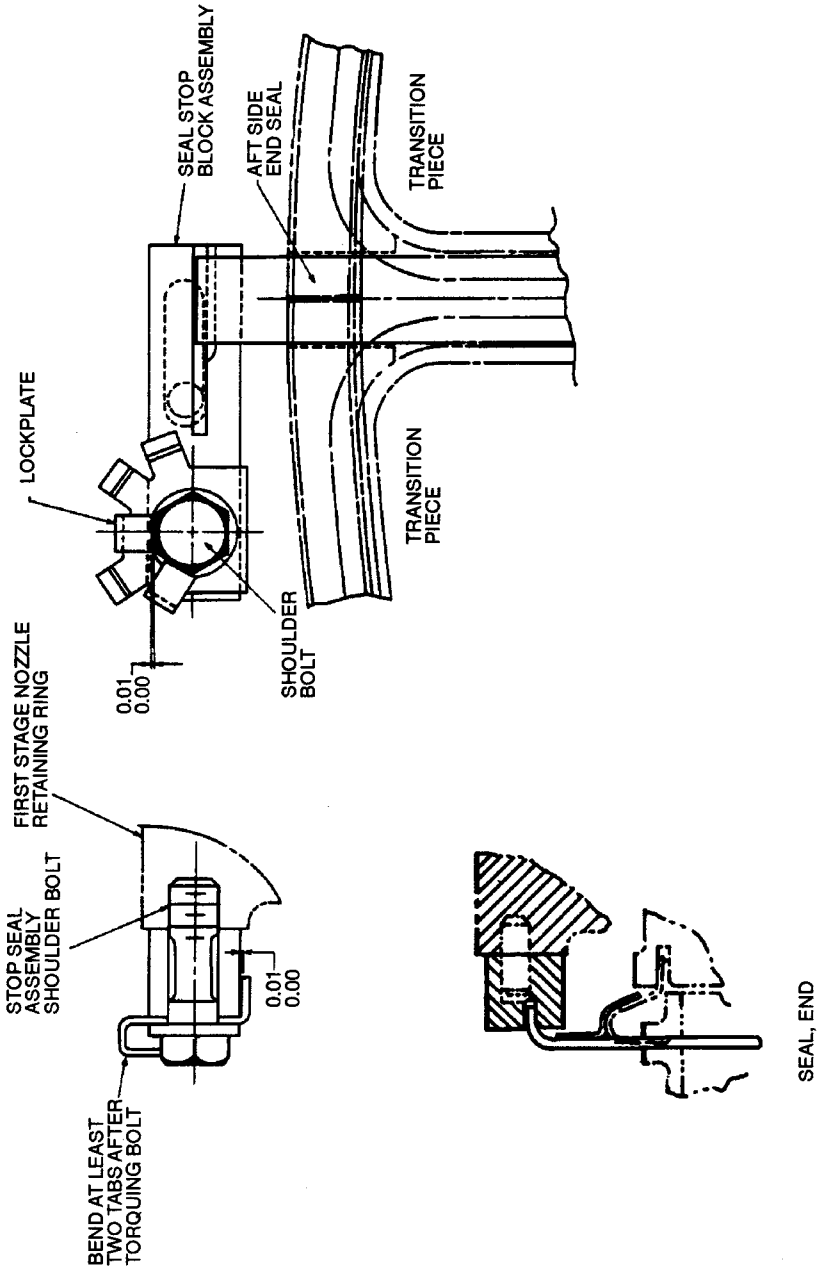



**Figure CI-R16. Assembly of End Seal and Stop Block to "B" Style Transition Piece**

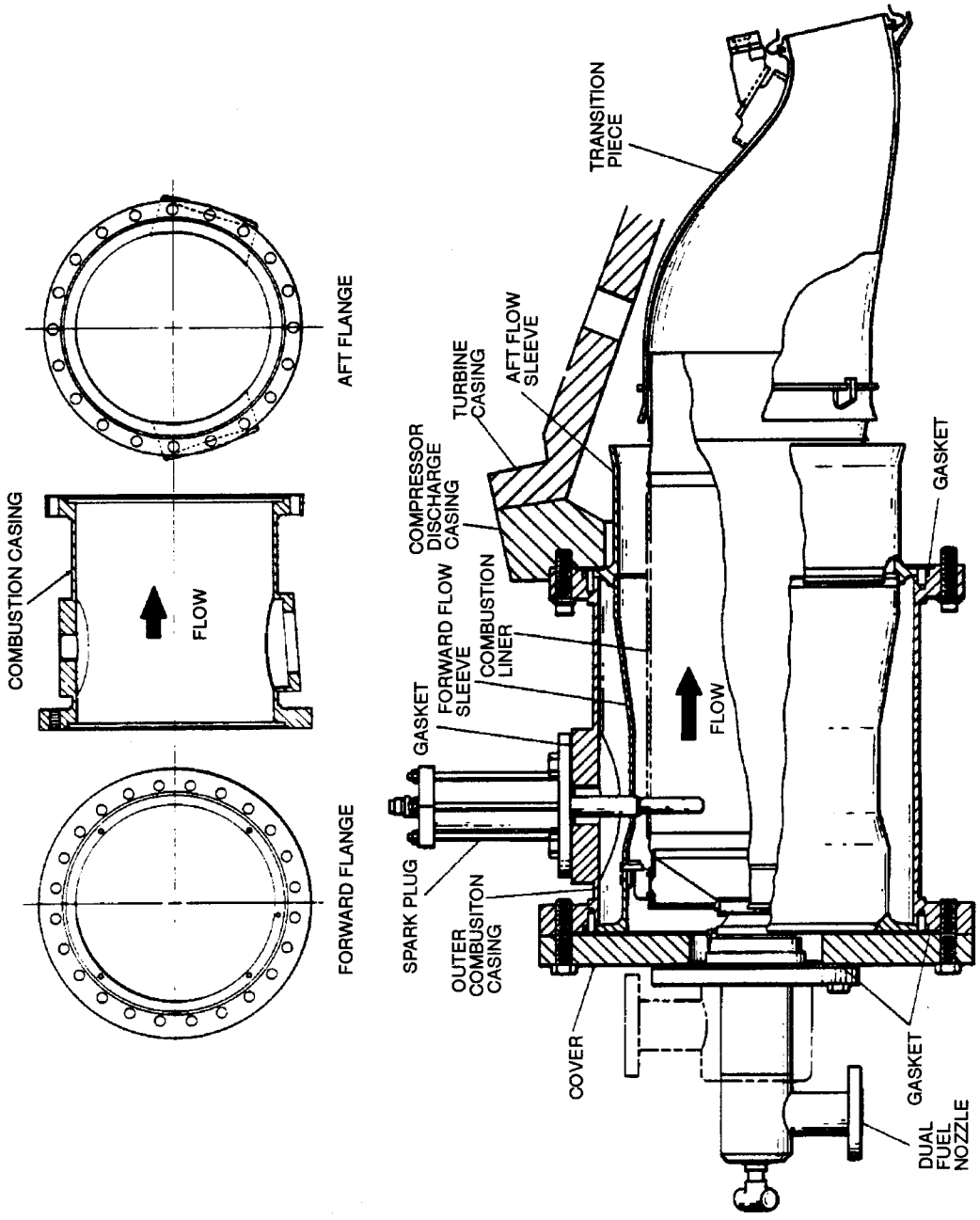



**Figure CI-R17. Outer Combustion Casing**

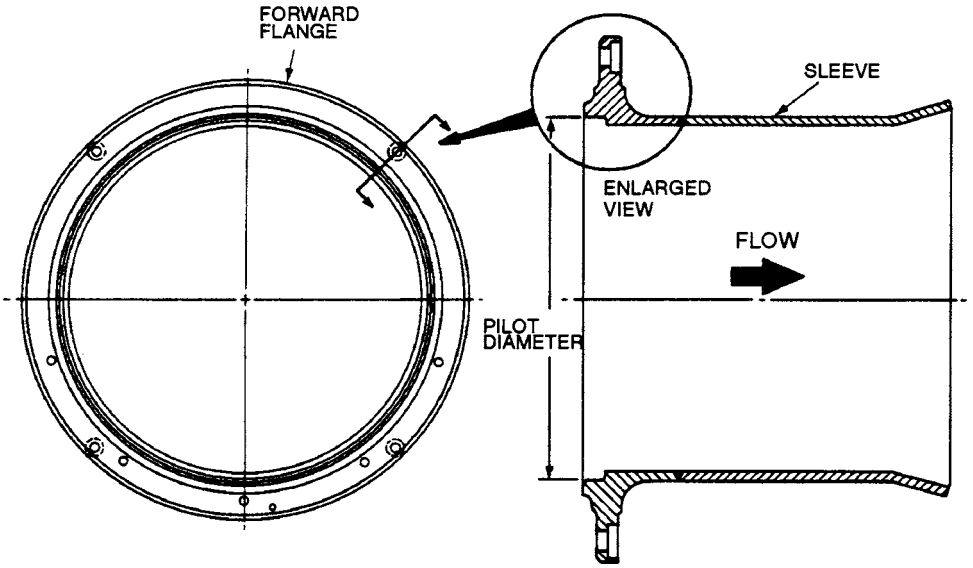





**Figure CI-R18. Aft Combustion Casing Flow Sleeve**

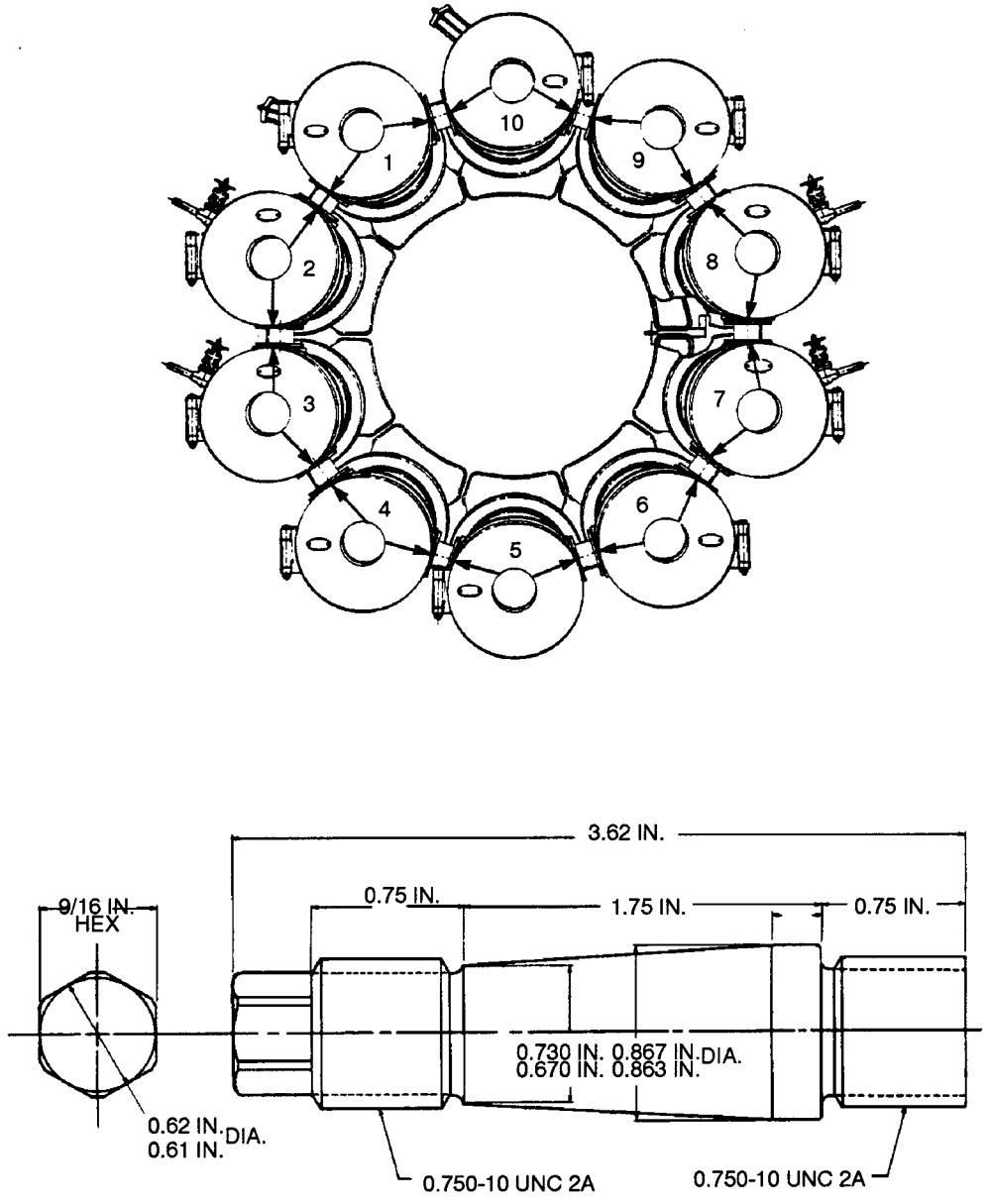



**Figure CI-R19. Outer Combustion Casing Alignment Pin and Assembly Locations**

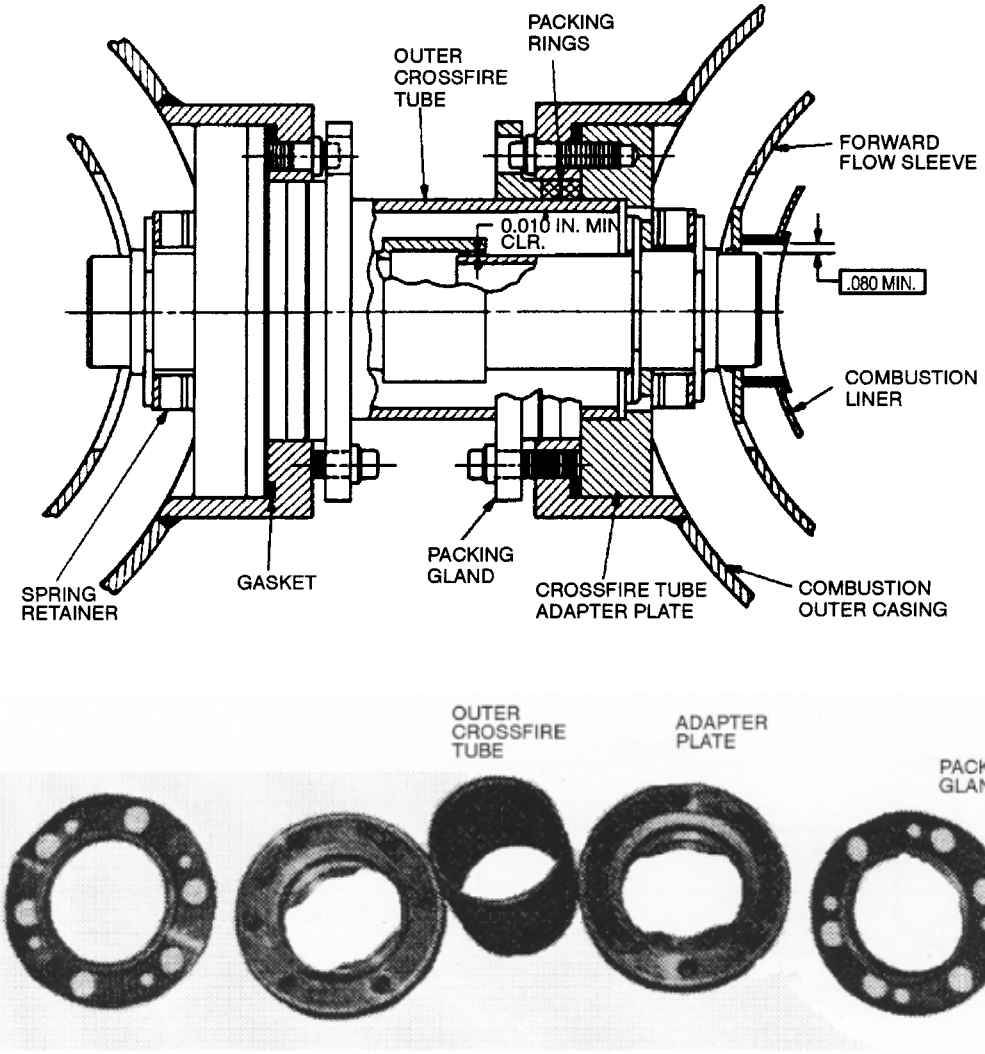



**Figure CI-R20. Crossfire Tube Assembly Cross Section**

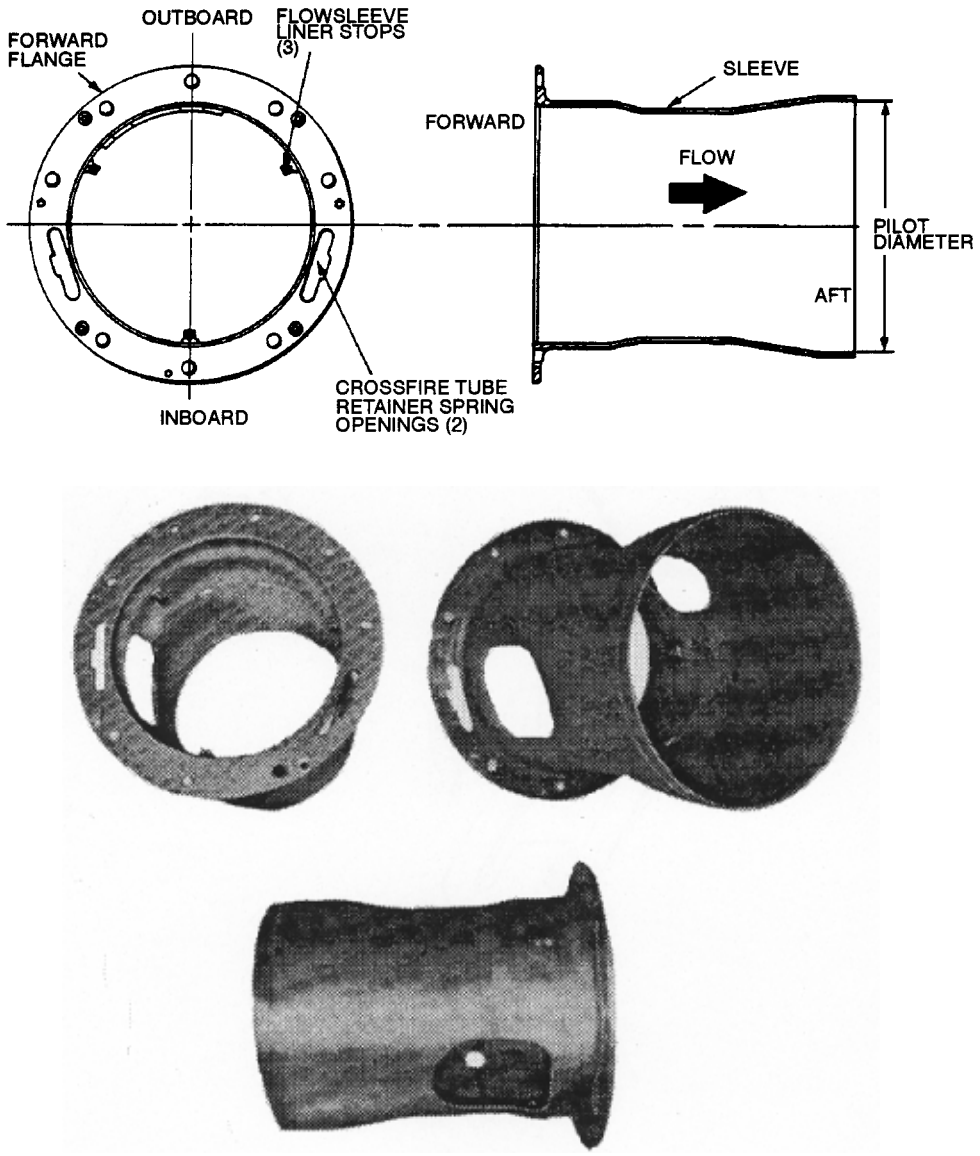



**Figure CI-R21. Forward Combustion Casing Flow Sleeve**

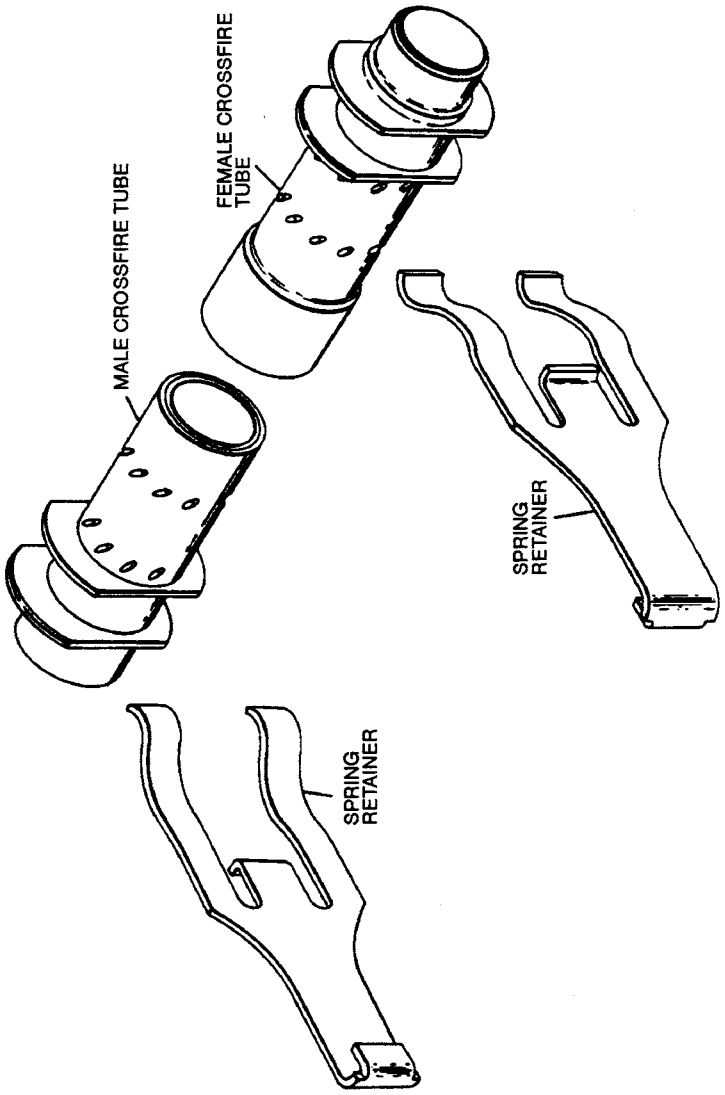



**Figure CI-R22. Crossfire Tubes and Spring Retainers**

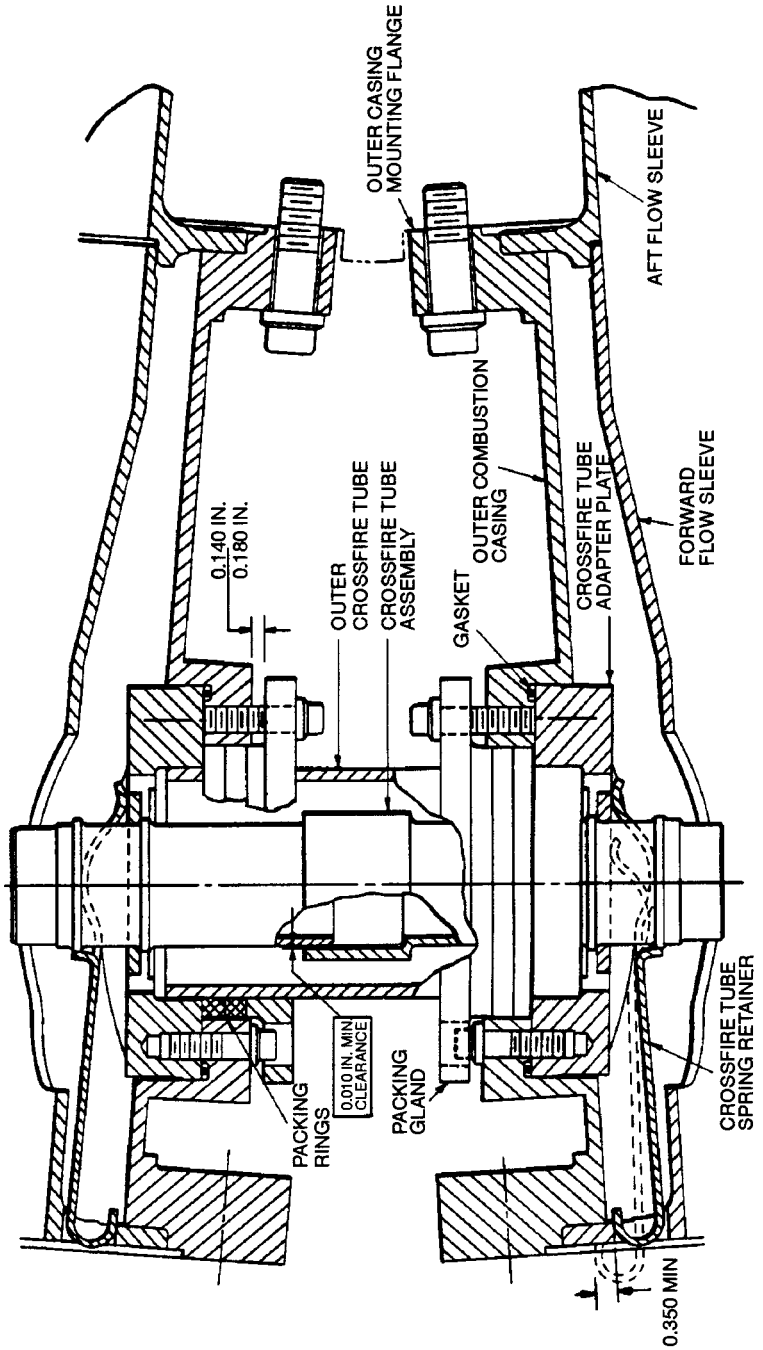



**Figure CI-R23. Crossfire Tube Assembly Clearances**

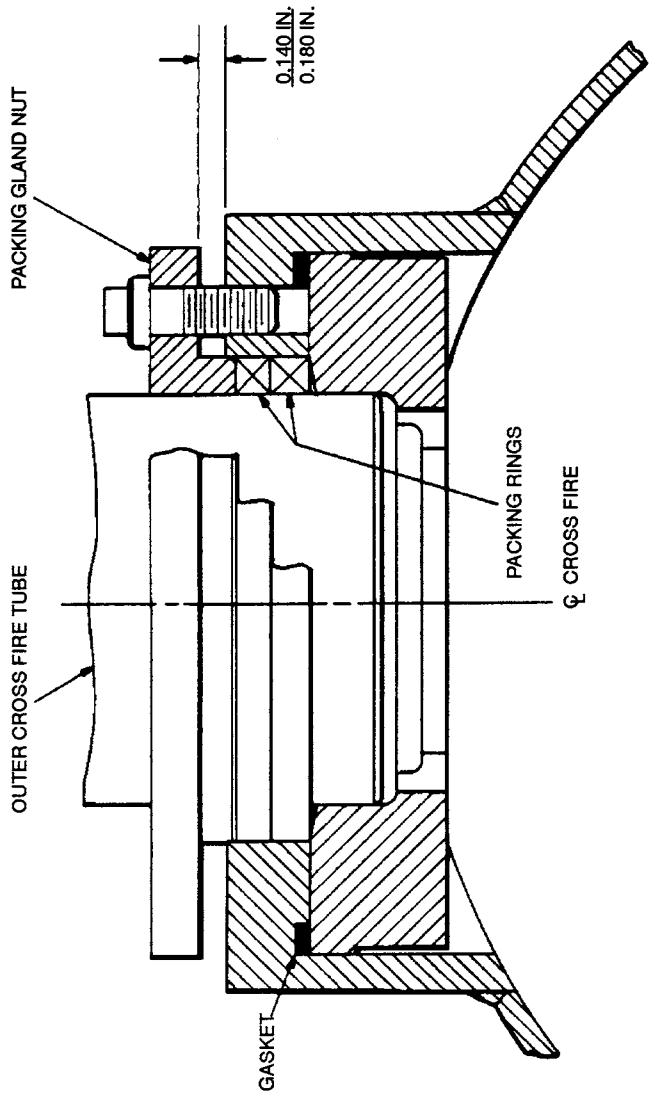



**Figure CI-R24. Assembly of Crossfire Tube Packing Rings and Packing Glands**

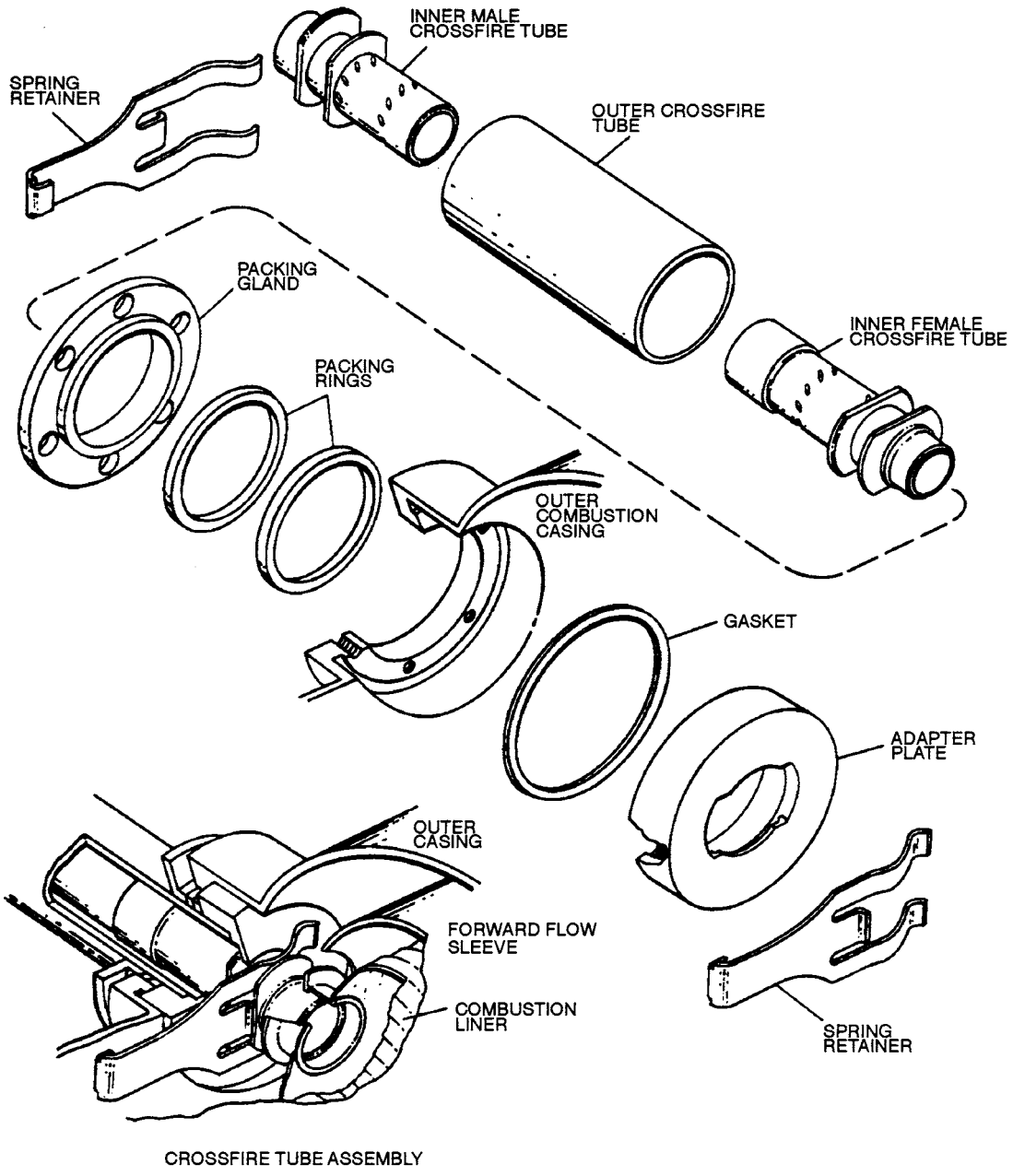



**Figure CI-R25. Combustion Components – Exploded View**

## OPERATION 3 - INSTALL OUTER COMBUSTION CASINGS AND AFT FLOW SLEEVES

1. Assemble the aft flow sleeve if removed to its mating outer casing. See Figs. CI-R17 and CI-R18.

2. Install a new gasket to the aft end of each combustion outer casing.

3. Refer to Fig. CI-R19 and install the two outer combustion casing alignment pins into the compressor discharge casing bulkhead number 5 position as indicated by the arrows.

4. Assemble number 5 outer casing to the compressor discharge casing utilizing the alignment pins. Align the outer casing scribe mark to the compressor discharge casing scribe mark placed during disassembly.

5. Coat all bolts with a light coating of Fel Pro C-102 anti-seize compound install and snug up the bolts. Do not torque bolts at this time. Remove the alignment pins and install the remaining two bolts.

6. Repeat Steps 1 through 4 for combustion casings 4, 6, 3, 7, 2, 8, 1, 9 and 10.

**OPERATION 4 - INSTALL INNER CROSSFIRE TUBE ASSEMBLIES AND CHECK CLEARANCES**

**Note:** Each outer combustion casing should have one adapter plate already in place. It was not removed when disassembly, Operation 14 in this procedure, was performed.

1. Temporarily install the remaining adapter plates into their respective combustion casing without the gasket. See Fig. CI-R20.

2. Install every other adapter plate bolt and finger tighten.

3. Install the forward flow sleeves to the combustion casing, install bolts and tighten. See Fig. CI-R21.

4. Install male and female crossfire tube on each combustion casing adapter plates. Align the crossfire tubes contoured flanges with the adapter plates contoured opening and move the tube radially outward and rotate 90° to seat the crossfire tube outer flange on the adapter plate.

5. Insert the crossfire tube spring retainers through the flange cutouts provided in each forward flow sleeve and seat each retainer inner tang against the flat portion of each crossfire tube flange. See Fig. CI-R22.

**Note:** It will be necessary to spring the retainers outer tang into the notch provided in each forward flow sleeve flange cutouts.

6. Using the crossfire tube clearance tool, verify that 0.010 inch minimum radial crossfire tube clearance has been achieved for each assembly. See Fig. CI-R23.

7. In instances where less clearance exists, the combustion casing with insufficient clearance will require minor realignment. Loosen the combustion casing bolts, realign the combustion casing to achieve crossfire tube clearance. After each crossfire tube assembly clearance is satisfied, final tighten and torque the outer combustion casing bolting.

8. Record the crossfire tube minimum and maximum clearances for each crossfire tube assembly.

9. After all crossfire tube clearances have been verified, remove each combustion casing spring retainer, forward flow sleeve, crossfire tubes and the adapter without the gasket.

**OPERATION 5 - INSTALL OUTER CROSSFIRE TUBES, ADAPTER PLATES, GASKETS, PACKING RINGS AND PACKING GLANDS**

1. Starting with the number five (5) position partially insert an outer crossfire tube through its mating crossfire tube opening.

2. Coat the inside and outside diameter of the packing rings with Vaseline and slide packing rings, packing glands into position on the tube, making sure the sequence of assembly of the parts is duplicated during this operation. See Fig. CI-R23.

3. From inside of the mating outer combustion casing crossfire tube opening, install one gasket and adapter plate. Coat the bolting with Fel Pro C-102 anti-seize compound. Install all bolts and torque to 33-37 lb. ft. in a staggered sequence.

4. Carefully slide the packing rings on each end of the outer crossfire tube into the packing ring cavities provided in the combustion casing.

5. Slide each packing ring into position. Coat the packing gland bolts with a light coating of Fel Pro C-102 anti-seize compound, install bolts and tighten the bolts to achieve a 0.140-0.180 inch gap between the packing glands and the crossfire tube flange of the combustion casing. See Fig. CI-R24.

6. Install the outer crossfire tubes, packing rings, packing gland gaskets and adapter plates in the remaining combustion casings per Steps 1 through 5.

**OPERATION 6 - INSTALL FORWARD FLOW SLEEVE, INNER CROSSFIRE TUBES, COMBUSTION LINERS AND SPRING RETAINERS (See Fig. CI-R25)**

**CAUTION: Inspect each outer combustion casing for foreign objects or debris prior to proceeding with assembly steps.**

1. Install each combustion casing forward flow sleeve.

2. Coat the flow sleeve bolts with a light coating of Fel Pro C-102 anti-seize compound. Install bolts and tighten.

3. Reinstall the inner male and female crossfire tubes per Operation 4, Step 4.

**Note:** In preparing for the next step, installation of the combustion liner, it is necessary to move each combustion casing crossfire tube radially outboard to provide space for the combustion liner assembly.

4. Install each combustion liner into the forward flow sleeve.

**CAUTION: Care should be exercised so that the combustion liners spring seals and crossfire tube air seals are not damaged. In addition it is important to align the combustion liner stops with the mating liner supports in the forward flow sleeve. The combustion liner should not be rotated once the liners spring seals are engaged into the transition piece.**

5. Seat each crossfire tube on its adapter plate by reaching through the combustion liners fuel nozzle opening and guiding each crossfire tube into position.

**CAUTION: Care should be exercised in this step to avoid damaging the combustion liners crossfire tubes air seal.**

6. Reinstall the crossfire tube spring retainers per Operation 4 Step 5.

**CAUTION: Verify that the amount that each spring retainer must be deflected prior to seating the spring retainers outer tang on the forward flow sleeves cutout notch. Minimum deflection allowed is 0.350 inch. See Fig. CI-R26.**

7. With the crossfire tube field tool provided, verify that 0.080 inch minimum clearance exists between each crossfire tube and the combustion liners crossfire tube collar. See Fig. CI-R27.

**CAUTION: Be sure to bottom the end of the clearance checking tool on the underside of the crossfire tube air seal to avoid a false reading. In the event that minimum clearances are not satisfied, examine the combustion liner and forward flow sleeve stops for damage or excessive wear.**

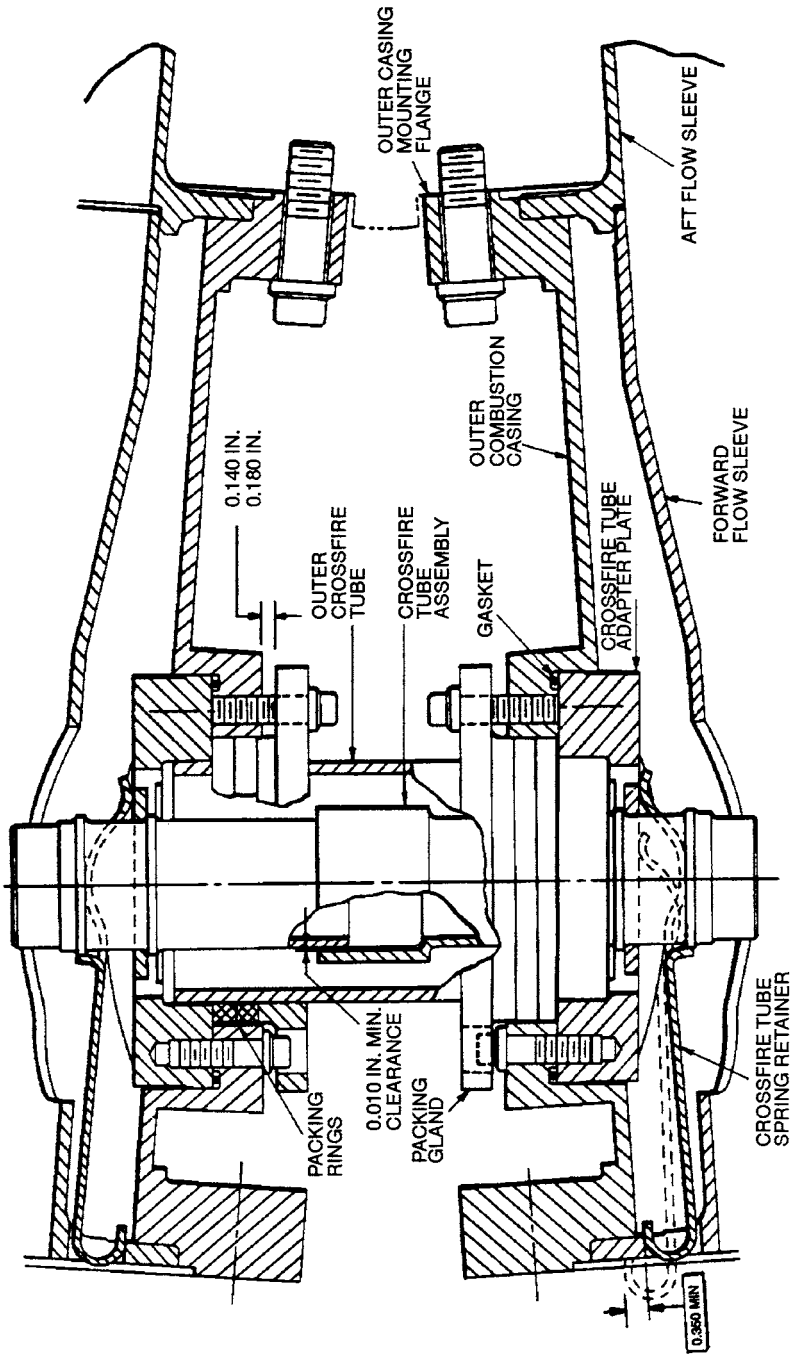



**Figure CI-R26. Crossfire Tube Spring Retainer Assembly Deflection Checks**

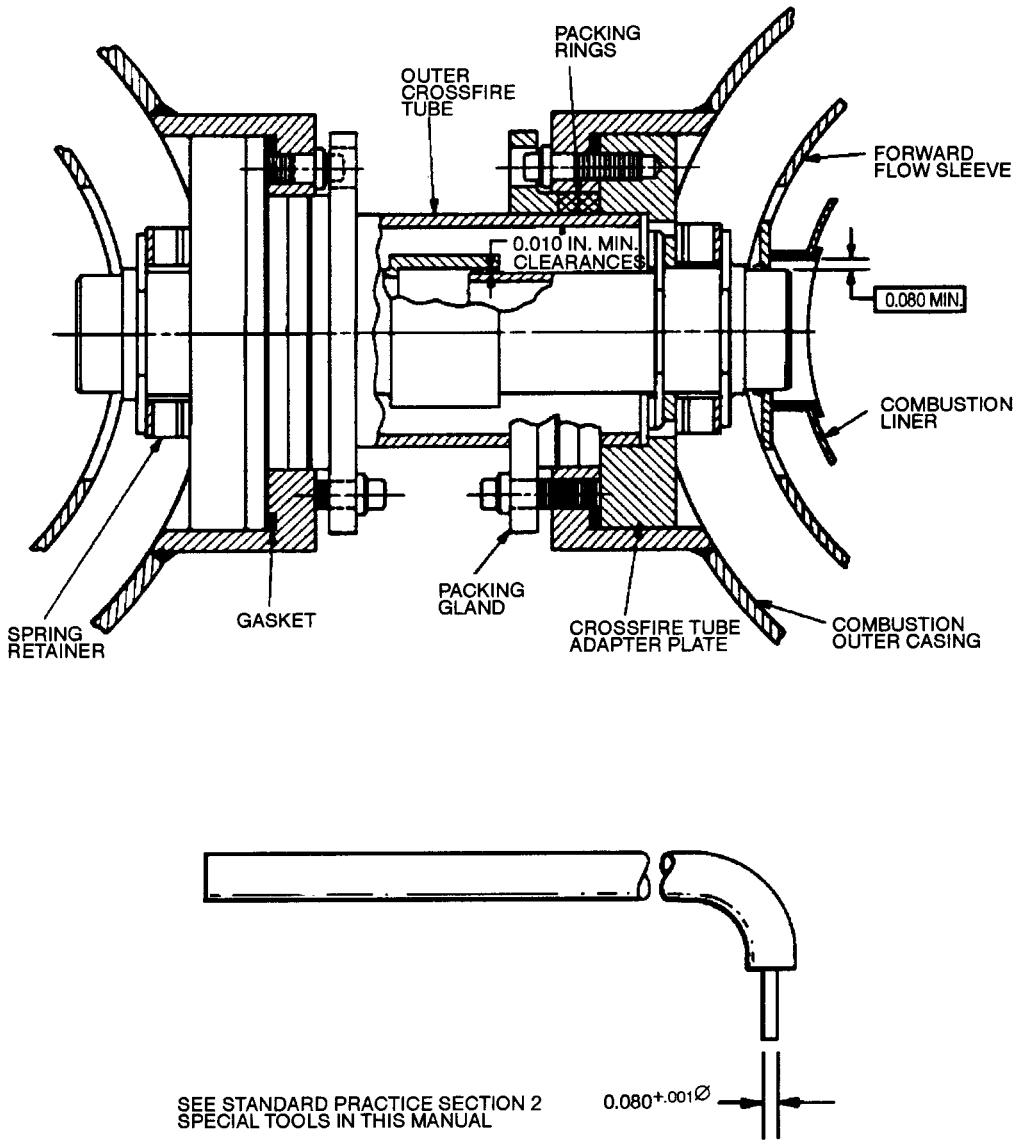



**Figure CI-R27. Crossfire Tube and Crossfire Tube Collar Clearance Checks**

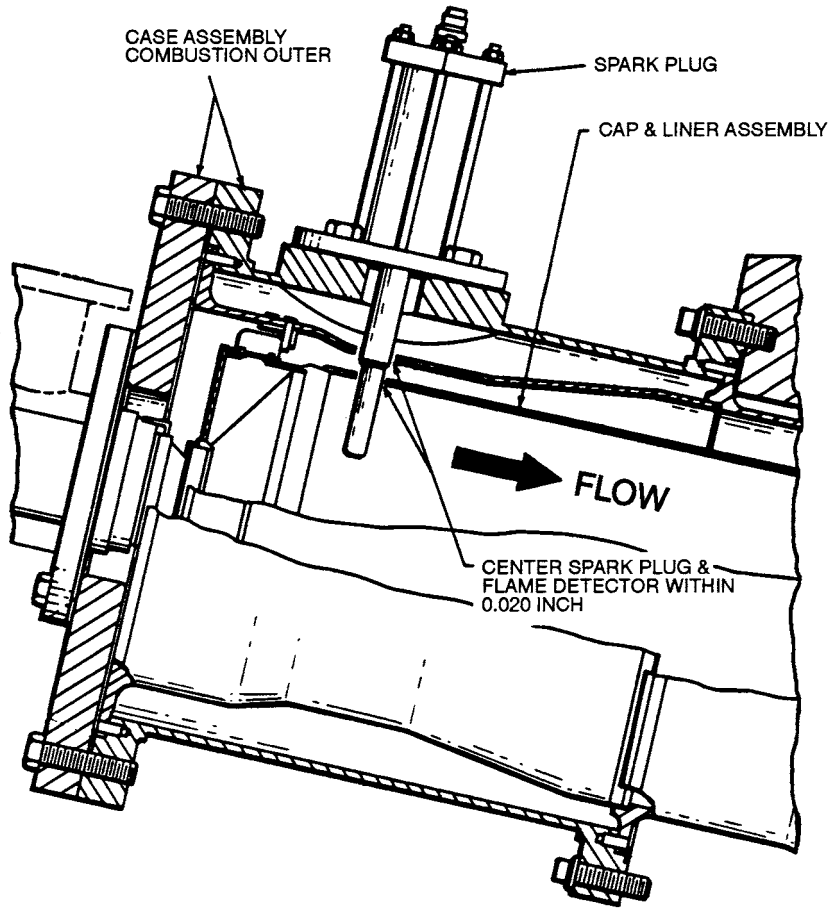



**Figure CI-R28. Spark Plug and Flame Detector Clearance Checks**

## OPERATION 7 - ASSEMBLE FLAME DETECTORS AND SPARK PLUGS TO OUTER COMBUSTION CASINGS

**Note:** Refer to the Combustion Chamber Arrangement drawing in the Parts Lists and Drawings volume of the Service Manual for proper chamber location for flame detectors and spark plugs. Dependent on which configuration applies, flame detectors can be located in chamber numbers 7 and 8 or 2, 3, 7 and 8. Spark plugs can be located in chamber numbers 1 and 2 or 1 and 10.

1. Install new gasket, assemble flame detector bodies with the valve handles in an upright position to their respective combustion casings.

2. Lightly coat bolts with Fel Pro C-102 anti-seize compound and bolt up flame detector bodies to outer combustion casing flanges.

**Note:** Center the flame detector probe to within 0.020 inches of the combustion liner holes centerline. See Fig. CI-R28.

3. Reassemble flame detectors to flame detector bodies and reconnect wiring to flame detectors.

4. Replace enclosure screens on flame detectors.

5. Reconnect power supply to flame detectors.

6. Install new gaskets and assemble spark plugs to their respective combustion casings.

7. Lightly coat bolts with Fel Pro C-102 anti-seize compound and bolt up spark plugs.

**Note:** Center the spark plug piston electrode assembly within 0.020 inch of the combustion liner holes centerline. See Fig. CI-R28.

8. Check electrode assembly for binding and correct if necessary.

9. Reconnect spark plug leads.

**CAUTION: Do not allow piston/electrode assembly to impact within the unit when making the above check.**

10. Lockwire connector to the spark plug.

**OPERATION 8 - CLOSE UP AND BOLT COMBUSTION CASING HINGED COVERS**

1. Prior to reclosing the hinged combustion casing cover, inspect the inside of the combustion casing for foreign objects and make a final check for security of installed pieces.

2. Install a new cover gasket in the groove of each combustion casing and close the cover.

3. Lightly coat the cover bolts with Fel Pro C-102 anti-seize compound and bolt up the combustion casing covers. Torque the bolts to 129-148 lb. ft. 180 degrees apart and work around the bolt circle in this manner.

**OPERATION 9 - RECONNECT FALSE START DRAIN LINES**

1. Reconnect the false start drain lines to combustion casing covers 3, 4, 5, 6 and 7.

2. Tighten the swagelok nut until the collar cannot be rotated by hand.

**OPERATION 10 - INSTALL 11TH STAGE COOLING AND SEALING AIR LINES**

1. On reassembly of piping, use new gaskets on all gasketed joints.

2. Assemble 11th stage piping removed being careful to replace orifices in their proper location from where they were removed at disassembly.

**Note:** In the event that the location identification of the orifices become lost, refer to the Reference Drawings volume of this Service Manual. The cooling and sealing air piping diagram in this section will give the proper location and orifice sizes required.

**OPERATION 11 - INSTALL FUEL NOZZLES**

1. Install new gasket on fuel nozzle assembly and place fuel nozzle on combustion casing cover from which it was removed.

**CAUTION: Do not impact fuel nozzle on liner during this operation.**

2. Lightly coat all bolting with Fel Pro C102 anti-seize compound.

3. Position fuel nozzle assembly on cover and bolt up fuel nozzle. Torque bolts to 60-68 lb. ft.

4. Keep fuel nozzle atomizing air and or fuel line fuel nozzle connection covered until the atomizing air and or fuel lines to the fuel nozzles are connected.

**Note:** On dual fuel units (if applicable) install purge line check valves also.

**OPERATION 12 - INSTALL FUEL OIL CHECK VALVES**

1. Lightly coat the male pipe threads of the outlet ends of the fuel check valves with Tefloc sealant X9366 so that the first thread is free of any sealant.

**CAUTION: Install the check valves to the fuel nozzles with the arrow in the direction of flow.**

2. Assemble the fuel check valves to the liquid fuel inlet fittings of the fuel nozzle assemblies. Each check valve is to be installed on the same fuel nozzle from which it was removed at disassembly.

**CAUTION: Securely hold the fuel nozzle fitting to prevent twisting the fuel tube when installing check valves.**

**OPERATION 13 - INSTALL LIQUID FUEL TUBING**

1. Remove protective covers from free ends of fuel lines, and insure that dirt and foreign objects have not entered the tubing.

2. Reinstall the fuel lines to the same combustion casings from which they were removed at disassembly.

3. Where applicable install liquid fuel purge lines also at this time.

**Note:** The end of the tube with its preswaged ferrules must be inserted into the fitting body until the front ferrule seats in the fitting. Tighten the nut by hand, then using a wrench continue tightening the nut until the collar cannot be rotated by hand.

**OPERATION 14 - INSTALL WATER INJECTION TUBING PIGTAILS (IF APPLICABLE)**

1. Remove protective covers from water injection tubing pigtail lines, fuel nozzles and water injection manifold.

2. Reinstall the water injection pigtail lines to the same fuel nozzle bodies from which they were removed at disassembly.

**Note:** The end of the tube with its preswaged ferrules must be installed into the fitting body until the front ferrule seats in the fitting. Tighten the nut by hand, then using a tubing wrench continue tightening the nut until the collar cannot be rotated by hand.

**OPERATION 15 - INSTALL ATOMIZING AIR PIPING**

1. Lightly coat all bolts with Fel Pro C-102 anti-seize compound.

2. Position the atomizing air line between the atomizing air ring manifold and the fuel nozzle assembly. Install new gaskets. Do not force flanges during assembly.

**CAUTION: During assembly of flanges parallelism between flanges must be maintained. If piping is not strain free within 0.125 inch, it must be corrected or replaced.**

3. Install all bolts finger tight and snug bolts up with wrench.

4. Torque all bolts to 60-68 lb. ft. using a crisscross bolt tightening sequence.

**OPERATION 16 - INSTALL FUEL GAS PIPING (IF APPLICABLE)**

1. Check to insure no foreign objects have entered the piping.

2. Install new gaskets and bolts into the flanges, use Fel Pro C-102 anti-seize compound on all bolts.

**CAUTION: During assembly of flanges parallelism between flanges must be maintained. If piping is not strain free within 0.125 inch, it must be corrected or replaced.**

3. Install all bolts finger tight and snug up bolts with a wrench.

4. Torque all bolts to 60-68 lb. ft. using a crisscross tightening sequence.

**OPERATION 17 - INSTALL STEAM INJECTION PIPING PIGTAILS (IF APPLICABLE)**

1. Remove protective covers from steam injection pigtails, steam injection manifold and combustion casing cover steam injection flanges.

2. Install the steam injection pigtails, using new gaskets, to the same combustion casing cover from which they were removed at disassembly. The matchmark tabs should correspond to matchmark on the manifold and the combustion casing cover.

3. Coat bolts with Fel Pro C-102 anti-seize compound.

4. Install all bolting in all the pigtail mating flanges and just snug bolting up.

**CAUTION: During assembly of flanges, parallelism between flanges must be maintained. If piping is not strain free within 0.125 inch, it must be corrected or replaced.**

5. Torque all bolting to 60-68 lb. ft. using a crisscross tightening sequence.

**OPERATION 18 - INSTALL TURBINE CASING BLANK COVER ON HAND HOLE PORT OR THE ATOMIZING AIR EXTRACTION MANIFOLD WHICHEVER APPLIES**

1. Install new gasket(s), coat the bolts with Fel Pro C-102 anti-seize compound.

2. Install blank cover or atomizing air extraction manifold to turbine casing.

3. Install all bolting and torque to 129-148 lb. ft. using a crisscross tightening sequence.

**OPERATION 19 - ASSEMBLE TURBINE COMPARTMENT ROOF AND SIDE PANELS**

1. Reassemble lagging side panel vertical beams, if any were removed during disassembly operation.

2. Attach four eyebolts to threaded receptacles on the roof. Rig the roof using a chainfall for lifting.

3. Apply new gasketing or caulking to roof section.

4. Lift the turbine roof over the turbine compartment and carefully lower, using the chainfall, into position.

5. Apply Fel Pro C-102 anti-seize compound to the bolting and bolt in place.

6. Reconnect the blower(s) motor leaks and conduit connection at underside of the roof.

7. Reconnect exhaust frame cooling piping flanges.

8. Reconnect (if applicable) VA14 valve, electrical wiring and conduit.

9. Reconnect CO2 piping and any other piping removed.

10. Make a complete visual inspection at topside and underside of roof to ensure all components have been assembled and reconnected.

11. Install turbine compartment lagging panels.

## STARTUP CHECKS

### OPERATION 20 - CLEANUP, VISUAL INSPECTION, PRESTART/STARTUP CHECKS

1. Clean up all fuel oil, lube oil and water spills.

2. Make a visual inspection of the unit, check to insure that the following has been completed.

    a. Check that all removed piping has been replaced.

    b. Check that all removed conduit has been replaced and electrical connections made. Confirm function of all devices.

    c. Confirm proper function of all devices.

    d. Correct any fuel, oil and water leaks observed.

    e. Inspect inlet and exhaust plenums.

3. Reconnect power to all systems disabled at start of job.

4. Rearm the CO2 system and manually reset and latch the CO2 vent dampers.

5. Perform precanking checks as follows:

**Note:** It is recommended that the following checks be made under the guidance of your GE Field Service Representative.

    a. Record all panel counter readings.

    b. Check operation of cooldown/emergency lube pumps.

    c. Check operation of auxiliary hydraulic pump.

    d. Make visual inspection to see if oil is flowing in bearing drains.

    e. Check atomizing air compressor (if applicable).

    f. Block out excitation and generator breaker equipment.

    g. Place unit on ratchet and observe for leaks, rubs, and control deficiencies.

    h. Check to insure fuel pump is primed.

6. Perform cranking checks as follows:

    a. Observe pickup rpm of speed sensors versus time.

    b. Inspect for casing air leaks.

c. Observe and record vibration channel outputs.

d. Observe and record crank speed.

e. Check starting means for proper operation.

f. Listen for rubbing noises.

g. Check temperature of lube oil in bearing drains or at the bearing header and tank.

h. Observe all panel and hydraulic manifold pressure gauge readings for abnormalities.

i. Trip unit using the emergency trip, and listen for abnormal noises from gas turbine and associated equipment during coastdown.

j. Observe proper drop-out rpm of speed sensors.

7. Initiate a start and perform the following zero to full speed, no load firing checks:

a. Observe applicable firing check parameters, relay sequences, exhaust temperature, vibration and speed.

b. Check spark plugs and flame detectors.

c. Observe and record acceleration rates.

d. Record full set of vibration readings.

e. Observe and record temperature suppression set point.

f. Note and record pick-up time of all appropriate electrical speed relays.

g. Check full speed-no load rpm.

h. Check digital set point.

i. Take a complete set of running data. See Running Inspections - Section 4 for this instruction.

j. Check for air, oil, fuel and water leaks.

k. Stake sight port pipe plugs in place, using two stakes on each plug, after shutdown.

l. Periodically recheck torque on the fuel nozzle mounting flange bolts after shutdown.

8. Perform load checks (if applicable):

   a. Take full set of load data.

   b. Check for air, oil, fuel, gas and water leaks.

9. Observe normal unloading and shut-down as follows:

   a. Note and record coastdown time.

   b. Listen for abnormal noises from gas turbine and associated equipment during coastdown.

   c. Observe unit vibration readings during coastdown.

   d. Check to ensure sight port plugs have been staked in two places, after shutdown.

   e. Periodically recheck torque on the fuel nozzle mounting flange bolts after shutdown.

10. AES/GT-FF Field Inspection Forms

    Submit a complete set of all AES/GT-FF Inspection Forms and operating data taken during the Combustion Inspection, including AES/GT-FF-6000 Site Information to your GE Field Service Representative.